

E3 Fiber Connect Defense Reviewer

Our module-by-module manual: the problem each part solves, its input, process and output, the data structures and algorithms inside, their time and space complexity, why we chose them, how to demonstrate them live — and a plain-language walk through the code.

 E3 Fiber

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Joshua Santos
1 Plan Selection · 2 Application · 3 Application Tracking

Justin Banaag
4 Application List · 5 Walk-in Override (New Application) · 6 Create Account

Dela Cruz Riceerich
7 Admin Login · 8 Accounts Management · 9 Admin/Staff Creation & Management · 10 Admin Registration & Approval

Aaron Sebastian
11 Billing: Client Billing List · 12 Billing: Bill Generation · 13 Payments (Payments Validation)

E3 Fiber Connect — fiber internet provider in Santa Maria, Bulacan · Built with HTML, CSS, Bootstrap 5 and JavaScript, with every data structure and algorithm written by hand · September 2026

E3 Fiber Connect · Defense Reviewer

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How to use this reviewer

Every module has the same layout, in the same order the panel asks the questions:

1 • Business process

What task or problem of the client (E3 Fiber Connect) the module solves.

2 • Input

The data the module needs — what the user types or picks, and the stored records it reads.

3 • Process

The data structure and algorithm used, step by step, with an Input → Process → Output picture.

4 • Output

What the module produces: records, screens, messages.

5 • Complexity analysis

Time and space complexity of each operation, and what n means.

6 • Justification

Why that data structure or algorithm, and why it fits the problem better than the alternatives.

7 • Demonstration

How to show it live, real screenshots, and a trace computed from the real code and data.

Code walkthrough

The actual functions from our files (with line numbers) and a basic explanation of each step.

Each member also has a **one-page summary** before their modules: a short pitch, the structures they must know, and likely panel questions with answers. All numbers in the traces were computed by running our real JavaScript on the sample data — they match the screenshots.

Running the demo

1. Open `index.html` by double-clicking it (Edge or Chrome). No server or internet is needed.
2. Staff demo accounts are on the sign-in page (one tap fills the form):

Role	E-mail	Password
Owner	maria.santos@e3fiberconnect.ph	owner123
Admin	juan.delacruz@e3fiberconnect.ph	e3admin
Support	paolo.reyes@e3fiberconnect.ph	support123
Pending registration	lorenzo.pascual@e3fiberconnect.ph	lorenzo2026

3. **Reloading the page resets all data** to the sample data (the arrays live in memory). The site asks before a reload if something was changed. Sample dates are relative to today, so bills are always “due this week”, “overdue”, and so on.
4. Every admin action can be undone (Undo button, the toast’s Undo, or Ctrl + Z).

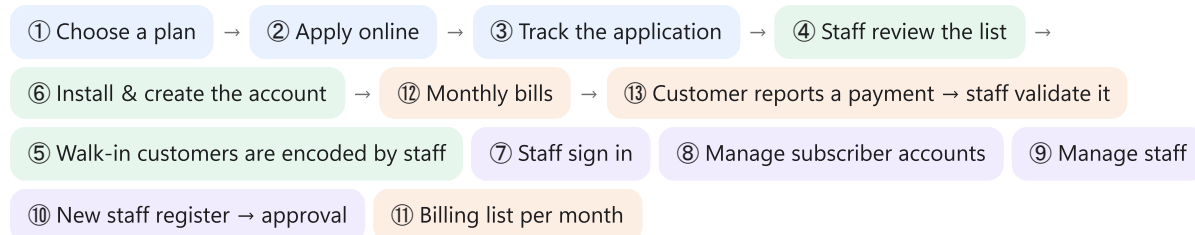
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System overview

E3 Fiber Connect is a fiber internet provider in Santa Maria, Bulacan. Before this system, applications were on paper, customers called to ask about their status, bills were computed by hand and online payments (GCash, Maya, bank) had to be checked one by one. Our website handles the whole customer life cycle and the staff work behind it.

The customer and staff flow



How the program is built

One page, many screens

Arrays live in the browser's memory, and memory is wiped whenever a new page opens — data cannot transfer between separate HTML files. So the whole site is **one** `index.html`: moving between screens only shows and hides sections (the part after # in the address picks the screen). The arrays survive every screen change.

Layers

dsa/ — our own arrays, strings, search, sort, stack, queue and hash table.

backend/ — the “database” (arrays) and business rules (applications, billing, payments ...).

ui/ — router, dialogs, notifications, forms.

views/ — one script per screen.

The data (arrays kept sorted by their key)

Array	Key	How it stays sorted	Used by modules
applications	referenceNo – E3-2026-004887	new numbers are always bigger → append O(1)	2, 3, 4, 5, 6
subscribers	accountNo (= reference no.)	installations finish in any order → sorted insert O(n)	6, 8, 11–13
bills	billId – BILL-202609-004872	sorted insert O(n)	8, 11, 12, 13
payments	paymentId – PAY-000002	append O(1)	13
staffMembers	id – STF-0002	append O(1)	7, 9, 10
registrations	registrationId – REG-000003	append O(1)	10
tickets, activityLog	ticketNo / time	append O(1)	support, activity

Two **hash tables** sit beside the arrays as indexes: `staffEmailIndex` (e-mail → staff id; modules 7, 9, 10) and `paymentReferenceIndex` (payment reference → payment ids; module 13). **Queues** give first-come-first-served order (review, payments, staff requests, support, oldest bill). **Stacks** power Undo, the Back button and open dialogs.

The project rules and how we met them

Rule from our instructor	How the code follows it
HTML, CSS, Bootstrap and JavaScript only; JS is the “backend”	No server, no framework, no database program. Bootstrap is used for the grid and the FAQ accordion.
Data cannot transfer between pages (arrays)	Single-page app with a hash router; no <code>localStorage</code> , no cookies, no server.
Procedural, not OOP	Only functions and plain records. No <code>class</code> , no <code>this</code> , no methods on data. A queue is the record { <code>items</code> , <code>front</code> , <code>count</code> , <code>capacity</code> } plus functions like <code>enqueue(queue, item)</code> .
No built-in helpers: <code>push</code> , <code>pop</code> , <code>shift</code> , <code>unshift</code> , <code>splice</code> , <code>sort</code> , <code>indexOf</code> , <code>includes</code> , <code>find</code> , <code>filter</code> , <code>map</code> ...	All replaced by our own functions in <code>assets/js/dsa/</code> (e.g. <code>arrayAppend</code> , <code>arrayInsertAt</code> , <code>linearSearch</code> , <code>binarySearch</code> , <code>insertionSort</code>). No regular expressions either.
State time and space complexity	Written above every function in the code, and measured live on the admin <i>Algorithms</i> page.

Data structures & algorithms primer

n = number of records · k = length of a text or number of results · b = number of hash buckets. All of these live in `assets/js/dsa/`.

Array (our “table”)

4871
[0]

4872
[1]

4874
[2]

4876
[3] new

`arrayAppend` writes at index `length`: $O(1)$. `arrayInsertAt` shifts the later items right: $O(n)$. `arrayRemoveAt` shifts left: $O(n)$. Reading `a[i]`: $O(1)$.

Linear search — $O(n)$

Starter
1st

Stream
2nd

Power
match

Pro

Ultimate

Check the items one by one until one matches. Works on any array (sorted or not). Best case $O(1)$, worst $O(n)$.

Binary search — $O(\log n)$

4871

4872

4873

4874

4875
middle

4876

4877

Needs a **sorted** array. Look at the middle; if the target is bigger keep the right half, else the left half. Each step halves the range: 17 records need at most 5 steps, 100,000 need about 17.

Lower bound + sorted insert (“put”) — $O(n)$

4872

4874

4875
goes here

4878
→ moves

4881
→ moves

A binary search finds the first position whose key is $\geq$ the new key ($O(\log n)$); then the later records shift one slot right ($O(n)$). The array stays sorted, so binary search keeps working.

Insertion sort — best $O(n)$, worst $O(n^2)$, stable

Take the next item and slide the bigger items to the right until its place is found. Very fast when the list is already almost sorted — our tables usually are — so it is our default.

Selection sort — always $O(n^2)$, not stable

For each position, scan the rest for the item that belongs there and swap it in. Always $n(n-1)/2$ comparisons, but at most $n-1$ swaps — the fewest moves.

Bubble sort — best $O(n)$, worst $O(n^2)$, stable

Walk the list swapping neighbours that are in the wrong order; after each pass the biggest item has “bubbled” to the end. Stops early when a pass makes no swap.

Naive string matching — $O(n \cdot m)$

Used by every search box: try every starting position of the text and compare letter by letter (“jua” finds “Juan Dela Cruz”), ignoring capital letters.

Stack — LIFO, push / pop / peek $O(1)$

Approve A

Reject B

Confirm P
top

Record { `items`, `top` }. Undo pushes every admin action and pops the newest (last in, first out). Also the Back button and open dialogs.

Queue — FIFO, circular buffer, enqueue / dequeue $O(1)$

PAY-2
[0] front

PAY-3
[1]

PAY-4
[2]

·
[3] rear

Record { `items`, `front`, `count`, `capacity` }; the rear is `(front + count) % capacity`, so nothing is shifted. First come, first served: review queue, payment validation, staff requests, support tickets, oldest bill first.

Hash table — separate chaining, $O(1)$ average

[0] empty

[1] key A → value

[2] key B → value → key C
→ value

[3] empty

The key is turned into a bucket number with the djb2 hash (`hash × 33 + character`, then the remainder). Keys that share a bucket (a collision) form a short chain. Grows when there are more than 0.75 keys per bucket.

Why procedural?

Every structure is a plain record plus functions that receive it (e.g. `stackPush(stack, item)`) instead of objects with methods. This follows the project rule and makes each step of the algorithm visible.

Joshua Santos

Module 1 Plan Selection · Module 2 Application · Module 3 Application Tracking

60-second pitch

My modules are what a new customer does before any staff member is involved. **Plan Selection** helps them pick the cheapest plan that is fast enough — a *linear search* through the plans in speed order — and lets them sort the plans with *selection sort*. **Application** collects and checks their details in five steps, then adds the application to the end of the `applications` array with the next reference number, which keeps the array sorted. **Application Tracking** uses that sorted order: a *binary search* finds any reference number in at most $\lceil \log_2(n+1) \rceil$ comparisons and shows the progress timeline with the name partly hidden.

Structures and algorithms I must be able to explain

Where	Structure / algorithm	Time	Space
Find a plan, the recommender	Linear search over the 5 plans	$O(n)$	$O(1)$
Plans in speed order	Insertion sort (already sorted $\rightarrow$ best case)	$O(n)$ best, $O(n^2)$ worst	$O(n)$ copy
Sort the plan cards	Selection sort	$O(n^2)$ always	$O(n)$ copy, $O(1)$ extra
Save an application	Array append (keeps the key order)	$O(1)$	$O(1)$
Duplicate check	Linear search on open applications	$O(n)$	$O(1)$
Track an application	Binary search on the reference number	$O(\log n)$	$O(1)$

Likely panel questions

Why linear search for the plans and not binary search?

There are only 5 plans; binary search would save at most 2 comparisons. The recommender also needs the first plan whose speed is at least a value — a single pass in speed order does exactly that.

Why is the applications array always sorted without sorting it?

Reference numbers come from a counter that only goes up (E3-2026-004886, ...004887). A new application always has the largest number, so appending it at the end keeps the order — $O(1)$, no sort needed.

What is the worst case of your binary search?

$\lceil \log_2(n+1) \rceil$ comparisons: 5 for our 17 applications, 14 for 10,000. A linear search would need up to n .

Why selection sort for the plan cards?

It makes the fewest swaps (at most $n-1$) and its cost is predictable: always $n(n-1)/2 = 10$ comparisons for 5 plans — the page prints that number under the cards.

How do you validate without regular expressions?

Each rule scans the characters with a loop — e.g. the e-mail must have exactly one “@”, allowed characters only and a dot after it; the mobile number is reduced to its digits and must be 11 digits starting with 09.

Why hide part of the name on the tracker?

Anyone who knows or guesses a reference number can open the tracker, so it shows “Ana R.” instead of the full name, and never the address or ID.

Plan Selection

Screens	Plans page (#/plans) and step 1 of the application (#/apply/power)
Files	backend/plans.js · views/public/plans.js · views/public/apply.js · dsa/search.js · dsa/sort.js
Data	PLANS — array of 5 plan records { id, name, speed, price, bestFor, devices, popular }
DSA	Linear search · insertion sort · selection sort

1 Business process

What task or problem from the client are you solving?

E3 sells five plans — Starter (50 Mbps, ₱800), Stream (70 Mbps, ₱1,000), Power (100 Mbps, ₱1,200), Pro (200 Mbps, ₱1,500) and Ultimate (300 Mbps, ₱2,000). Customers often don't know which speed they need, so they either overpay or pick a plan that is too slow and complain later. This module **recommends the cheapest plan that is still fast enough** for the household, lets the customer **sort and compare** the plans, and carries the chosen plan into the application form.

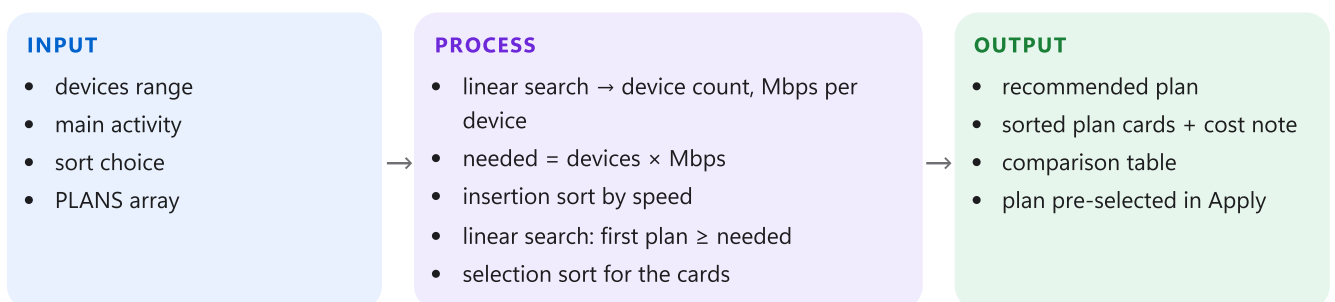
2 Input

What data is needed by the module?

Input	Where it comes from	Values
Number of devices	segmented control	1–3, 4–6, 7–10, 11+ (counted as 3, 6, 10, 14 devices)
Main activity	segmented control	Browsing 4 · Streaming 8 · Work & school 10 · Gaming 16 Mbps per device
Sort choice	drop-down	speed or price, low→high or high→low (e.g. price-desc)
Chosen plan	"Choose" button / radio card	plan id, e.g. power (sent in the address #/apply/power)
Stored data	PLANS, DEVICE_RANGES, PLAN_ACTIVITIES	fixed arrays of records

3 Process

What data structure and/or algorithm did you use? How does it work step by step?



Recommender (recommendPlan):

- Find the chosen device range and activity with a **linear search** (findDeviceRange, findActivity).
- Compute the speed needed: devices × Mbps per device (e.g. 10 × 10 = 100 Mbps).
- Put the plans in speed order with **insertion sort**. They are stored in speed order already, so this is its best case: $n-1 = 4$ comparisons, 0 moves.
- Linear search** from the slowest plan: stop at the first plan whose speed $\geq$ the needed speed — that is also the cheapest one that is fast enough.
- If no plan is fast enough, recommend the fastest plan (Ultimate).

Sorting the cards (selectionSort): copy the array; for each position scan the rest for the plan that belongs there (e.g. the most expensive) and swap it in; count comparisons and swaps and print them under the cards.

Choosing: the button opens `#!/apply/<id>`; the Apply screen checks the id with `findPlan` (linear search) and pre-selects that plan in step 1.

4 Output

What result does the module produce?

The recommended plan with the reason ("About 100 Mbps for 7–10 devices ... found by checking 3 plans"), the plan cards in the chosen order with "Sorted with selection sort: 10 comparisons, 2 swaps", a feature comparison table, and the chosen plan already selected on the application form with its price summary.

5 Complexity analysis

What is the time complexity? What is the space complexity?

Operation	Time	Space	Note (n = number of plans = 5)
<code>findPlan</code> , <code>findActivity</code> , <code>findDeviceRange</code>	$O(n)$	$O(1)$	linear search
Insertion sort by speed	$O(n)$ here (best case); $O(n^2)$ worst	$O(n)$ copy, $O(1)$ extra	plans are stored in speed order
Find the first plan fast enough	$O(n)$	$O(1)$	stops early
recommendPlan overall	$O(n)$	$O(n)$	
Selection sort of the cards	$O(n^2)$ in every case — $n(n-1)/2 = 10$ comparisons	$O(n)$ copy, $O(1)$ extra	at most $n-1$ swaps
Comparison table	$O(n \cdot r)$	$O(n \cdot r)$	$r = 7$ feature rows

6 Justification

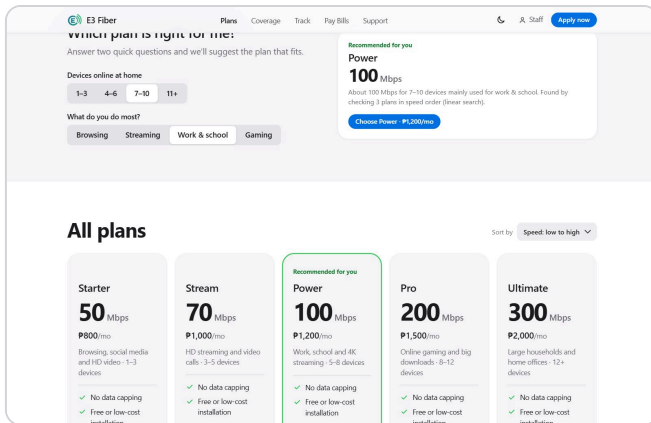
Why did you choose that data structure or algorithm? Why is it appropriate?

- **Array of records:** plans are a small, fixed list that is shown in order — an array is the simplest structure and is read in $O(1)$ by index.
- **Linear search:** with 5 items it is as fast as anything else, needs no sorted key, and naturally answers "the *first* plan that is fast enough". Binary search would need the plans sorted by the searched field and saves at most 2 comparisons.
- **Insertion sort** for the speed order: the plans are already in that order, which is insertion sort's best case (linear time).
- **Selection sort** for the cards: at most $n-1$ swaps and a fixed, easy-to-explain number of comparisons, which we display to the user. For 5 items $O(n^2)$ is only 10 comparisons.

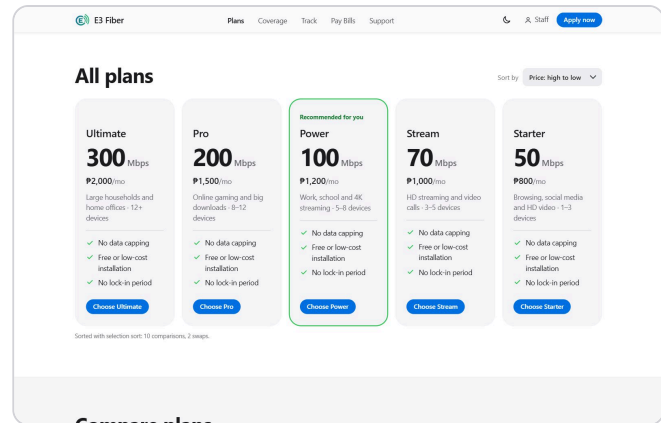
7 Demonstration

Show that the module works correctly — the input, the process and the output.

- 1 Open **Plans**. Choose "7–10" devices and "Work & school" → the recommender shows **Power**.
- 2 Change the sort to "Price: high to low" → Ultimate comes first and the note shows the selection-sort cost.
- 3 Press "Choose Power" → the application opens on step 1 with Power selected.



Input 7–10 devices + Work & school → output Power (100 Mbps needed).



Selection sort by price, high to low, with its cost under the cards.

Trace — the recommender (computed by our code)

Input: 7–10 devices (counted as 10) used mainly for work & school (10 Mbps each). **Needed speed** = $10 \times 10 = 100$ Mbps. Insertion sort put the plans in speed order with 4 comparisons and 0 moves — they were already in order, the best case $O(n)$.

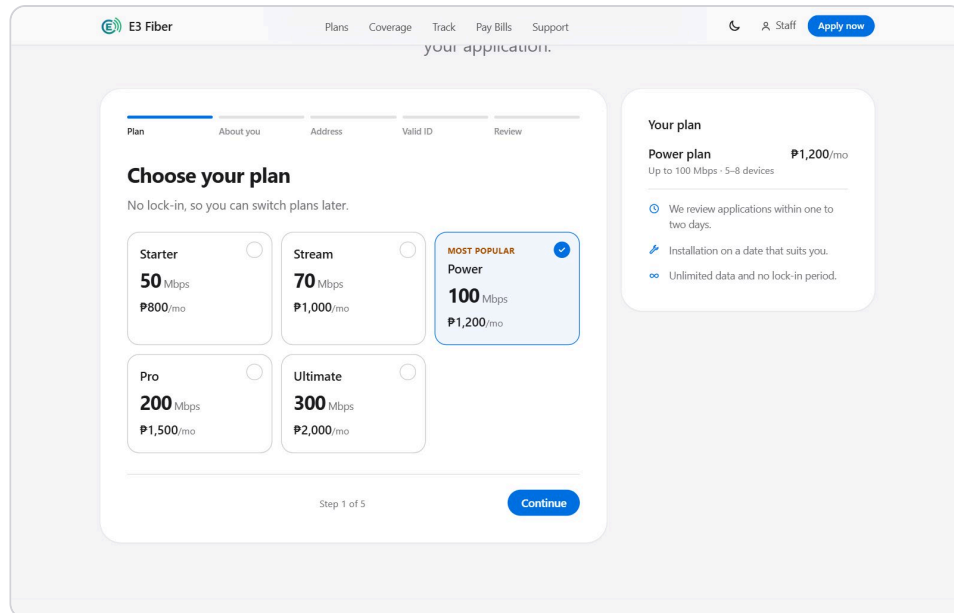
Check	Plan	Speed	Price	Speed $\geq$ 100 Mbps?
1	Starter	50 Mbps	₱800	✗ too slow
2	Stream	70 Mbps	₱1,000	✗ too slow
3	Power	100 Mbps	₱1,200	✓ fast enough → recommended
4	Pro	200 Mbps	₱1,500	not checked — search already stopped
5	Ultimate	300 Mbps	₱2,000	not checked — search already stopped

Output: Power, found after checking 3 of 5 plans (linear search stops at the first match).

Trace — selection sort by price, high to low

Pass	Work	Most expensive of the rest	Array after the pass (blue = sorted part)
start	—	—	Starter ₱800 Stream ₱1,000 Power ₱1,200 Pro ₱1,500 Ultimate ₱2,000
pass 1	4 comparisons	Ultimate → swapped into position 1	Ultimate ₱2,000 Stream ₱1,000 Power ₱1,200 Pro ₱1,500 Starter ₱800
pass 2	3 comparisons	Pro → swapped into position 2	Ultimate ₱2,000 Pro ₱1,500 Power ₱1,200 Stream ₱1,000 Starter ₱800
pass 3	2 comparisons	Power already in place	Ultimate ₱2,000 Pro ₱1,500 Power ₱1,200 Stream ₱1,000 Starter ₱800
pass 4	1 comparison	Stream already in place	Ultimate ₱2,000 Pro ₱1,500 Power ₱1,200 Stream ₱1,000 Starter ₱800

Total: **10 comparisons** (= $n(n-1)/2 = 5 \cdot 4/2$) and **2 swaps**. The real `selectionSort` reports 10 comparisons and 2 swaps — the same numbers shown under the plan cards.



Output: the chosen plan arrives pre-selected in step 1 of the application.

Code walkthrough

assets/js/backend/plans.js · lines 85–106

```

85  /**
86   * recommendPlan – the slowest (cheapest) plan that is still fast enough.
87   * needed speed = devices × Mbps per device for the main activity; then a
88   * linear search over the plans in speed order stops at the first plan whose
89   * speed is at least that much (or returns the fastest plan).
90   * Time O(n) (plans are already in speed order, so insertion sort is O(n)) · Space O(n)
91   */
92  function recommendPlan(rangeId, activityId) {
93      const range = findDeviceRange(rangeId);           // 1. the answers: how many devices ...
94      const activity = findActivity(activityId);         // ... and what they are mostly used for
95      const needed = range.devices * activity.mbpsPerDevice; // 2. speed needed = devices × Mbps per device
96      const bySpeed = insertionSort(PLANS, 'speed', 'asc'); // 3. plans from slowest (cheapest) to fastest
97      let checked = 0;
98      for (let i = 0; i < bySpeed.length; i++) {        // 4. linear search: the first plan fast enough
99          checked++;
100         if (bySpeed[i].speed >= needed) {
101             return { plan: bySpeed[i], neededMbps: needed, checked: checked, range: range, activity: activity };
102         }
103     }
104     // 5. no plan is fast enough → recommend the fastest one
105     return { plan: bySpeed[bySpeed.length - 1], neededMbps: needed, checked: checked, range: range, activity:
activity };
106 }

```

1. Find what the customer picked. `findDeviceRange` and `findActivity` are linear searches over two tiny lists.
2. Multiply the number of devices by the speed each device needs.
3. `insertionSort` returns the plans from slowest (cheapest) to fastest. It works on a copy, so `PLANS` is not changed.
4. The loop is a linear search: the first plan with enough speed is returned, together with how many plans were checked (shown on screen).
5. If the loop ends without a match, the last (fastest) plan is recommended.

assets/js/dsa/sort.js · lines 109–135

```

109  /**
110   * selectionSort – for each position, scans the unsorted part for the item
111   * that belongs there (the smallest, or largest when descending) and swaps it in.
112   * Always makes n(n-1)/2 comparisons but at most n-1 swaps.
113   * Time: O(n²) in every case · Space: O(1) extra (plus the copy) · Stable: no
114   */
115  function selectionSort(records, field, order) {
116      const list = arrayCopy(records);                // sort a copy, so the table keeps its own order
117      const direction = order === 'desc' ? -1 : 1;

```

```

118 let comparisons = 0;
119 let swaps = 0;
120 for (let i = 0; i < list.length - 1; i++) { // 1. fill position i (0, 1, 2 ...) with the right item
121     let best = i;
122     for (let j = i + 1; j < list.length; j++) { // 2. scan everything after it
123         comparisons++;
124         if (compareForOrder(list[j][field], list[best][field], direction) < 0) {
125             best = j; // remember the item that should come first
126         }
127     }
128     if (best !== i) { // 3. swap that item into position i
129         arraySwap(list, i, best);
130         swaps++;
131     }
132 }
133 recordRun('Selection sort', comparisons, swaps);
134 return list;
135 }

```

1. `arrayCopy` makes a copy (our replacement for `slice`) so the original order stays; `direction` is `-1` for “high to low”.
2. The outer loop fills position `i`; the inner loop scans the rest and remembers in `best` the item that should come first.
3. `compareForOrder` compares two values in the chosen direction; empty values always go last.
4. `arraySwap` exchanges two slots (no built-in helper). Only one swap per position → at most $n-1$ swaps.
5. `recordRun` saves the comparison and swap counts, which the page prints under the cards.

Application (online)

Screens	Apply wizard (#/apply) → Application received (#/submitted/<ref>)
Files	backend/applications.js (validateApplication, findOpenApplicationFor, submitApplication) · views/public/apply.js · backend/utlis.js (validators)
Data	applications — array sorted by referenceNo; BARANGAYS (24, sorted); ID_TYPES
DSA	Character-by-character validation · binary search (barangay) · linear search (duplicates, ID types) · array append · queue (review queue)

1 Business process

What task or problem from the client are you solving?

Before, customers filled in paper forms at the office and details were often incomplete or unreadable. The online application lets anyone in Santa Maria apply from home, **checks every field before it is saved** (valid e-mail and mobile, 18 years or older, a barangay in the service area, a valid ID, consent), **blocks duplicate applications** from the same person, and gives a **reference number** that the customer uses to track the application — and that later becomes the account number.

2 Input

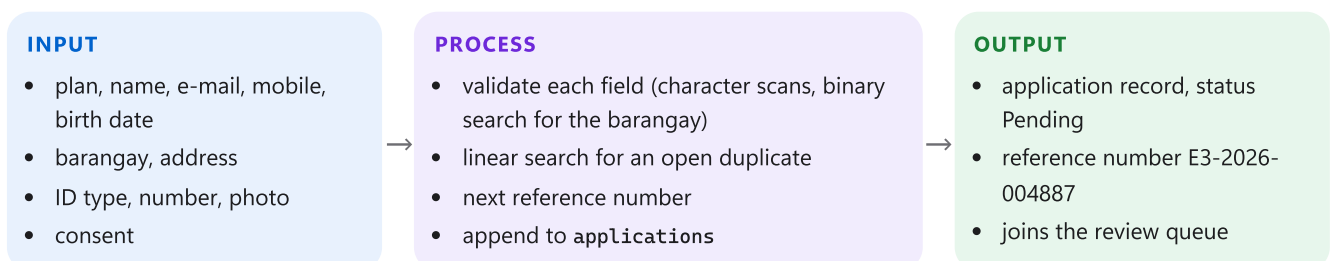
What data is needed by the module?

Step	Field	Rule checked
1 Plan	Plan	must exist in PLANS
2 About you	Full name	first and last name, 3–80 characters
	E-mail	one "@", allowed characters only, a dot in the domain
	Mobile	11 digits starting with 09 (" +63 917..." is accepted and tidied)
	Birth date	real date, not in the future, age 18–120
3 Address	Barangay	one of the 24 barangays of Santa Maria (coming-soon ones join the waitlist)
	Street, landmark	street 5–150 characters; landmark optional, ≤ 100
4 Valid ID	ID type, ID number	type from the list of 11; number 4–30 characters
	ID photo	JPG, PNG or PDF, at most 10 MB
	Consent	must be ticked

Stored data read: applications (to find open duplicates), BARANGAYS, ID_TYPES and the reference counter.

3 Process

What data structure and/or algorithm did you use? How does it work step by step?



- Wizard:** “Continue” validates only the current step with the same `validateApplication` the backend uses, and shows the message under each wrong field (`stepForField` is a linear search that finds which step owns a field).

- 2 **Validate everything** again on submit: text checks loop over the characters (no regular expressions); age uses our date maths (ageOn); the barangay is found by **binary search** on the 24 barangays sorted by name; the ID type by linear search.
- 3 **Duplicate check** (findOpenApplicationFor): a **linear search** over the applications that are still Pending, Approved or For Installation for the same e-mail or mobile number.
- 4 **Reference number**: the counter goes up by one → E3-2026-004887.
- 5 **Save**: arrayAppend puts the record at the end. It has the largest number, so the array stays sorted — no sort is needed and binary search keeps working.
- 6 The record starts with status Pending and one history step; it is automatically part of the staff's **review queue** (first come, first served).

4 Output

What result does the module produce?

A new application record { referenceNo, fullName, email, contactNumber, birthDate, barangay, completeAddress, landmark, planId, idType, idNumber, idPhotoName, status: "Pending", submittedAt, source: "online", history: [...] }, the "Application received" screen with the reference number and a summary, and inline error messages when something is wrong.

5 Complexity analysis

What is the time complexity? What is the space complexity?

Operation	Time	Space
Field checks (e-mail, mobile, name ...)	$O(k)$ per field, k = its length	$O(k)$
Barangay look-up (binary search, $B = 24$)	$O(\log B)$	$O(1)$
ID type look-up (linear, 11 types)	$O(t)$	$O(1)$
Duplicate check over n applications	$O(n)$	$O(1)$
Next reference number + append	$O(1)$	$O(1)$ (one new record)
submitApplication overall	$O(n)$ (the duplicate scan dominates)	$O(1)$ extra

6 Justification

Why did you choose that data structure or algorithm? Why is it appropriate?

- **Append to a sorted array**: reference numbers only grow, so appending is $O(1)$ and keeps the array sorted — this is what makes the tracker's binary search possible without ever sorting.
- **Linear search for duplicates**: the rule depends on the status (only open applications count), on two fields (e-mail or mobile) and on text normalisation; a single pass handles all of that. A hash table would need an index per field and updates on every status change — not worth it for our size.
- **Binary search for the barangay**: the list is fixed and sorted by name, so the look-up is $O(\log B)$.
- **Hand-written character checks**: required by the project rules (no regular expressions) and they give a precise message for each kind of mistake.
- The wizard reuses the backend validator, so the screen and the saved data follow exactly the same rules.

7 Demonstration

Show that the module works correctly — the input, the process and the output.

- 1 Apply → step 2: type a one-word name, a double "@" e-mail, a short mobile number and a 2012 birth date → press Continue → every mistake is explained under its field.
- 2 Fix them, finish steps 3–4, review in step 5 and submit → the reference number appears.

Input with mistakes → the validator's messages.

Step 5 — everything valid, ready to submit.

Output: the application is saved with reference number E3-2026-004887.

Trace — validation and saving (computed by our code)

Field	What was typed	Validator result
Full name	Ana	X Enter the first and last name.
E-mail	ana.reyes@gmail.com	X Enter an e-mail like name@example.com.
Mobile	0917 555	X Enter an 11-digit mobile number like 0917 555 0199.
Birth date	2012-05-10	X The applicant must be at least 18 years old.
Barangay		X Choose a barangay in Santa Maria.
Street address		X Enter the house number and street.
ID type		X Choose an ID type.
ID number		X Enter the ID number.
ID photo	id.exe	X Upload a JPG, PNG or PDF file.
Consent	false	X Please read the Privacy Notice and give your consent to continue.

With the corrected data the validator returns **0 errors**, so `submitApplication` continues: duplicate check → next number → append.

The new application **E3-2026-004887** (status Pending) is appended at index 16 — it has the largest number, so the table is still sorted: ✓ **verified**.

Code walkthrough

assets/js/backend/applications.js · lines 186–242

```
186 /**
187  * submitApplication - validate, then add a new application with the next
188  * reference number. Online applications come from customers; walk-ins are
189  * typed by staff (and can be undone).
190  * Time O(n) (duplicate check) + O(1) append · Space O(1)
191  */
192 function submitApplication(data, source, actor) {
193   const errors = validateApplication(data, source); // 1. check every field
194   if (hasAnyErrors(errors)) {
195     return { ok: false, errors: errors };
196   }
197   const duplicate = findOpenApplicationFor(data.email, data.contactNumber); // 2. one open application per person
198   if (duplicate) {
199     return { ok: false, errors: { email: 'An application for this e-mail or mobile number is already in progress'
200 (' + duplicate.referenceNo + ').' } } };
201   }
202   counters.reference = counters.reference + 1; // 3. the next reference number
203   const now = nowISO();
204   const referenceNo = formatReferenceNo(readClock().year, counters.reference);
205   const barangay = findBarangay(data.barangay);
206   const byName = source === 'online' ? 'Online form' : nameOfActor(actor) + ' (walk-in)';
207   const application = {
208     referenceNo: referenceNo,
209     fullName: collapseSpaces(data.fullName),
210     email: trimText(data.email),
211     contactNumber: normalizeMobile(data.contactNumber),
212     birthDate: data.birthDate,
213     city: SERVICE_CITY,
214     barangay: barangay.name,
215     completeAddress: collapseSpaces(data.completeAddress),
216     landmark: collapseSpaces(data.landmark),
217     planId: data.planId,
218     customPrice: data.planId === 'custom' ? Number(data.customPrice) : null,
219     idType: data.idType,
220     idNumber: trimText(data.idNumber),
221     idPhotoName: textOf(data.idPhotoName),
222     status: 'Pending',
223     submittedAt: now,
224     consentAt: now, // proof of consent (Data Privacy Act, RA 10173)
225     source: source,
226     notes: '',
227     installDate: null,
228     installSlot: null,
229     rejectReason: '',
230     history: [{ status: 'Pending', at: now, by: byName, note: source === 'online' ? 'Application received online'
231 : 'Walk-in application at the office' }],
232   };
233   arrayAppend(applications, application); // 4. the new number is the largest → the table stays sorted
234   if (source === 'online') { // 5. log it (walk-ins typed by staff can also be undone)
235     logActivity('application', 'New online application ' + referenceNo + ' from ' + application.fullName,
236 'Customer');
237   } else {
238     logActivity('application', 'Walk-in application ' + referenceNo + ' for ' + application.fullName,
239 nameOfActor(actor));
240     pushUndo('Create walk-in application ' + referenceNo, [insertOperation('applications', referenceNo)],
241 nameOfActor(actor));
242   }
243   markDataChanged();
244   return { ok: true, application: application };
245 }
```

1. Step 1 calls the validator; if any field has a message, the errors are returned to the form and nothing is saved.
2. Step 2 refuses a second open application for the same e-mail or mobile number (linear search, below).

- Step 3 builds the next reference number from the counter and the current year.
- The record is built with tidy values (`collapseSpaces`, `normalizeMobile`) and one history step.
- Step 4 appends it — the key order stays sorted. Step 5 writes the activity log; a walk-in (staff) application also gets an Undo step.

assets/js/backend/applications.js · lines 79–97

```

79  /**
80   * findOpenApplicationFor – an application still in progress with the same
81   * e-mail or mobile number (stops duplicate applications). Linear search.
82   * Time O(n) · Space O(1)
83   */
84  function findOpenApplicationFor(email, mobile) {
85    const wantedEmail = toLowerText(trimText(email));
86    const wantedMobile = digitsOnly(normalizeMobile(mobile));
87    for (let i = 0; i < applications.length; i++) {
88      const app = applications[i];
89      if (linearSearchValue(OPEN_APPLICATION_STATUSES, app.status) === -1) {
90        continue;
91      }
92      if (toLowerText(app.email) === wantedEmail || (wantedMobile !== '' && digitsOnly(app.contactNumber) ===
wantedMobile)) {
93        return app;
94      }
95    }
96    return null;
97  }

```

- The typed e-mail is lower-cased and the mobile reduced to its digits, so “JUAN@...” and “juan@...” match.
- The loop visits every application (linear search) but skips finished ones using `linearSearchValue` on the list of open statuses.
- The first open application with the same e-mail or mobile is returned; otherwise `null`.

assets/js/backend/applications.js · lines 118–137 (part of `validateApplication`)

```

118  const name = collapseSpaces(data.fullName);
119  if (name.length < 3 || textFind(name, ' ') === -1) {
120    errors.fullName = 'Enter the first and last name.';
121  } else if (name.length > 80) {
122    errors.fullName = 'Use 80 characters or fewer.';
123  }
124  if (!isValidEmail(data.email)) {
125    errors.email = 'Enter an e-mail like name@example.com.';
126  }
127  if (!isValidMobile(data.contactNumber)) {
128    errors.contactNumber = 'Enter an 11-digit mobile number like 0917 555 0199.';
129  }
130  if (!isValidISODate(data.birthDate)) {
131    errors.birthDate = 'Enter the birth date.';
132  } else if (data.birthDate > today) {
133    errors.birthDate = 'The birth date can’t be in the future.';
134  } else if (ageOn(data.birthDate, today) < 18) {
135    errors.birthDate = 'The applicant must be at least 18 years old.';
136  } else if (ageOn(data.birthDate, today) > 120) {
137    errors.birthDate = 'Check the birth year.';

```

- The name must have a space (first and last name) and at most 80 characters.
- `isValidEmail` and `isValidMobile` scan the text character by character (no regular expressions).
- The birth date must be a real date, not in the future, and the applicant at least 18 (`ageOn` counts whole years with our calendar maths).

Application Tracking

Screens	Track page (<code>#/track</code> , <code>#/track/E3-2026-004887</code>)
Files	<code>backend/applications.js</code> (<code>trackApplication</code> , <code>applicationProgress</code>) · <code>views/public/track.js</code> · <code>dsa/search.js</code> · <code>backend/utils.js</code>
Data	<code>applications</code> — sorted by <code>referenceNo</code> ; each has a <code>history</code> array of steps
DSA	Binary search · character scans (format check) · linear scan of the history

1 Business process

What task or problem from the client are you solving?

Applicants used to call or visit the office to ask “what happened to my application?”. With the tracker they type their reference number and immediately see where it is — received, under review, installation scheduled (with the date), connected — or why it was not approved, **without exposing their personal details** to someone who guesses a number.

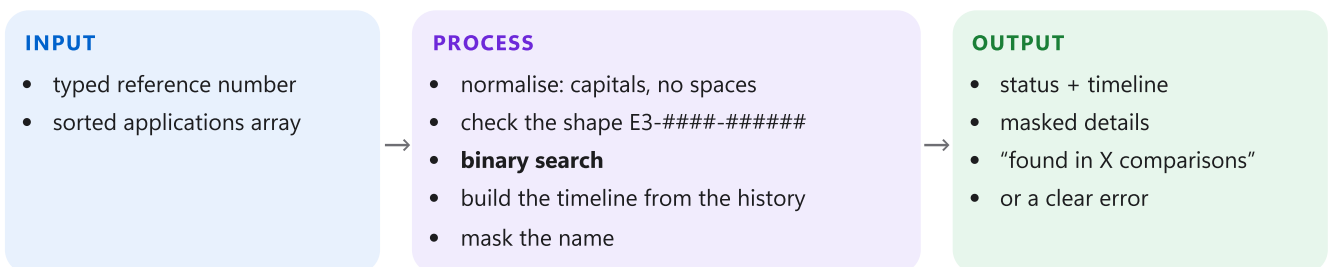
2 Input

What data is needed by the module?

- The reference number typed by the customer, in any form (“e3-2026-004887”) or given in the address (`#/track/E3-2026-004887`).
- The `applications` array (sorted by reference number) and each application’s `history` and status.

3 Process

What data structure and/or algorithm did you use? How does it work step by step?



- 1 `normalizeReference` turns the input into capitals and removes spaces.
- 2 `isValidReference` checks the shape E3-####-##### character by character (`hasCodeShape`); a wrong shape shows “Use the format E3-2026-004879.”.
- 3 **Binary search** (`findApplication`): compare the middle reference of the range with the target; if the middle is smaller keep the right half, if bigger keep the left half; repeat until equal or the range is empty.
- 4 `applicationProgress` turns the status and history into 4 steps (done / current / upcoming) or a 2-step “not approved” path.
- 5 `maskName` shows “Ana R.” instead of the full name; the address and ID are never shown.

4 Output

What result does the module produce?

A result card with the status badge, the timeline, the plan, barangay and dates, a note such as “Found with binary search in ... comparisons among 17 applications” (the exact count is in the trace below), a message for the installation date or rejection reason — or an error: wrong format, or “We couldn’t find E3-2026-999999”.

5 Complexity analysis

What is the time complexity? What is the space complexity?

Operation	Time	Space
Normalise and check the format (k = 14 characters)	O(k)	O(k)
Binary search over n applications	O(log n) — at most $\lceil \log_2(n+1) \rceil$ comparisons	O(1)
Timeline (4 steps × h history entries)	O(h)	O(1)
Linear search (for comparison)	O(n)	O(1)

17 applications → at most 5 comparisons; 1,000 → 10; 1,000,000 → 20.

6 Justification

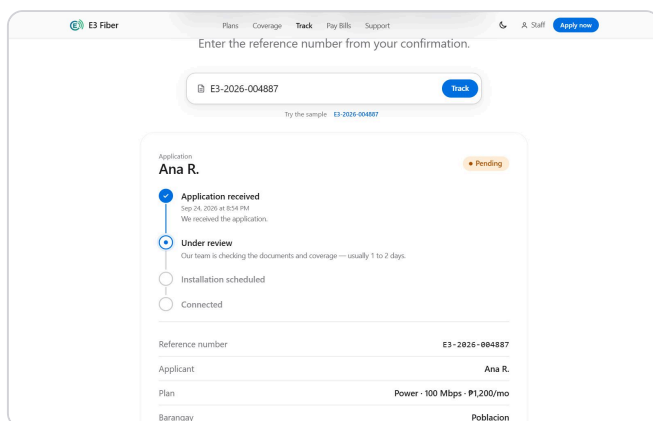
Why did you choose that data structure or algorithm? Why is it appropriate?

- The tracker looks up **one record by an exact key** — the classic case for binary search.
- The array is **already sorted for free** (Module 2 appends increasing numbers), so binary search costs no extra work and no extra memory.
- A hash table would also find it in O(1) on average, but it needs extra memory, must be kept in step with every change, and gives no order. Binary search on the existing array is fast enough (5 steps) and simpler.
- Checking the format first means garbage input never reaches the search, and the customer gets a helpful message.

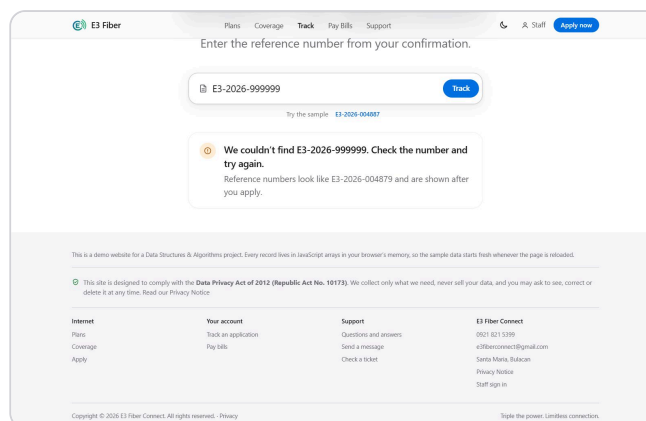
7 Demonstration

Show that the module works correctly — the input, the process and the output.

- From “Application received”, press “Track this application” (or open Track and type the number in small letters).
- The result shows the timeline and the binary-search note. Then type E3-2026-999999 → not found.



Found — masked name, timeline and the binary-search note.



A number that does not exist.

Trace — binary search (computed by our code)

Search for **E3-2026-004887** among **17** applications sorted by reference number (the input was typed in small letters with spaces; `normalizeReference` fixed it first).

004871 [0]	004872 [1]	004873 [2]	004874 [3]	004875 [4]	004876 [5]	004877 [6]	004878 [7]	004879 [8] step 1	004880 [9]	004881 [10]	004882 [11]
004883 [12] step 2	004884 [13]	004885 [14] step 3	004886 [15] step 4	004887 [16] step 5							

Step	low	high	middle	Reference at middle	Compare with the target
1	0	16	8	E3-2026-004879	smaller → go right
2	9	16	12	E3-2026-004883	smaller → go right

Step	low	high	middle	Reference at middle	Compare with the target
3	13	16	14	E3-2026-004885	smaller → go right
4	15	16	15	E3-2026-004886	smaller → go right
5	16	16	16	E3-2026-004887	✓ equal — found

Found in **5 comparisons** (the real `trackApplication` reported 5). A linear search would have needed all 17 — this record is the last one, its worst case. Binary search never needs more than $\lceil \log_2(17 + 1) \rceil = 5$.

Not found: E3-2026-999999

Step	low	high	middle	Reference at middle	Compare
1	0	16	8	E3-2026-004879	smaller → go right
2	9	16	12	E3-2026-004883	smaller → go right
3	13	16	14	E3-2026-004885	smaller → go right
4	15	16	15	E3-2026-004886	smaller → go right
5	16	16	16	E3-2026-004887	smaller → go right

After 5 comparisons low (17) passed high (16): the range is empty, so the tracker answers “We couldn’t find E3-2026-999999”.

Code walkthrough

assets/js/backend/applications.js · lines 36–55

```

36  /**
37   * trackApplication - public tracker: tidy the typed reference, check its shape,
38   * then binary search. Returns the application (or null) with the search cost.
39   * Time O(log n)
40   */
41  function trackApplication(input) {
42    const referenceNo = normalizeReference(input);           // 1. tidy it: capitals, no spaces
43    if (!isValidReference(referenceNo)) {                   // 2. it must look like E3-2026-004879
44      return { ok: false, referenceNo: referenceNo, error: 'Use the format E3-2026-004879.' };
45    }
46    const application = findApplication(referenceNo);       // 3. binary search on the sorted table
47    return {                                               // 4. the result + how many comparisons it took
48      ok: application !== null,
49      referenceNo: referenceNo,
50      application: application,
51      comparisons: dsaLastRun.comparisons,
52      n: applications.length,
53      error: application ? '' : 'We couldn't find ' + referenceNo + '. Check the number and try again.',
54    };
55  }

```

1. Tidy the input first, so small letters and spaces don’t matter.
2. Reject anything that doesn’t have the reference shape before searching.
3. `findApplication` is `binarySearch(applications, 'referenceNo', ...)`.
4. The result also carries `dsaLastRun.comparisons` — the number the page shows under the result.

assets/js/dsa/search.js · lines 76–102

```

76  /**
77   * binarySearch - index of the record whose `field` equals `value`, or -1.
78   * The records MUST be sorted by `field` in ascending order. Each step looks at
79   * the middle record and discards the half that cannot contain the value.
80   * Time O(log n) - about 17 steps for 100,000 records · Space O(1)
81   */
82  function binarySearch(sortedRecords, field, value) {
83    let low = 0;                                           // 1. start with the whole table as the range
84    let high = sortedRecords.length - 1;

```

```

85   let comparisons = 0;
86   while (low <= high) {                                // 2. repeat while the range still has records
87       const middle = Math.floor((low + high) / 2);      // 3. look at the middle record
88       const middleValue = sortedRecords[middle][field];
89       comparisons++;
90       if (middleValue === value) {                      // 4. found it
91           recordRun('Binary search', comparisons, 0);
92           return middle;
93       }
94       if (middleValue < value) {
95           low = middle + 1; // the value can only be in the right half
96       } else {
97           high = middle - 1; // the value can only be in the left half
98       }
99   }
100   recordRun('Binary search', comparisons, 0);
101   return -1;
102 }

```

1. low and high mark the part of the table that can still contain the target.
2. The loop runs while that part is not empty.
3. `Math.floor((low + high) / 2)` is the middle position; its key is compared with the target.
4. Equal → found. Smaller → the target can only be to the right (`low = middle + 1`). Bigger → to the left (`high = middle - 1`).
5. If the loop ends, the key is not in the table and `-1` is returned. The comparisons are recorded either way.

assets/js/backend/utils.js · lines 542–562

```

542  /**
543   * hasCodeShape - checks text against a shape like "E3-####-#####"
544   * ("#" = any digit, anything else must match exactly).
545   * Time O(n) · Space O(1)
546   */
547  function hasCodeShape(value, shape) {
548      const text = textOf(value);
549      if (text.length !== shape.length) {
550          return false;
551      }
552      for (let i = 0; i < shape.length; i++) {
553          if (shape[i] === '#') {
554              if (!isDigitChar(text[i])) {
555                  return false;
556              }
557          } else if (text[i] !== shape[i]) {
558              return false;
559          }
560      }
561      return true;
562  }

```

1. The shape "E3-####-#####" uses # for "any digit"; every other character must match exactly.
2. A different length fails immediately; otherwise one pass compares each position.

Justin Banaag

Module 4 Application List · Module 5 Walk-in Override (New Application) · Module 6 Create Account

60-second pitch

My modules are the staff side of an application. The **Application List** filters the applications with a *linear search*, searches them with *naive string matching*, and sorts them with the algorithm the staff picks — *insertion*, *selection* or *bubble sort* — showing the real number of comparisons. Pending applications form a *queue*, so the oldest one is reviewed first. The **Walk-in override** lets staff encode an application for a customer at the office, with a custom price and an ID checked in person; every such action can be undone with a *stack*. **Create Account** turns an installed application into a subscriber account: a *binary search (lower bound)* finds where the new account belongs in the sorted subscribers array and the later records shift one place — a *sorted insert*.

Structures and algorithms I must be able to explain

Where	Structure / algorithm	Time	Space
Filter by status	Linear search (collect all matches)	$O(n)$	$O(k)$
Search box	Naive string matching on 5 fields	$O(n \cdot L \cdot m)$	$O(k)$
Sort the list	Insertion / selection / bubble sort	$O(n^2)$ worst (insertion & bubble $O(n)$ best)	$O(n)$ copy
Review next	Queue (circular buffer), peek	$O(n)$ to build, $O(1)$ peek	$O(p)$
Undo a walk-in	Stack push / pop	$O(1)$ push, $O(n)$ removal	$O(1)$
Create account	Lower bound (binary search) + shift	$O(\log n) + O(n) = O(n)$	$O(1)$

Likely panel questions

Why offer three sorting algorithms?

To compare them on real data: the caption shows each one's comparisons and moves. Insertion sort is the default because our lists are almost sorted (best case $O(n)$); selection sort makes the fewest swaps; bubble sort stops early when a pass has no swap.

Isn't $O(n^2)$ too slow?

For the size of E3's lists (tens to a few hundred records) it takes well under a millisecond — the Algorithms page benchmark sorts 1,000 records in a few milliseconds. We also filter and search first, so only the matching rows are sorted.

Why a queue for reviewing?

Fairness: applications are reviewed first come, first served. Our queue is a circular buffer, so taking the front is $O(1)$ — no shifting like `shift()` would do.

What exactly does the walk-in "override"?

Three online rules: staff may set a custom monthly price (₱300–₱20,000), the ID photo is optional because the original ID is checked in person (a required checkbox), and the application is marked "walk-in". Consent under the Data Privacy Act is still required: staff tick that the customer agreed to the Privacy Notice. Every other rule is the same validator.

Why not just append the new subscriber at the end?

Installations don't finish in reference-number order. Appending could break the sorted order and binary search would give wrong answers. The sorted insert keeps the array sorted.

How does Undo remove the account again?

Create account pushes one undo entry with two operations (restore the application, remove the subscriber). Undo pops it and reverses both — finding the records by binary search.

Application List

Screens	Admin → Applications (#/admin/applications), application detail
Files	backend/applications.js (listApplications, countApplicationsByStatus, buildReviewQueue) · views/admin/applications.js · dsa/search.js · dsa/sort.js
Data	applications array (sorted by reference number)
DSA	Linear search · naive string matching · insertion / selection / bubble sort · queue

1 Business process

What task or problem from the client are you solving?

The E3 staff need one place to see every application and act on it: which are waiting, which are approved or scheduled, find one applicant fast, order the list by date, name or plan price, and process them in a fair order. Without it, staff had to go through paper forms. The list also shows the next action for each application (Approve / Reject, Schedule, Create account).

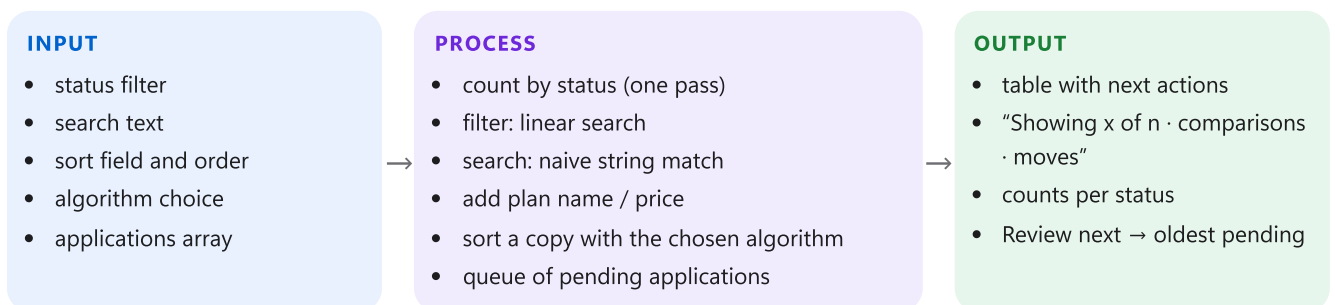
2 Input

What data is needed by the module?

Input	Values
Status filter (segmented control)	All · Pending · Approved · Scheduled · Installed · Rejected
Search text	matched against name, e-mail, reference number, barangay and mobile
Sort	Newest / Oldest first · Name A–Z / Z–A · Plan price high / low
Sorting algorithm	Insertion sort · Selection sort · Bubble sort
Stored data	the <code>applications</code> array; PLANS for names and prices

3 Process

What data structure and/or algorithm did you use? How does it work step by step?



- 1** `countApplicationsByStatus` walks the array once and counts each status (the numbers on the filter buttons and the sidebar badge).
- 2** **Filter** — `linearSearchAll` collects the applications with the chosen status (skipped for "All").
- 3** **Search** — `textSearchRecords`: for every record and each of 5 fields, lower-case the text and look for the query letter by letter (naive string matching); one matching field is enough.
- 4** **Rows** — each match gets its plan name and monthly price (linear search in the 5 plans).
- 5** **Sort** — `sortRecords` sorts a *copy* with the chosen algorithm, counting comparisons and moves; the copy keeps the table in reference order for binary search.
- 6** The cost is shown under the table and logged in the Algorithms page (a ring-buffer queue of the last 40 operations).
- 7** **Review next** — `buildReviewQueue` enqueues the pending applications in arrival order; the button opens the front one (`queuePeek`).

4 Output

What result does the module produce?

The filtered, searched and sorted table (name, reference, plan, barangay, date, status and the action buttons), the counts on each filter, the caption under the table ("Showing x of n applications · the algorithm used · its comparisons and moves · its average Big-O · time", as in the screenshots below), and the detail page of the application at the front of the review queue.

5 Complexity analysis

What is the time complexity? What is the space complexity?

Operation	Time	Space
Count by status	$O(n)$	$O(1)$
Filter (linear search)	$O(n)$	$O(k)$ matches
Search (5 fields, text length L, query length m)	$O(n \cdot L \cdot m)$	$O(k)$
Insertion sort	best $O(n)$ · average / worst $O(n^2)$	$O(n)$ copy, $O(1)$ extra
Selection sort	$O(n^2)$ always	$O(n)$ copy, $O(1)$ extra
Bubble sort	best $O(n)$ · average / worst $O(n^2)$	$O(n)$ copy, $O(1)$ extra
Build the review queue / peek	$O(n)$ / $O(1)$	$O(p)$ pending

6 Justification

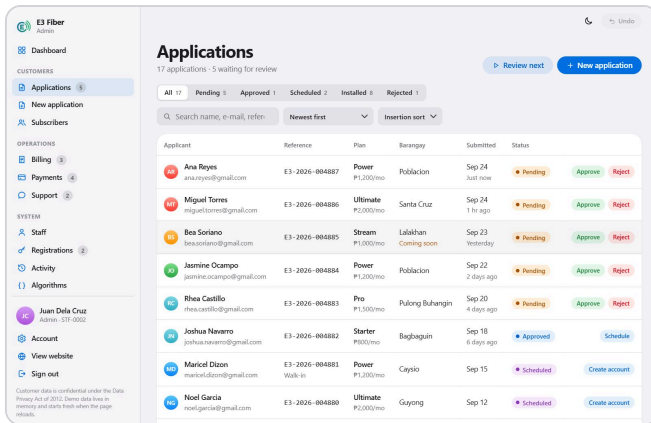
Why did you choose that data structure or algorithm? Why is it appropriate?

- **Linear search to filter:** the filter can be any status, and statuses are not the sort key — every record must be looked at once, which is exactly $O(n)$.
- **Naive string matching:** staff type part of a name or number ("jua", "4883"), so we need "contains", not "equals"; with short fields the simple algorithm is fast and easy to verify.
- **Three sorts:** insertion sort is the default because the table is stored in date order — "Oldest first" is its best case ($n-1$ comparisons). Selection sort guarantees the fewest swaps; bubble sort stops early on sorted data. Showing their counts lets us prove the complexity live.
- **Sorting a copy:** the stored table must stay sorted by reference number, otherwise binary search (tracking, detail pages, Undo) would break.
- **Queue:** first come, first served, with $O(1)$ access to the front.

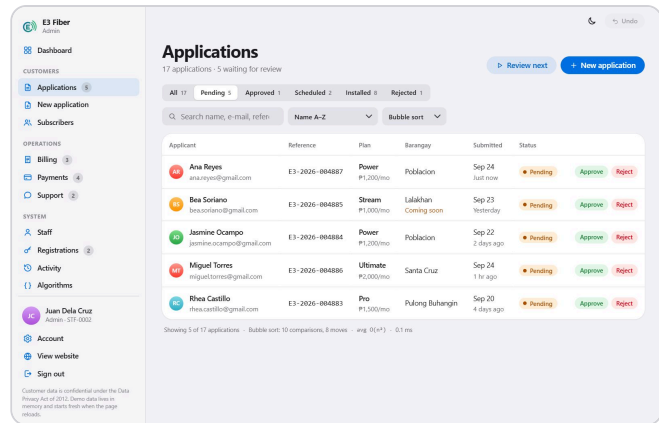
7 Demonstration

Show that the module works correctly — the input, the process and the output.

- 1 Sign in as Admin → Applications. Note the counts and the caption under the table.
- 2 Click "Pending", sort "Name A–Z", choose "Bubble sort" → the rows and the caption change.
- 3 Type part of a name in the search box; press "Review next" to open the oldest pending application.



All applications, newest first with insertion sort.



Pending only, Name A-Z with bubble sort — see the caption.

Trace — the cost of each algorithm (computed by our code)

Sorting all **16** sample applications (the table is stored oldest → newest):

Algorithm	Newest first	Oldest first	Name A-Z
Insertion sort	120 comp. · 120 moves	15 comp. · 0 moves	77 comp. · 66 moves
Selection sort	120 comp. · 8 moves	120 comp. · 0 moves	120 comp. · 14 moves
Bubble sort	120 comp. · 120 moves	15 comp. · 0 moves	117 comp. · 66 moves

“Newest first” reverses the stored order — the **worst case** for insertion and bubble sort: $n(n-1)/2 = 120$ comparisons. “Oldest first” is already in order — their **best case**, $n-1 = 15$ comparisons. Selection sort always makes 120 comparisons but the fewest moves (swaps).

Filter + search example: status *Pending* (4 of 16 by linear search) + search “poblacion” (naive string matching on 5 fields) → Jasmine Ocampo (E3-2026-004884).

Review queue (FIFO): Rhea Castillo (front) → Jasmine Ocampo → Bea Soriano → Miguel Torres — “Review next” opens the front one.

Code walkthrough

assets/js/backend/applications.js · lines 262–282

```
262 /**
263  * listApplications – filter by status (linear search), search the text
264  * (naive string matching), then sort with the chosen algorithm.
265  * options: { status, query, sortField, sortOrder, algorithm }
266  * Time  $O(n \cdot L \cdot m)$  search +  $O(n^2)$  sort · Space  $O(n)$ 
267  */
268 function listApplications(options) {
269   const status = options.status || 'all';
270   const base = status === 'all' ? applications : linearSearchAll(applications, 'status', status); // 1. filter by status
271   const found = textSearchRecords(base, ['fullName', 'email', 'referenceNo', 'barangay', 'contactNumber'], options.query || ''); // 2. search box
272   const rows = [];
273   for (let i = 0; i < found.length; i++) { // 3. add the plan name and price to each row
274     arrayAppend(rows, applicationRow(found[i]));
275   }
276   const started = stopwatchStart();
277   const sorted = sortRecords(rows, options.sortField || 'submittedAt', options.sortOrder || 'desc', options.algorithm || 'insertion'); // 4. sort
278   const ms = stopwatchMs(started);
279   const stats = { algorithm: dsaLastRun.algorithm, comparisons: dsaLastRun.comparisons, moves: dsaLastRun.moves, ms: ms, n: rows.length };
280   logOperation(stats.algorithm, 'Applications list', rows.length, stats.comparisons, stats.moves, ms); // 5. show the cost
281   return { rows: sorted, total: applications.length, stats: stats };
282 }
```

1. If a status is chosen, `linearSearchAll` keeps only those applications.
2. `textSearchRecords` keeps the ones whose name, e-mail, reference, barangay or mobile contains the search text.
3. `applicationRow` adds the plan name and price, so the list can also be sorted by price.
4. `sortRecords` runs the chosen algorithm; `stopwatchStart/Ms` time it and `dsaLastRun` holds its comparisons and moves.
5. `logOperation` adds a line to the Algorithms page log; the function returns the rows and the numbers for the caption.

assets/js/dsa/sort.js · lines 137–166

```

137 /**
138  * insertionSort - grows a sorted part at the front: each new item is taken out
139  * and the larger items slide one place right until its spot is found.
140  * Very fast on lists that are already almost sorted (our tables usually are).
141  * Time: best O(n) (already sorted), average and worst O(n²)
142  * Space: O(1) extra (plus the copy) · Stable: yes
143  */
144 function insertionSort(records, field, order) {
145     const list = arrayCopy(records);           // sort a copy, so the table keeps its own order
146     const direction = order === 'desc' ? -1 : 1;
147     let comparisons = 0;
148     let shifts = 0;
149     for (let i = 1; i < list.length; i++) {      // 1. items before i are already sorted
150         const current = list[i];                // 2. take out the next item
151         let j = i - 1;
152         while (j >= 0) {
153             comparisons++;
154             if (compareForOrder(list[j][field], current[field], direction) > 0) {
155                 list[j + 1] = list[j];          // 3. a bigger item slides one place right
156                 shifts++;
157                 j--;
158             } else {
159                 break;                          // found the spot
160             }
161         }
162         list[j + 1] = current;                  // 4. drop the item into the gap
163     }
164     recordRun('Insertion sort', comparisons, shifts);
165     return list;
166 }

```

1. Everything before position `i` is already sorted; `current` is the next item to place.
2. While the item on the left is bigger, it slides one place to the right (a “shift” — counted as a move).
3. When a smaller-or-equal item is found, the loop stops (that is why a sorted list costs only $n-1$ comparisons) and `current` drops into the gap.

assets/js/dsa/search.js · lines 199–225

```

199 /**
200  * textSearchRecords - records where any of `fields` contains `query`,
201  * ignoring capital letters ("jua" finds "Juan Dela Cruz"). An empty query
202  * returns every record. Linear search + naive string matching per field.
203  * Time O(n · f · L · m), n records, f fields, L field length, m query length
204  * Space O(k) for the k matches
205  */
206 function textSearchRecords(records, fields, query) {
207     const needle = toLowerText(trimText(query));
208     const matches = [];
209     if (needle === '') {
210         for (let i = 0; i < records.length; i++) {
211             arrayAppend(matches, records[i]);
212         }
213         return matches;
214     }
215     for (let i = 0; i < records.length; i++) {      // 1. every record (linear search)
216         for (let f = 0; f < fields.length; f++) {    // 2. every searchable field of it
217             if (textContains(toLowerText(records[i][fields[f]]), needle)) { // 3. naive string match, ignoring
capitals
218                 arrayAppend(matches, records[i]);
219                 break; // one matching field is enough
220             }

```

```

221     }
222   }
223   recordRun('Linear search (text)', records.length, 0);
224   return matches;
225 }

```

1. The query is trimmed and lower-cased once; an empty query keeps every record.
2. For each record, each field is lower-cased and `textContains` (naive string matching) looks for the query.
3. `break` stops checking the other fields as soon as one matches, so a record is added only once.

assets/js/backend/applications.js · lines 296–309

```

296  /**
297   * buildReviewQueue - a QUEUE of pending applications, oldest first.
298   * The table is already in submission order, so one pass enqueues them FIFO.
299   * Time O(n) · Space O(p) for p pending applications
300   */
301  function buildReviewQueue() {
302    const queue = createQueue(8);
303    for (let i = 0; i < applications.length; i++) {
304      if (applications[i].status === 'Pending') {
305        enqueue(queue, applications[i]);
306      }
307    }
308    return queue;
309  }

```

1. A new circular-buffer queue is created.
2. The table is in submission order, so enqueueing the pending ones in one pass gives first-come-first-served order.

Walk-in Override (New Application)

Screens	Admin → New application (#/admin/applications/new) → application detail
Files	views/admin/new-application.js · backend/applications.js (validateApplication with source “walk-in”, submitApplication) · backend/history.js (Undo)
Data	applications array · the Undo stack · the review queue
DSA	Same validation and append as Module 2 · stack (Undo) · queue (review order)

1 Business process

What task or problem from the client are you solving?

Not every customer can apply online — some come to the office, some are seniors without e-mail access at home, some are offered a special price by the owner. Staff need to **encode the application for them**, with a few rules **overridden**: a custom monthly price, the ID checked in person (so no photo upload is required) and the application marked as a walk-in. Because staff type quickly at the counter, every walk-in can be **undone** right after.

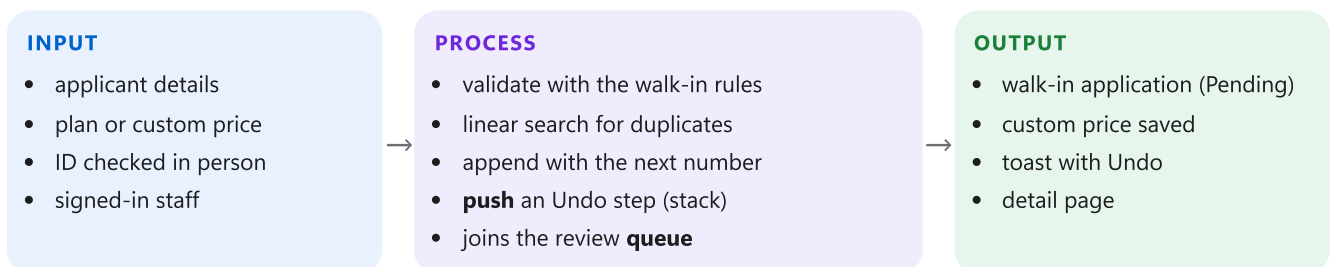
2 Input

What data is needed by the module?

Input	Rule for walk-ins
Plan	one of the 5 plans or “Custom” with a monthly price from ₱300 to ₱20,000
Name, e-mail, mobile, birth date, barangay, address, ID type and number	same rules as the online form (Module 2)
Scanned ID	optional (JPG, PNG or PDF if given)
“I checked the original ID”	required checkbox
Customer agreed to the Privacy Notice (RA 10173)	required checkbox
Signed-in staff member	recorded as the author (“Juan Dela Cruz (walk-in)”)

3 Process

What data structure and/or algorithm did you use? How does it work step by step?



- 1 The summary box shows the price and the **queue position** the application will get (the review queue size + 1).
- 2 `validateApplication(data, 'walk-in')`: for “Custom” it checks the price range instead of looking up a plan; in the walk-in branch the photo is optional and the ID checkbox is required.
- 3 `submitApplication` runs the same duplicate check (linear search), takes the next reference number and **appends** — the array stays sorted.
- 4 The history says “Walk-in application at the office” by the staff member; `pushUndo` puts an “insert” operation on the **Undo stack**.

- If staff press Undo, `undoLastAction` pops the newest entry (LIFO) and removes the record (found by binary search).

4 Output

What result does the module produce?

A new application with `source`: "walk-in" and, if chosen, `customPrice`; it appears in the Applications list (marked "Walk-in") and at the rear of the review queue; a toast "Created walk-in application for ..." with an Undo button; the detail page of the new application.

5 Complexity analysis

What is the time complexity? What is the space complexity?

Operation	Time	Space
Validation (walk-in rules)	$O(k)$ per field	$O(k)$
Duplicate check	$O(n)$	$O(1)$
Append the application	$O(1)$	$O(1)$
Push an Undo step	$O(1)$ ($O(n)$ only when the 25-step limit drops the bottom)	$O(1)$
Undo: pop + find (binary search) + remove	$O(1) + O(\log n) + O(n)$	$O(1)$
Queue position (build the review queue)	$O(n)$	$O(p)$

6 Justification

Why did you choose that data structure or algorithm? Why is it appropriate?

- One validator with a "source" switch:** online and walk-in applications follow the same rules except the three overrides, so a mistake can never be fixed in one form and forgotten in the other.
- Stack for Undo:** the action to reverse is always the most recent one — last in, first out. Push, pop and peek (for the button label) are all $O(1)$. The stack keeps only 25 steps and is cleared at sign-in, so one person can never undo another person's work.
- Queue:** a walk-in joins the same first-come-first-served review queue as online applications — fair to both.
- Append:** the new number is the largest, so the table stays sorted without sorting.

7 Demonstration

Show that the module works correctly — the input, the process and the output.

- Admin → New application: choose "Custom", type ₱950, fill in the applicant, leave the ID box unticked → Create → an error asks to confirm the ID.
- Tick the box → Create → the detail page opens; the application is marked walk-in; press Undo in the toast to remove it again.

Input: custom price ₱950; queue position shown.

Process: the walk-in rule "confirm the original ID".

Output: the walk-in application, ready for review.

Trace — walk-in rules, queue and Undo (computed by our code)

Field	Walk-in rule	Result for the first try
Plan	Plan “custom” allowed; price must be ₱300 – ₱20,000	✗ Enter a monthly price from ₱300 to ₱20,000.
Scanned ID	Optional (the ID is seen in person)	✓ no photo needed
“I checked the original ID”	Required	✗ Confirm that you checked the original ID.
Customer consent (Data Privacy Act)	Required	✗ Confirm the customer agreed to the Privacy Notice.

The same data through the *online* rules would fail on: `idPhoto`, `consent` — that is the “override”.

After fixing the price (₱950) and ticking both boxes: **E3-2026-004887**, source **walk-in**, monthly price ₱950, first history step “Walk-in application at the office” by Juan Dela Cruz (walk-in). Review queue: 4 → 5 (it joined at the rear).

Undo stack top: “Create walk-in application E3-2026-004887” (1 step). Pressing Undo pops it and removes the record: ✓ **application removed**.

Code walkthrough

assets/js/backend/applications.js · lines 109–115 (part of `validateApplication`)

```

109  if (source === 'walk-in' && data.planId === 'custom') {
110      const price = Number(data.customPrice);
111      if (!(price >= 300 && price <= 20000)) {
112          errors.customPrice = 'Enter a monthly price from ₱300 to ₱20,000.';
113      }
114  } else if (!findPlan(data.planId)) {
115      errors.planId = 'Choose a plan.';

```

1. For a walk-in with the “Custom” plan, `Number(...)` reads the typed price and it must be from 300 to 20,000.
2. Everyone else must choose a real plan (`findPlan`, linear search).

assets/js/backend/applications.js · lines 161–180 (part of `validateApplication`)

```

161  if (source === 'online') {
162      if (!data.idPhotoName) {
163          errors.idPhoto = 'Upload a photo of the ID.';
164      } else if (!isAllowedIdPhoto(data.idPhotoName, data.idPhotoType)) {
165          errors.idPhoto = 'Upload a JPG, PNG or PDF file.';

```

```

166     } else if (data.idPhotoSize > MAX_ID_PHOTO_BYTES) {
167         errors.idPhoto = 'The file must be 10 MB or smaller.';
168     }
169     if (!data.consent) {
170         errors.consent = 'Please read the Privacy Notice and give your consent to continue.';
171     }
172 } else {
173     if (data.idPhotoName && !isAllowedIdPhoto(data.idPhotoName, data.idPhotoType)) {
174         errors.idPhoto = 'Upload a JPG, PNG or PDF file.';
175     }
176     if (!data.idChecked) {
177         errors.idChecked = 'Confirm that you checked the original ID.';
178     }
179     if (!data.consent) { // Data Privacy Act: walk-in customers consent too
180         errors.consent = 'Confirm the customer agreed to the Privacy Notice.';

```

1. Online: the ID photo and the consent box are required.
2. Walk-in (the `else`): a photo is optional but must be a JPG, PNG or PDF if given, and staff must confirm they checked the original ID and that the customer agreed to the Privacy Notice (Data Privacy Act).

assets/js/backend/history.js · lines 66–76

```

66  /**
67   * pushUndo - remember an action so it can be reversed. When the stack already
68   * holds UNDO_LIMIT entries, the oldest (bottom) one is dropped first.
69   * Time O(1), O(n) only when the bottom is dropped · Space O(1)
70   */
71  function pushUndo(label, operations, actorName) {
72      if (stackSize(undoStack) >= UNDO_LIMIT) {
73          stackRemoveBottom(undoStack);
74      }
75      stackPush(undoStack, { label: label, operations: operations, at: nowISO(), actor: actorName || 'System' });
76  }

```

1. If the stack already holds 25 steps, the oldest (the bottom) is dropped first.
2. `stackPush` puts the new step on top: its label (“Create walk-in application ...”), the operations to reverse, the time and who did it.

assets/js/backend/history.js · lines 157–180

```

157  /**
158   * undoLastAction - pop the newest entry and reverse its operations, last one
159   * first (an action that changed two tables is unwound in reverse order).
160   * Returns the entry, null when there is nothing to undo, or
161   * { blocked: true, label, error } when it can't be undone (it stays on the stack).
162   * Time O(k · n) for k operations · Space O(1)
163   */
164  function undoLastAction(actorName) {
165      const top = stackPeek(undoStack); // 1. look at the newest entry first
166      if (!top) {
167          return null;
168      }
169      const blocked = undoBlockedReason(top); // 2. would reversing it break a customer's report or request?
170      if (blocked !== '') {
171          return { blocked: true, label: top.label, error: 'Can't undo "' + top.label + '": ' + blocked };
172      }
173      const entry = stackPop(undoStack); // 3. pop it and reverse its operations, newest first
174      for (let i = entry.operations.length - 1; i >= 0; i--) {
175          revertOperation(entry.operations[i]);
176      }
177      logActivity('undo', 'Undid: ' + entry.label, actorName);
178      markDataChanged();
179      return entry;
180  }

```

1. `stackPop` takes the newest step (last in, first out); nothing to undo → `null`.
2. The operations are reversed from the last to the first; an “insert” is reversed by removing that record (binary search + `arrayRemoveAt`).

3. The undo itself is written to the activity log.

Create Account

Screens	Application detail → “Installed — create account” sheet → subscriber page
Files	backend/subscribers.js (previewSubscriberAccount, validateAccountDetails, createSubscriberAccount, createSubscriberFromApplication) · backend/applications.js (completeInstallation) · views/admin/applications.js
Data	subscribers — array sorted by accountNo; the application being installed
DSA	Binary search (find the application, check no account exists) · lower bound + array insert (sorted insert) · stack (Undo)

1 Business process

What task or problem from the client are you solving?

After the technician installs the fiber, the customer must become a **subscriber**: an account that can be billed every month and can pay online. The account must be created only for an application that is scheduled for installation, only after the connection was tested, and only once. It gets the account number (the same as the reference number, so the customer keeps one number), the billing day (the day it was connected) and, optionally, the modem’s serial number.

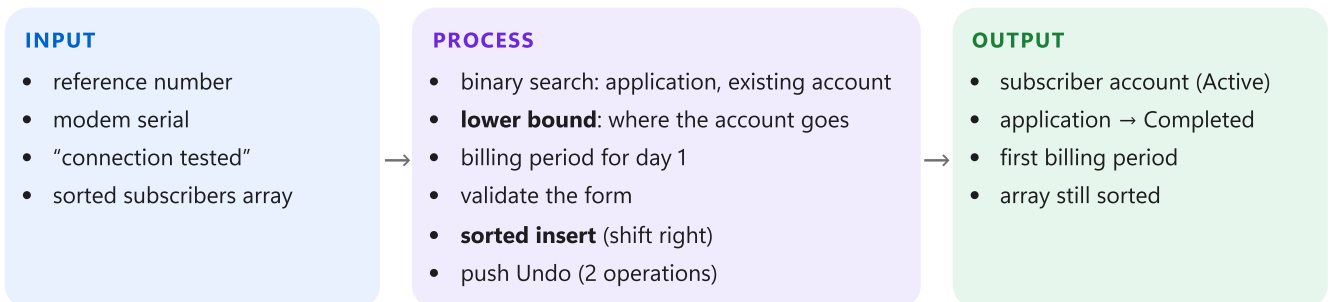
2 Input

What data is needed by the module?

Input	Rule
The application (reference number)	must exist and be “For Installation”; no account with that number yet
Modem (ONU) serial number	optional; 6–24 letters, digits or dashes; saved in capitals
“The fiber is installed and the connection was tested”	required checkbox
Stored data	applications, subscribers (sorted), the plan price, today’s date

3 Process

What data structure and/or algorithm did you use? How does it work step by step?



- 1 Preview** (previewSubscriberAccount): binary search finds the application; it must be “For Installation”; a second binary search confirms there is no account with that number yet.
- 2 Lower bound**: a binary search for the first position whose account number is $\geq$ the new one — the sheet shows that position and the neighbours.
- 3 Billing**: billingPeriod computes the first period (today to the day before the same date next month) and the due date (15 days after the start).
- 4 Validate** the form (validateAccountDetails): the test checkbox and the serial format.
- 5 Create** (completeInstallation): the application becomes Completed; createSubscriberFromApplication builds the record and sortedInsert puts it in place — arrayInsertAt moves the later accounts one slot right.

- 6 One Undo step holds two operations: restore the application and remove the new account.

4 Output

What result does the module produce?

A subscriber record { `accountNo`, `fullName`, `email`, `contactNumber`, `barangay`, `completeAddress`, `planId`, `customPrice`, `status`: "Active", `since`, `applicationRef`, `modemSerial` } at the correct position of the sorted array, the application marked "Completed" (its timeline shows "Connected"), the subscriber page (with the modem serial), and a toast with Undo. From now on Billing (Module 12) bills this account and the customer can use Pay Bills.

5 Complexity analysis

What is the time complexity? What is the space complexity?

Operation	Time	Space
Find the application / check the account (binary search)	$O(\log a) + O(\log n)$	$O(1)$
Lower bound (insert position)	$O(\log n)$	$O(1)$
Validate the form (serial of length k)	$O(k)$	$O(1)$
Shift the later records (<code>arrayInsertAt</code>)	$O(n - \text{position})$, $O(n)$ worst	$O(1)$
Create account overall	$O(n)$	$O(1)$ extra (one new record)

a = number of applications, n = number of subscribers.

6 Justification

Why did you choose that data structure or algorithm? Why is it appropriate?

- **The subscribers array must stay sorted** by account number: Pay Bills, billing, the subscriber page and Undo all find accounts by binary search.
- **Installations finish in any order** — in our demo Maricel (...4881) is installed before Noel (...4880). Appending Noel would break the order, so we insert him in place.
- **Lower bound + shift ($O(n)$)** is cheaper than appending and re-sorting ($O(n \log n)$ at best, $O(n^2)$ with our simple sorts), and finding the spot is only $O(\log n)$.
- A linked list could insert without shifting, but finding the spot would then be $O(n)$ and binary search would be impossible — the array is the better fit because we search far more often than we insert.

7 Demonstration

Show that the module works correctly — the input, the process and the output.

- 1 Open Noel Garcia's application (Scheduled) → "Installed — create account". The sheet shows the account number, billing day, first bill and *where* the account goes.
- 2 Press "Create account" without the checkbox → error. Type the serial, tick the box → Create.
- 3 Open the subscriber page: status Active, modem serial saved.

Create subscriber account ✕

Noel Garcia · Brgy. Guyong. Creating the account completes the installation — you can undo it right after.

Account number E3-2026-004880 (same as the application)

Plan Ultimate · 300 Mbps · ₱2,000/mo

Billing day The 24th of every month

First bill Sep 24 – Oct 23 · ₱2,000 · due Oct 9, 2026

Pay Bills sign-in Account number + mobile 0998 *** 0150

Where the account goes

... E3-2026-004878 **E3-2026-004880** E3-2026-004881

Subscribers are kept sorted by account number. Binary search found position 9 of 10 in 3 comparisons; the 1 account after it move one place to the right.

Modem (ONU) serial number (optional)

☐ The fiber is installed and the connection was tested.

Cancel Create account

Process: the insert position found by binary search.

Create subscriber account ✕

Noel Garcia · Brgy. Guyong. Creating the account completes the installation — you can undo it right after.

Account number E3-2026-004880 (same as the application)

Plan Ultimate · 300 Mbps · ₱2,000/mo

Billing day The 24th of every month

First bill Sep 24 – Oct 23 · ₱2,000 · due Oct 9, 2026

Pay Bills sign-in Account number + mobile 0998 *** 0150

Where the account goes

... E3-2026-004878 **E3-2026-004880** E3-2026-004881

Subscribers are kept sorted by account number. Binary search found position 9 of 10 in 3 comparisons; the 1 account after it move one place to the right.

Modem (ONU) serial number (optional)

☐ The fiber is installed and the connection was tested.

ⓘ Confirm that the fiber is installed and the connection was tested.

Cancel Create account

Validation: the connection test must be confirmed.

E3 Fiber Admin

Dashboard

CUSTOMERS

Applications 6

New application

Subscribers

OPERATIONS

Billing 3

Payments 4

Support 2

SYSTEM

Staff

Registrations 2

Activity

Algorithms

Juan Dela Cruz
Admin · STF-0002

Account

View website

Sign out

Customer data is confidential under the Data Privacy Act of 2012. Demo data lives in memory and starts fresh when the page reloads.

Application

Noel Garcia

E3-2026-004880 Since Sep 24, 2026 · Brgy. Guyong Active

Change plan Suspend Terminate

Account

E-mail noel.garcia@gmail.com

Mobile 0998 555 0150

Address 45 Rizal Ave., Purok 4, Brgy. Guyong

Landmark Beside Guyong Barangay Hall

Application E3-2026-004880

Modem serial ZTEG-C8A1F2B7

Billing history

Newest period first

Period	Due	Amount	Status
No bills yet			

Create one from Billing.

Balance

₱0

All paid up · Next period starts Oct 24, 2026

Paid so far ₱0

Plan

Ultimate

300 Mbps

₱2,000/mo

Payment reports

No payment reports.

Output: the new subscriber account.

Trace — the sorted insert (computed by our code)

Maricel Dizon (E3-2026-004881) was installed first, so her account is already at the end. Now Noel Garcia's account **E3-2026-004880** — an *older* number — must go in front of it.

1 · Where does it go? (lowerBound = binary search for the first key ≥ 004880)

Step	low	high	middle	Account at middle	Decision
1	0	9	4	E3-2026-004875	smaller → low = 5
2	5	9	7	E3-2026-004878	smaller → low = 8
3	8	9	8	E3-2026-004881	not smaller → high = 8

Position **8** (the sheet shows it as "position 9 of 10"), found in 3 comparisons; the real preview reported 3. Neighbours: E3-2026-004878 before, E3-2026-004881 after.

2 · Insert (arrayInsertAt shifts the later records one slot right)

004871 [0]	004872 [1]	004873 [2]	004874 [3]	004875 [4]	004876 [5]	004877 [6]	004878 [7]	004880 [8]
---------------	---------------	---------------	---------------	---------------	---------------	---------------	---------------	---------------

Before: 9 accounts. The yellow record moves one slot to the right (1 shift).

004871 [0]	004872 [1]	004873 [2]	004874 [3]	004875 [4]	004876 [5]	004877 [6]	004878 [7]	004880 [8]	004881 [9]
---------------	---------------	---------------	---------------	---------------	---------------	---------------	---------------	---------------	---------------

After: 10 accounts, still sorted — Noel at index 8, application status now Completed, modem serial saved as ZTEG-C8A1F2B7.

3 • The account details

Item	Value
Account number	E3-2026-004880 (same as the reference number)
Billing day	day 24 of every month — the day the account was created (anniversary billing)
First bill	2026-09-24 → 2026-10-23, due 2026-10-09, ₱2,000
Form errors if left wrong	Use 6–24 letters, digits or dashes, as printed on the modem. Confirm that the fiber is installed and the connection was tested.

Code walkthrough

assets/js/backend/subscribers.js · lines 58–92

```

58  /**
59   * previewSubscriberAccount - what "Create account" will do, before it happens:
60   * the account number, the billing day and first bill, and WHERE the record will
61   * go in the sorted subscribers array (lowerBound = binary search for the spot).
62   * Time O(log n) · Space O(1)
63   */
64  function previewSubscriberAccount(referenceNo) {
65    const app = findApplication(referenceNo);
66    if (!app) {
67      return { ok: false, error: 'Application not found.' };
68    }
69    if (app.status !== 'For Installation') {
70      return { ok: false, error: 'Schedule the installation first.' };
71    }
72    if (findSubscriber(referenceNo)) {
73      return { ok: false, error: 'This customer already has a subscriber account.' };
74    }
75    const since = nowISO();
76    const position = lowerBound(subscribers, 'accountNo', referenceNo);
77    const comparisons = dsaLastRun.comparisons;
78    const clock = readClock();
79    return {
80      ok: true,
81      application: app,
82      accountNo: referenceNo,
83      monthly: planPrice(app.planId, app.customPrice),
84      billingDay: clock.day,
85      firstPeriod: billingPeriod({ since: since }, clock.year, clock.month),
86      position: position,
87      total: subscribers.length,
88      comparisons: comparisons,
89      before: position > 0 ? subscribers[position - 1].accountNo : '',
90      after: position < subscribers.length ? subscribers[position].accountNo : '',
91    };
92  }

```

1. Three guards: the application exists, it is scheduled for installation, and no account exists yet (two binary searches).
2. `lowerBound` finds the insert position; `dsaLastRun.comparisons` is how many steps it took.
3. `billingPeriod` gives the first period and due date; `before` / `after` are the neighbouring account numbers shown in the sheet.

assets/js/backend/subscribers.js · lines 32–56

```

32  /**
33   * createSubscriberFromApplication - open an account for a completed
34   * application ("put" with sortedInsert). Returns { subscriber, position }.
35   * Time O(log n) search + O(n) shift · Space O(1)

```

```

36  */
37  function createSubscriberFromApplication(app, sinceISO, extras) {
38      const subscriber = {
39          accountNo: app.referenceNo,
40          fullName: app.fullName,
41          email: app.email,
42          contactNumber: app.contactNumber,
43          barangay: app.barangay,
44          completeAddress: app.completeAddress,
45          landmark: app.landmark,
46          planId: app.planId,
47          customPrice: app.customPrice,
48          status: 'Active',
49          since: sinceISO,
50          statusChangedAt: sinceISO,
51          applicationRef: app.referenceNo,
52          modemSerial: extras && extras.modemSerial ? extras.modemSerial : '',
53      };
54      const position = sortedInsert(subscribers, 'accountNo', subscriber);
55      return { subscriber: subscriber, position: position };
56  }

```

1. The subscriber copies the application's details; the account number is the reference number.
2. `sortedInsert` = `lower bound + arrayInsertAt`; it returns the position, which the trace and Undo use.

assets/js/dsa/search.js · lines 129–149

```

129  /**
130   * lowerBound - the first position whose 'field' is >= 'value' in a sorted array
131   * (records.length when every key is smaller). This is where 'value' belongs.
132   * Time O(log n) · Space O(1)
133   */
134  function lowerBound(sortedRecords, field, value) {
135      let low = 0; // 1. the answer is somewhere in 0 ... n
136      let high = sortedRecords.length;
137      let comparisons = 0;
138      while (low < high) { // 2. halve the range until one position is left
139          const middle = Math.floor((low + high) / 2);
140          comparisons++;
141          if (sortedRecords[middle][field] < value) {
142              low = middle + 1; // middle is smaller → the spot is to the right
143          } else {
144              high = middle; // middle is bigger or equal → the spot is here or left
145          }
146      }
147      recordRun('Binary search (position)', comparisons, 0);
148      return low; // 3. the first position whose key is ≥ value
149  }

```

1. Like binary search, but it looks for the first key $\geq$ the value, so it also works when the key is not in the array yet.
2. "Smaller" moves `low` past the middle; otherwise `high` becomes the middle. When they meet, that is the position.

assets/js/dsa/arrays.js · lines 68–86

```

68  /**
69   * arrayInsertAt - put an item at a position (replaces splice(index, 0, item)).
70   * Items from 'index' onward move one slot to the right.
71   * Time O(n - index), O(n) worst · Space O(1)
72   */
73  function arrayInsertAt(array, index, item) {
74      let position = index;
75      if (position < 0) { // keep the position inside the array
76          position = 0;
77      }
78      if (position > array.length) {
79          position = array.length;
80      }
81      for (let i = array.length; i > position; i--) { // 1. from the end, move each item one slot right
82          array[i] = array[i - 1];
83      }
84      array[position] = item; // 2. the gap is now at 'position' → write the item

```

```
85     return array.length;  
86 }
```

1. From the end of the array down to the position, each item moves one slot right (our replacement for `splice`).
2. The new item is written into the gap. The number of moves is $(n - \text{position})$ — zero when inserting at the end.

Dela Cruz Riceerich

Module 7 Admin Login · Module 8 Accounts Management (Subscribers) · Module 9 Admin/Staff Creation & Management · Module 10 Admin Registration & Approval

60-second pitch

My modules decide **who can do what**. **Admin Login** finds the staff member by e-mail in a *hash table* in $O(1)$ on average, compares a salted password hash, and pauses sign-in after 5 wrong tries. **Accounts Management** lists, searches and sorts subscribers, finds one by *binary search*, changes plans and statuses through a strict set of allowed moves, and settles the oldest bill first with a *queue*. **Staff Management** creates accounts with a one-time temporary password, keeps e-mails unique through the same hash table, and enforces role rules. **Registration & Approval** lets new employees ask for access themselves; the requests wait in a first-in-first-out *queue* until an Owner or Admin approves one, which creates the staff account with the password the person chose.

Structures and algorithms I must be able to explain

Where	Structure / algorithm	Time	Space
Find staff by e-mail (sign-in, duplicates)	Hash table, djb2, separate chaining	$O(1)$ average	$O(n + b)$
Find staff by id / subscriber by account	Binary search	$O(\log n)$	$O(1)$
Lists (subscribers, staff, requests)	Linear filter · string search · chosen sort	$O(n^2)$ worst	$O(n)$
Oldest bill first	Queue (after insertion sort by due date)	$O(b + k^2)$	$O(k)$
Staff requests	Queue (circular buffer)	$O(r)$ build, $O(1)$ peek	$O(p)$
Every change	Undo stack	$O(1)$ push / pop	$O(1)$ per step

Likely panel questions

Why a hash table for e-mails?

Sign-in and “is this e-mail taken?” happen often; a hash table answers in $O(1)$ on average instead of scanning every staff member. The e-mail is lower-cased first, so capital letters don’t matter.

What is a collision and how do you handle it?

Two keys landing in the same bucket. Each bucket is a small array (a chain); we compare the keys in that chain one by one. The table grows (about $\times 2$ buckets, rehashing every key) when there are more than 0.75 keys per bucket, so chains stay short.

Do you store passwords?

No — only a salted hash (the salt is the staff id or the request id). Sign-in hashes what was typed and compares the results. Our hash is a teaching hash; a real system would check passwords on a server with a slow algorithm such as bcrypt.

Why can’t an Admin suspend the Owner, or you suspend yourself?

Role rules in `canManageMember`: nobody changes their own account (you could lock yourself out) and an Admin can’t change an Owner. Only an Owner can add another Owner.

Why doesn’t the registration form say “e-mail already has an account”?

That would let strangers test which e-mails exist. The form accepts the request; the reviewer sees the hash-table check fail and rejects it as a duplicate.

Why is the balance not stored in the subscriber record?

It is always computed from the bills (sum of unpaid bills), so it can never disagree with them after a payment, an Undo or a new bill.

Admin Login

Screens	Staff sign-in (#/admin/login); every admin screen is guarded
Files	backend/auth.js (signIn, currentStaff) · backend/staff.js (findStaffByEmail, staffPasswordMatches) · dsa/hashtable.js · views/admin/login.js · ui/router.js (route guards)
Data	staffEmailIndex (hash table e-mail → id) · staffMembers (sorted by id) · authState { staffId, failedByEmail (hash table e-mail → wrong tries), pausedUntil }
DSA	Hash table (djb2 + separate chaining) · binary search · salted password hash · per-e-mail counter (a second hash table) + pause

1 Business process

What task or problem from the client are you solving?

Customer data, bills and payments must only be seen and changed by E3 staff. Each employee signs in with their own account and role (Owner, Admin, Support), so the system knows who did what (the activity log and Undo use it). The login must stop password guessing, refuse suspended or deleted accounts, force people with a temporary password to choose their own, and tell someone whose registration is still waiting what is happening.

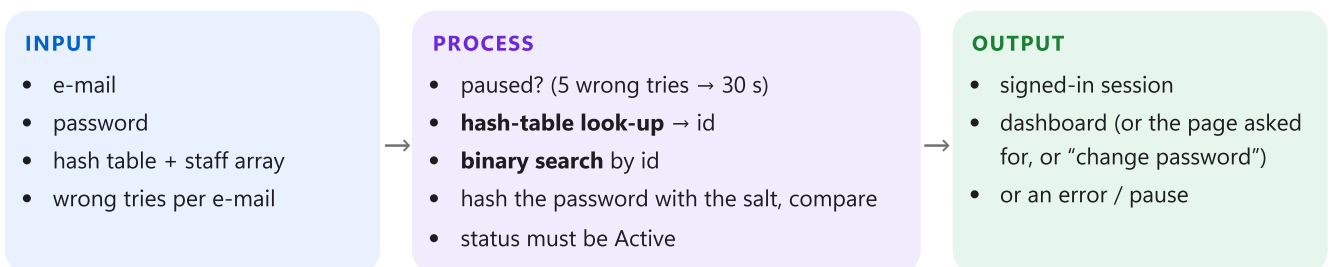
2 Input

What data is needed by the module?

Input	Detail
E-mail	any capitals / spaces — tidied before the look-up
Password	compared as a salted hash, never stored as typed
Stored	the e-mail hash table, the staff records (id, role, status, passwordHash, passwordSalt, mustChangePassword), the registrations (for waiting requests), the wrong-try count per e-mail and the pause time

3 Process

What data structure and/or algorithm did you use? How does it work step by step?



- 1 If sign-in is paused (after 5 wrong passwords in a row), say how many seconds are left.
- 2 `findStaffByEmail`: lower-case and trim the e-mail, **djb2-hash** it to a bucket, compare the keys in that bucket's chain, get the staff id, then **binary search** the staff array by id.
- 3 `staffPasswordMatches`: hash the typed password with the member's salt and compare with the stored hash.
- 4 Not a staff password (or no account)? Check for a waiting or rejected **registration** with the same e-mail and password and explain its status — checked the same way whether or not a staff account exists, so nobody can learn which e-mails are staff. Otherwise count the wrong try for that e-mail in a second **hash table**; the 5th for one e-mail — or the 10th overall, which stops someone trying a few passwords on every e-mail — pauses sign-in for 30 seconds.
- 5 A suspended or deleted account is refused.
- 6 Success: reset that e-mail's counter, remember the staff id, clear the Undo stack, record the sign-in time and log it. The router then opens the requested page; a temporary password forces the Account page first.

4 Output

What result does the module produce?

A signed-in session (`authState.staffId`) — the sidebar shows the name and role and every guarded screen opens; or a clear message (“The e-mail or password is incorrect.”, “This account is suspended ...”, “Too many attempts ...”, “Your request REG-... is still waiting ...”) with a shake of the card. Signing out clears the session and the Undo stack.

5 Complexity analysis

What is the time complexity? What is the space complexity?

Step	Time	Space
Hash the e-mail (k characters)	$O(k)$	$O(1)$
Look in the bucket’s chain	$O(1)$ average, $O(n)$ worst (all keys in one bucket)	$O(1)$
Binary search by id	$O(\log n)$	$O(1)$
Password hash (400 rounds over p characters)	$O(p)$ — a constant amount of work per try	$O(1)$
Waiting-registration check (r requests)	$O(r)$ (only when the staff password did not match)	$O(1)$
Count a wrong try for that e-mail (hash table)	$O(1)$ average	$O(1)$ per e-mail tried
signIn overall	$O(1)$ average (plus $O(\log n)$ and the fixed hash cost)	$O(1)$; the hash table itself $O(n + b)$

6 Justification

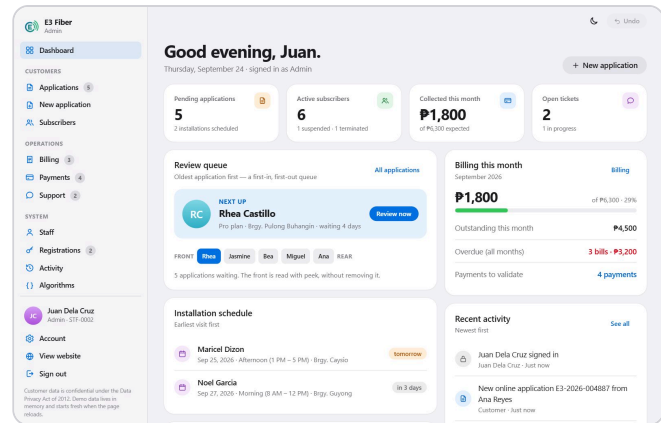
Why did you choose that data structure or algorithm? Why is it appropriate?

- **Hash table:** e-mail is the key people type; staff are stored by id. A hash table maps e-mail $\rightarrow$ id in $O(1)$ on average, where a linear search would be $O(n)$ — and the same table answers “is this e-mail already used?” for Modules 9 and 10.
- **Separate chaining** handles collisions simply (a short array per bucket), and resizing at load 0.75 keeps chains short.
- **Binary search by id** reuses the staff array’s sorted order (ids only grow, records are appended).
- **Salted hash:** passwords are never kept in readable form; the salt makes equal passwords look different.
- **A counter per e-mail + a pause time:** the simplest data that stops unlimited guessing. Keeping the counts in a hash table (e-mail $\rightarrow$ wrong tries) means signing in to your own account does not reset the count for someone else’s.

7 Demonstration

Show that the module works correctly — the input, the process and the output.

- 1 Open a staff page (e.g. `#/admin/applications`) $\rightarrow$ you are sent to sign-in.
- 2 Type the Admin e-mail with a wrong password $\rightarrow$ “The e-mail or password is incorrect.” (the card shakes).
- 3 Tap the Admin demo chip $\rightarrow$ Sign in $\rightarrow$ you return to the page you asked for.



Output: signed in as Juan Dela Cruz (Admin).

Wrong password → error, no hint about which part was wrong.

Trace — the e-mail hash table and a sign-in (computed by our code)

The staff e-mails are stored in a hash table with 17 buckets (djb2 hash, then remainder):

E-mail (key)	djb2 hash	Bucket	Value = staff id
maria.santos@e3fiberconnect.ph	1,390,595,461	1,390,595,461 mod 17 = 0	STF-0001 · Owner · Active
juan.delacruz@e3fiberconnect.ph	3,956,422,187	3,956,422,187 mod 17 = 15	STF-0002 · Admin · Active
paolo.reyes@e3fiberconnect.ph	2,202,796,774	2,202,796,774 mod 17 = 14	STF-0003 · Support · Active
grace.lim@e3fiberconnect.ph	714,487,751	714,487,751 mod 17 = 4	STF-0004 · Support · Suspended
daniel.tan@e3fiberconnect.ph	1,847,100,051	1,847,100,051 mod 17 = 3	STF-0005 · Admin · Deleted

[0] STF-0001	[1] empty
[2] empty	[3] STF-0005
[4] STF-0004	[5] empty
[6] empty	[7] empty
[8] empty	[9] empty
[10] empty	[11] empty
[12] empty	[13] empty
[14] STF-0003	[15] STF-0002
[16] empty	

Signing in as juan.delacruz@e3fiberconnect.ph: hash → bucket **15** → compare with the 1 key(s) in that bucket → id **STF-0002** → binary search on the staff table by id (3 comparisons) → password check:

Password typed	hashPassword(typed, salt "STF-0002")	Stored hash	Match?
e3admin	4c11d5e5	4c11d5e5	✓ signed in
wrong-password	336680c5	4c11d5e5	✗ "The e-mail or password is incorrect."

Wrong try	Message
1	The e-mail or password is incorrect.
2	The e-mail or password is incorrect.

Wrong try	Message
3	The e-mail or password is incorrect.
4	The e-mail or password is incorrect.
5	Too many attempts. Sign-in is paused for 30 seconds.
6	Too many attempts. Try again in 30 seconds.

Code walkthrough

assets/js/backend/auth.js · lines 27–69

```

27  /**
28   * signIn - check the e-mail (hash-table look-up) and the password hash.
29   * Only Active accounts may sign in.
30   * Time O(1) average + one password hash (400 fixed rounds, O(k) for k characters)
31   */
32  function signIn(email, password) {
33    const now = Date.now();
34    if (authState.pausedUntil > now) { // 1. paused after 5 wrong tries?
35      const seconds = Math.ceil((authState.pausedUntil - now) / 1000);
36      return { ok: false, error: 'Too many attempts. Try again in ' + pluralize(seconds, 'second') + '.' };
37    }
38    const emailKey = toLowerText(trimText(email));
39    const member = findStaffByEmail(email); // 2. hash-table look-up by e-mail, O(1)
40    if (!member || !staffPasswordMatches(member, password)) { // 3. compare the salted password hash
41      // Not a staff password - maybe a registration that is waiting or was not approved.
42      // It is only mentioned when the password the person chose for it matches, and it
43      // is checked whether or not a staff account exists, so the answer never reveals
44      // which e-mails belong to staff.
45      const request = registrationForSignIn(email, password);
46      if (request) {
47        return { ok: false, error: registrationStatusMessage(request), registration: request };
48      }
49      const tries = (hashGet(authState.failedByEmail, emailKey) || 0) + 1; // 4. count wrong tries for this e-
50      authState.failedInARow = authState.failedInARow + 1; // ... and in a row on any e-mail
51      if (tries >= SIGN_IN_MAX_ATTEMPTS || authState.failedInARow >= SIGN_IN_MAX_TOTAL_ATTEMPTS) { // 5th for one
52        hashPut(authState.failedByEmail, emailKey, 0);
53        authState.failedInARow = 0;
54        authState.pausedUntil = now + SIGN_IN_PAUSE_MS;
55        return { ok: false, error: 'Too many attempts. Sign-in is paused for 30 seconds.' };
56      }
57      hashPut(authState.failedByEmail, emailKey, tries);
58      return { ok: false, error: 'The e-mail or password is incorrect.' };
59    }
60    if (member.status !== 'Active') { // 5. suspended or deleted accounts can't sign in
61      return { ok: false, error: 'This account is ' + toLowerText(member.status) + '. Ask the owner to reactivate it.' };
62    }
63    hashPut(authState.failedByEmail, emailKey, 0); // 6. success: this e-mail's count starts again;
64    authState.staffId = member.id;
65    clearUndoHistory(); // Undo only ever reverses your own actions from this sign-in
66    member.lastSignIn = nowISO();
67    logActivity('auth', member.fullName + ' signed in', member.fullName);
68    return { ok: true, staff: member };
69  }

```

- Step 1 checks the pause time; `Math.ceil` turns milliseconds into whole seconds for the message.
- Step 2 is the hash-table look-up (below). Step 3: `staffPasswordMatches` hashes the typed password with the stored salt and compares.
- When it doesn't match (or there is no account), a waiting or rejected registration is explained — only if *its* password matches. Step 4 then counts the wrong try for that e-mail in `authState.failedByEmail` (`hashGet` / `hashPut`); the 5th for that e-mail, or the 10th overall (`failedInARow`), sets `pausedUntil` 30 seconds ahead. The message never says whether the e-mail or the password was wrong.

4. Step 5 refuses suspended/deleted accounts; step 6 records the session, clears Undo and logs the sign-in.

assets/js/backend/staff.js · lines 29–33

```
29  /** findStaffByEmail - hash-table look-up by e-mail, then by id. O(1) average */
30  function findStaffByEmail(email) {
31      const id = hashGet(staffEmailIndex, toLowerText(trimText(email)));
32      return id === null ? null : findStaffById(id);
33  }
```

1. The e-mail is trimmed and lower-cased, then hashGet returns the staff id stored under it (or null).
2. findStaffById is a binary search on the staff array sorted by id.

assets/js/dsa/hashtable.js · lines 96–104

```
96  /**
97   * hashGet - the value stored under `key`, or null.
98   * Time O(1) average · Space O(1)
99   */
100 function hashGet(table, key) {
101     const bucket = table.buckets[hashText(key, table.buckets.length)]; // 1. jump straight to the key's bucket
102     const position = hashFindInBucket(bucket, key);                      // 2. check the few keys in that bucket
103     return position === -1 ? null : bucket[position].value;
104 }
```

1. hashText turns the key into a bucket number (djb2: start at 5381, multiply by 33 and add each character, then take the remainder).
2. hashFindInBucket compares the few keys in that bucket; a missing key returns null.

assets/js/backend/staff.js · lines 40–47

```
40  /**
41   * staffPasswordMatches - hash the typed password with the member's salt and
42   * compare it with the stored hash (the password itself is never stored).
43   * Time O(r · n) for the hash · Space O(1)
44   */
45  function staffPasswordMatches(member, password) {
46      return hashPassword(password, member.passwordSalt) === member.passwordHash;
47  }
```

1. The same hash function with the same salt always gives the same result, so equal results mean the right password.

Accounts Management (Subscribers)

Screens	Admin → Subscribers (#/admin/subscribers) and a subscriber's page (#/admin/subscribers/<account>)
Files	backend/subscribers.js (listSubscribers, findSubscriber, setSubscriberStatus, changeSubscriberPlan) · backend/billing.js (unpaidBillsQueue, settleOldestBill) · views/admin/subscribers.js, subscriber-detail.js
Data	subscribers (sorted by account number) · bills · the Undo stack
DSA	Binary search · linear filter + string search · chosen sort · queue (oldest bill first) · stack (Undo)

1 Business process

What task or problem from the client are you solving?

Once customers are connected, the office must manage their accounts: find a subscriber quickly, see who owes the most, look at one account's bills, balance, payments and tickets, **change the plan** (the next bill uses the new price), **suspend** an account for non-payment, **reactivate** it, or **terminate** it — and settle a bill paid at the office, always the oldest one first.

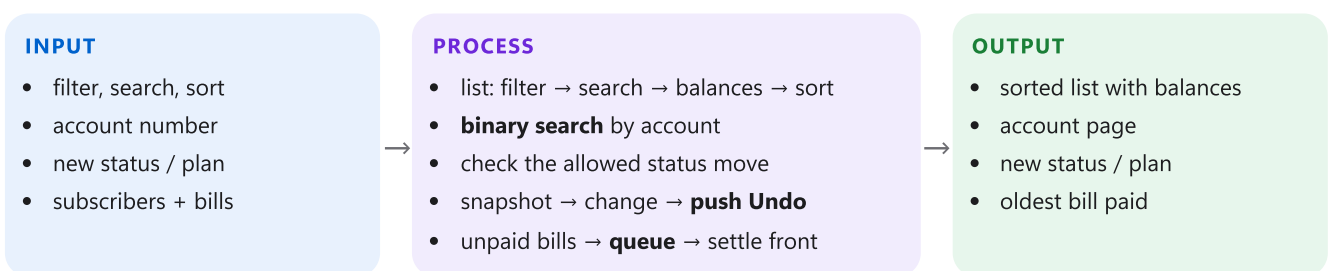
2 Input

What data is needed by the module?

Input	Values / rules
Status filter · search · sort · algorithm	All / Active / Suspended / Terminated · name, e-mail, account, barangay, mobile · newest, oldest, name, balance, plan price
Account number (the page address)	e.g. E3-2026-004876
Status change	Active → Suspended; Active or Suspended → Terminated (final)
Plan change	another plan, or Custom ₱300–₱20,000; not the same plan
Stored data	subscribers, bills, payment reports, tickets

3 Process

What data structure and/or algorithm did you use? How does it work step by step?



- List** (listSubscribers): filter by status (linear search), search the text, compute each row's **balance** by summing its unpaid bills, then sort with the chosen algorithm.
- Open an account**: findSubscriber = binary search on the sorted account numbers; its bills are collected (linear) and ordered newest first (insertion sort).
- Status change** (setSubscriberStatus): binary search → check the move is allowed → copy the old values → change → push one Undo step → log.
- Plan change** (changeSubscriberPlan): the plan must exist (or the custom price be in range) and differ from the current one; bills already issued keep their amount.
- Settle at the office**: unpaidBillsQueue collects the account's unpaid bills, sorts them by due date and enqueues them; settleOldestBill dequeues the front (the oldest) and marks it paid.

4 Output

What result does the module produce?

The subscribers table (plan, balance in red when overdue, status) with the sorting caption; the account page with bills, balance, “next to settle”, payment reports and tickets; updated status or plan with a toast offering Undo; and the oldest unpaid bill marked paid.

5 Complexity analysis

What is the time complexity? What is the space complexity?

Operation	Time	Space
List (n subscribers, B bills): filter + search + balances	$O(n \cdot B)$ for the balances	$O(n)$
Sort the rows	$O(n^2)$ worst (insertion/bubble $O(n)$ best)	$O(n)$
Find one account (binary search)	$O(\log n)$	$O(1)$
Status or plan change	$O(\log n)$	$O(1)$
Oldest-bill queue for one account (k unpaid bills)	$O(B + k^2)$	$O(k)$
Settle the front bill	$O(1)$ dequeue + $O(\log B)$ to update	$O(1)$

6 Justification

Why did you choose that data structure or algorithm? Why is it appropriate?

- **Binary search by account number:** every subscriber page, Pay Bills sign-in, bill and payment check looks up one account by its exact number — the array is kept sorted by Module 6, so each look-up is $O(\log n)$.
- **Allowed-moves check (a small state machine):** prevents impossible changes such as reactivating a terminated account.
- **Queue for settling:** payments clear the oldest debt first (first in, first out) — fair to the customer and standard accounting practice.
- **Balances computed, not stored:** always correct after payments, new bills or Undo.
- **Undo stack:** a wrong suspension can be reversed immediately.

7 Demonstration

Show that the module works correctly — the input, the process and the output.

- 1 Subscribers → sort “Balance: high to low” with selection sort → Patricia Gonzales (suspended, two overdue bills) is first.
- 2 Open Patricia’s page: two unpaid bills, “Next to settle” is the oldest one.
- 3 Open Kristine’s page → Change plan → choose another plan → Save → press Undo.

Subscriber	Account	Plan	Barangay	Since	Balance	Status
Patricia Gonzales	E3-2026-004876	Power P1,200/mo	Catmon	Jul 27, 2026	P2,400	Suspended
Kristine Ramos	E3-2026-004874	Starter P800/mo	Poblacion	Jul 22, 2026	P1,600	Active
Carlo Mendoza	E3-2026-004877	Pro P1,300/mo	Manggahan	Jul 30, 2026	P1,500	Active
Ramon Villareal	E3-2026-004872	Power P1,200/mo	Silangan	Jul 16, 2026	P1,200	Active
Mark Bautista	E3-2026-004875	Stream P1,000/mo	Bulac	Jul 25, 2026	P1,000	Active
Liza Fernandez	E3-2026-004871	Starter P800/mo	Santa Clara	Jul 12, 2026	P0	Active
Jerome Santiago	E3-2026-004873	Power P1,200/mo	Parada	Jul 19, 2026	P0	Terminated
Angelica Cruz	E3-2026-004878	Stream P1,000/mo	San Vicente	Aug 2, 2026	P0	Active
Noel Garcia	E3-2026-004880	Ultimate P2,000/mo	Gugyong	Sep 24, 2026	P0	Active

Sorted by balance with selection sort.

Account	Balance
E3-2026-004876 Since Jul 27, 2026 - Brgy. Catmon	P2,400
Next to settle: July 2026 - P1,200	
Payments clear the oldest bill first (first in, first out).	
Mark oldest bill paid	
Plan	Power 100 Mbps P1,200/mo
Payment reports	

One account: status, bills, balance, oldest bill first.

Change plan

Kristine Ramos is on Starter · 50 Mbps · ₱800/mo. The next bill uses the new price; bills already issued stay the same.

Starter 50 Mbps ₱800/mo	Stream 70 Mbps ₱1,000/mo
MOST POPULAR Power 100 Mbps ₱1,200/mo	Pro 200 Mbps ₱1,500/mo
Ultimate 300 Mbps ₱2,000/mo	Custom ₱ any price Special offer · staff only

CancelSave plan

Change plan — the next bill uses the new price.

Trace — sorting, searching and the bill queue (computed by our code)

Sorting the 8 subscribers by balance, high to low (each row's balance is summed from the bills first):

Algorithm	Comparisons	Moves
Insertion sort	19	15
Selection sort	28	3
Bubble sort	27	15

Top rows	Account	Balance	Status
Patricia Gonzales	E3-2026-004876	₱2,400	Suspended
Kristine Ramos	E3-2026-004874	₱1,600	Active
Carlo Mendoza	E3-2026-004877	₱1,500	Active
Ramon Villareal	E3-2026-004872	₱1,200	Active

Counts by status (one pass): 6 Active · 1 Suspended · 1 Terminated. Opening Patricia's page finds E3-2026-004876 by binary search in 2 comparisons.

Her unpaid bills as a queue (oldest due date at the front):

202607-004876 · P1,200
FRONT · due 2026-08-11

"Mark oldest bill paid" dequeues the front bill, so the oldest debt is always settled first (FIFO).

Code walkthrough

```

214 /**
215  * setSubscriberStatus - suspend, reactivate or terminate an account.
216  * Allowed: Active → Suspended, Suspended → Active, Active/Suspended → Terminated.
217  * Time O(log n)
218  */
219 function setSubscriberStatus(accountNo, status, actor) {
220   const subscriber = findSubscriber(accountNo);           // 1. binary search by account number
221   if (!subscriber) {
222     return { ok: false, error: 'Subscriber not found.' };
223   }
224   if (subscriber.status === 'Terminated') {
225     return { ok: false, error: 'This account is already terminated.' };
226   }
227   const allowed = (subscriber.status === 'Active' && (status === 'Suspended' || status === 'Terminated'))
228     || (subscriber.status === 'Suspended' && (status === 'Active' || status === 'Terminated'));
229   if (!allowed) {                                         // 2. only the allowed moves
230     return { ok: false, error: 'That change is not allowed.' };
231   }
232   const verbs = { Active: 'Reactivate', Suspended: 'Suspend', Terminated: 'Terminate' };
233   const before = snapshotFields(subscriber, ['status', 'statusChangedAt']); // 3. remember the old values
234   subscriber.status = status;                             // 4. change the status
235   subscriber.statusChangedAt = nowISO();
236   // 5. one undo step (stack push)
237   pushUndo(verbs[status] + ' ' + subscriber.fullName, [updateOperation('subscribers', accountNo, before),
nameOfActor(actor)]);
238   logActivity('subscriber', verbs[status] + 'd ' + subscriber.fullName + ' (' + accountNo + ')',
nameOfActor(actor));
239   markDataChanged();
240   return { ok: true, subscriber: subscriber };
241 }

```

1. Binary search for the account; a terminated account can never change again.
2. Only four moves are allowed: Active → Suspended or Terminated, Suspended → Active or Terminated.
3. snapshotFields copies the old status and date so Undo can restore them.
4. The change is made, one Undo step is pushed and the activity log records who did it.

```

243 /**
244  * changeSubscriberPlan - move an account to another plan (or a custom price).
245  * Bills already issued keep their amount; the next bill uses the new price.
246  * Time O(log n)
247  */
248 function changeSubscriberPlan(accountNo, planId, customPrice, actor) {
249   const subscriber = findSubscriber(accountNo);
250   if (!subscriber) {
251     return { ok: false, error: 'Subscriber not found.' };
252   }
253   if (subscriber.status === 'Terminated') {
254     return { ok: false, error: 'A terminated account can't change plans.' };
255   }
256   let price = null;
257   if (planId === 'custom') {
258     price = Number(customPrice);
259     if (!(price >= 300 && price <= 20000)) {
260       return { ok: false, error: 'Enter a monthly price from ₱300 to ₱20,000.' };
261     }
262   } else if (!findPlan(planId)) {
263     return { ok: false, error: 'Choose a plan.' };
264   }
265   if (planId === subscriber.planId && price === subscriber.customPrice) {
266     return { ok: false, error: 'That is already the current plan.' };
267   }
268   const before = snapshotFields(subscriber, ['planId', 'customPrice']);
269   const oldLabel = planName(subscriber.planId);
270   subscriber.planId = planId;
271   subscriber.customPrice = price;
272   pushUndo('Change ' + subscriber.fullName + ' to ' + planName(planId), [updateOperation('subscribers',
accountNo, before)], nameOfActor(actor));
273   logActivity('subscriber', 'Changed ' + subscriber.fullName + ' from ' + oldLabel + ' to ' + planName(planId),

```

```

nameOfActor(actor));
274   markDataChanged();
275   return { ok: true, subscriber: subscriber };
276 }

```

1. A custom price must be ₱300–₱20,000; a normal plan must exist (`findPlan`).
2. Choosing the current plan again is refused; otherwise the old plan is snapshotted for Undo and the new one saved.

assets/js/backend/billing.js · lines 166–184

```

166 /**
167  * unpaidBillsQueue – the account's unpaid bills as a QUEUE, oldest due date
168  * first, so payments always settle the oldest debt first (FIFO).
169  * Time  $O(n + k^2)$  · Space  $O(k)$ 
170  */
171 function unpaidBillsQueue(accountNo) {
172   const unpaid = [];
173   for (let i = 0; i < bills.length; i++) {
174     if (bills[i].accountNo === accountNo && bills[i].status === 'Unpaid') {
175       arrayAppend(unpaid, bills[i]);
176     }
177   }
178   const ordered = insertionSort(unpaid, 'dueDate', 'asc');
179   const queue = createQueue(4);
180   for (let i = 0; i < ordered.length; i++) {
181     enqueue(queue, ordered[i]);
182   }
183   return queue;
184 }

```

1. One pass collects the account's unpaid bills.
2. Insertion sort orders them by due date (oldest first), then they are enqueued in that order.

assets/js/backend/billing.js · lines 213–223

```

213 /**
214  * settleOldestBill – take the front of the unpaid-bills queue (dequeue) and pay it.
215  * Time  $O(n + k^2)$  · Space  $O(k)$ 
216  */
217 function settleOldestBill(accountNo, actor) {
218   const oldest = dequeue(unpaidBillsQueue(accountNo));
219   if (!oldest) {
220     return { ok: false, error: 'There are no unpaid bills.' };
221   }
222   return settleBill(oldest.billId, actor, 'Office payment');
223 }

```

1. `dequeue` takes the front of the queue — the oldest unpaid bill — and `settleBill` marks it paid (with its own Undo step).

Admin/Staff Creation & Management

Screens	Admin → Staff (#/admin/staff) · “Add staff” sheet · temporary-password sheet
Files	backend/staff.js (validateNewStaff, addStaff, generateTempPassword, setStaffStatus, resetStaffPassword, listStaff) · views/admin/staff.js · dsa/hashtable.js
Data	staffMembers (sorted by id) · staffEmailIndex (hash table)
DSA	Hash table (uniqueness) · array append · binary search · sort by role rank · stack (Undo)

1 Business process

What task or problem from the client are you solving?

The Owner and Admins must give new employees an account with the right role, reset forgotten passwords, and suspend, reactivate or delete accounts when people leave — with rules so nobody can take over the system: e-mails are unique, only the Owner can create another Owner, an Admin can’t change the Owner, and nobody can change their own account. The temporary password is shown **once** and must be changed at the first sign-in.

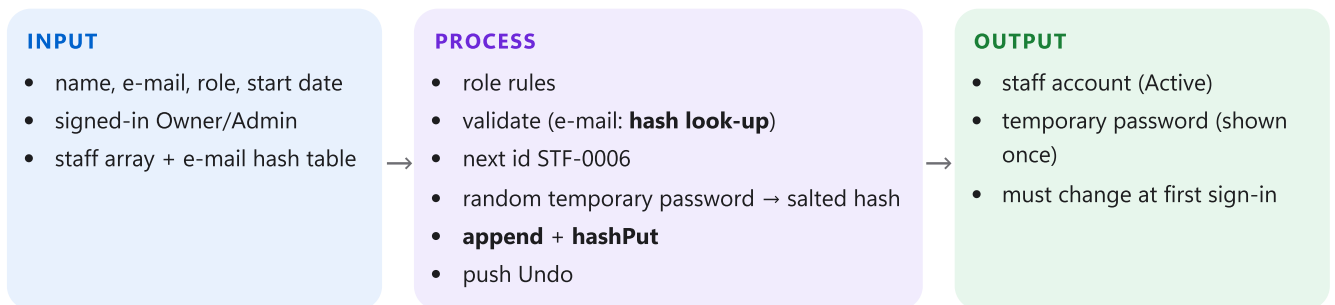
2 Input

What data is needed by the module?

Input	Rule
Full name	first and last name, ≤ 80 characters
Work e-mail	valid, and not already used (hash-table look-up, ignoring capitals)
Role	Owner (only if you are the Owner), Admin or Support
Start date	a real date within the next year
Signed-in member (actor)	must be an Owner or Admin
Later actions	Suspend · Reactivate · Reset password · Delete (on another member)

3 Process

What data structure and/or algorithm did you use? How does it work step by step?



- Only an Owner or Admin may add staff (canManageStaff).
- validateNewStaff checks each field; isStaffEmailTaken is one hash-table look-up.
- The counter gives the next id; ids only grow, so **appending** keeps the array sorted for binary search.
- generateTempPassword picks 10 random characters (no look-alikes such as l, I, O, 0, 1), with at least two digits; only its salted hash is stored.
- hashPut adds e-mail → id to the hash table (it grows when the load passes 0.75); one Undo step can remove the account again.
- Managing:** setStaffStatus finds the member by binary search, applies the role rules and the allowed moves (Active ↔ Suspended, → Deleted), and pushes Undo; resetStaffPassword issues a new temporary password.

- 7 **Listing:** filter, search and sort (by role rank Owner → Admin → Support, name, start date or last sign-in) with the chosen algorithm.

4 Output

What result does the module produce?

A new staff record { id, fullName, email, role, status: "Active", startDate, lastSignIn: null, passwordHash, passwordSalt, mustChangePassword: true }, the e-mail in the hash table, the credential sheet with the one-time password, the updated staff list, and status changes with Undo.

5 Complexity analysis

What is the time complexity? What is the space complexity?

Operation	Time	Space
Validate (e-mail uniqueness by hash)	$O(1)$ average + $O(k)$ for the text	$O(1)$
Temporary password (10 characters)	$O(1)$	$O(1)$
Append the record	$O(1)$	$O(1)$
hashPut	$O(1)$ average; $O(n + b)$ on the rare resize	$O(1)$
Suspend / reactivate / delete / reset	$O(\log n)$	$O(1)$
List with sort	$O(n^2)$ worst	$O(n)$

6 Justification

Why did you choose that data structure or algorithm? Why is it appropriate?

- **Hash table for e-mails:** uniqueness must be checked on every new account and every registration, and the same table serves the sign-in — $O(1)$ average instead of $O(n)$.
- **Append with increasing ids:** $O(1)$ and keeps binary search by id valid.
- **Role rank** (Owner 1, Admin 2, Support 3) turns "sort by role" into a number sort, so the same generic sorting functions work.
- **Temporary password shown once, stored hashed:** the admin never keeps a readable copy, and the employee must replace it.
- A deleted member keeps their record (status Deleted), so the e-mail stays reserved and the history still shows who did what.

7 Demonstration

Show that the module works correctly — the input, the process and the output.

- 1 Staff → Add staff: Paula Ignacio, paula.ignacio@e3fiberconnect.ph, Support → Add → the one-time password appears.
- 2 Try the same e-mail in capitals → "That e-mail already has an account."
- 3 Sign out and in as Paula with the temporary password → she must choose a new password first.

Add staff

Full name

Paula Ignacio

Work e-mail

paula.ignacio@e3fiberconnect.ph

Role

Support

Start date

09/24/2026

Cancel

Add staff

Account created

Paula Ignacio can now sign in as Support with this temporary password.

E-mail

paula.ignacio@e3fiberconnect.ph

Password

yb2dzR3HPW

Share it privately. It is shown only once, and they'll be asked to choose their own password when they first sign in.

Copy

Done

Output: the temporary password, shown once.

Input: the new member's details.

E3 Fiber Admin

Dashboard

CUSTOMERS

Applications 6

New application

Subscribers

OPERATIONS

Billing 3

Payments 4

Support 2

SYSTEM

Staff

Registrations 2

Activity

Algorithms

Juan Dela Cruz Admin - ST1-0002

Account

View website

Sign out

Staff

4 active accounts - e-mails indexed in a hash table for sign-in

2 staff requests waiting for approval

New employees asked for an account. Approving a request creates their sign-in.

Review requests

All 6 Active 4 Suspended 1 Deleted 1

Search name, e-mail, role

Role

Insertion sort

Name	Role	Last sign-in	Status
Maria Santos maria.santos@e3fiberconnect.ph	Owner Since Nov 3, 2025	Yesterday	Active
Juan Dela Cruz juan.delacruz@e3fiberconnect.ph	Admin Since Jan 5, 2026	Just now	Active
Daniel Tan daniel.tan@e3fiberconnect.ph	Admin Since Mar 20, 2026	Aug 10, 2026	Deleted
Paolo Reyes paulo.reyes@e3fiberconnect.ph	Support Since Feb 16, 2026	2 days ago	Active
Grace Lim grace.lim@e3fiberconnect.ph	Support Since Mar 2, 2026	Aug 25, 2026	Suspended
Paula Ignacio paula.ignacio@e3fiberconnect.ph	Support Temporary password	Never	Active

Showing 6 of 6 staff accounts · Insertion sort: 7 comparisons, 2 moves · avg: O(n²) · < 0.01 ms

The staff list with the new member ("Temporary password").

Trace — adding a member and the role rules (computed by our code)

Step	Result
Validate (hash-table look-up for the e-mail)	✓ no errors
New id (counter + 1, appended → ids stay sorted)	STF-0006
E-mail stored in lower case	paula.ignacio@e3fiberconnect.ph
Temporary password	10 random characters, shown once; stored only as hash f3aac235
Must change it at first sign-in	✓ yes
hashPut(e-mail → id)	bucket 1: paula.ignacio@e3fiberconnect.ph → STF-0006
Hash table size	5 → 6 keys in 17 buckets (load $0.35 \leq 0.75$, so no resize)
Rule checked	Message
Same e-mail in capitals	✗ That e-mail already has an account.
An Admin adding an Owner	✗ Only the Owner can add another Owner.
Changing your own account	✗ You can't change this account.
An Admin changing the Owner	✗ You can't change this account.

Code walkthrough

assets/js/backend/staff.js · lines 124–158

```
124 /**
125  * addStaff - create an account with a temporary password. Returns the password
126  * once, so it can be shown to the admin (it is only stored as a hash).
127  * Time O(1) append + O(1) hash put · Space O(1)
128  */
129 function addStaff(data, actor) {
130   if (!canManageStaff(actor)) { // 1. only Owners and Admins add staff
131     return { ok: false, errors: { fullName: 'Only an Owner or Admin can add staff.' } };
132   }
133   const errors = validateNewStaff(data, actor); // 2. check the fields (e-mail through the hash table)
134   if (hasAnyErrors(errors)) {
135     return { ok: false, errors: errors };
136   }
137   counters.staff = counters.staff + 1; // 3. the next staff id, e.g. STF-0006
138   const id = formatStaffId(counters.staff);
139   const tempPassword = generateTempPassword(); // 4. a random temporary password
140   const member = {
141     id: id,
142     fullName: collapseSpaces(data.fullName),
143     email: toLowerText(trimText(data.email)),
144     role: data.role,
145     status: 'Active',
146     startDate: data.startDate,
147     lastSignIn: null,
148     passwordHash: hashPassword(tempPassword, id),
149     passwordSalt: id,
150     mustChangePassword: true,
151   };
152   arrayAppend(staffMembers, member); // 5. ids increase → the table stays sorted
153   hashPut(staffEmailIndex, member.email, id); // 6. index the e-mail for sign-in
154   pushUndo('Add staff ' + member.fullName, [insertOperation('staffMembers', id)], nameOfActor(actor)); // 7. undo
155   logActivity('staff', 'Added ' + member.fullName + ' as ' + member.role, nameOfActor(actor));
156   markDataChanged();
157   return { ok: true, staff: member, tempPassword: tempPassword };
158 }
```

1. Steps 1–2: permission, then field checks (the e-mail goes through the hash table).
2. Steps 3–4: next id and a random temporary password; the record stores only `hashPassword(tempPassword, id)`.
3. Steps 5–7: append (sorted by id), index the e-mail, push one Undo step, log it, and return the temporary password once so the screen can show it.

assets/js/backend/staff.js · lines 160–184

```
160 /**
161  * setStaffStatus - suspend, reactivate or delete a staff account.
162  * Time O(log n)
163  */
164 function setStaffStatus(id, status, actor) {
165   const member = findStaffById(id);
166   if (!member) {
167     return { ok: false, error: 'Staff member not found.' };
168   }
169   if (!canManageMember(actor, member)) {
170     return { ok: false, error: 'You can't change this account.' };
171   }
172   const allowed = (member.status === 'Active' && (status === 'Suspended' || status === 'Deleted'))
173     || (member.status === 'Suspended' && (status === 'Active' || status === 'Deleted'));
174   if (!allowed) {
175     return { ok: false, error: 'That change is not allowed.' };
176   }
177   const verbs = { Active: 'Reactivate', Suspended: 'Suspend', Deleted: 'Delete' };
178   const before = snapshotFields(member, ['status']);
179   member.status = status;
180   pushUndo(verbs[status] + ' ' + member.fullName, [updateOperation('staffMembers', id, before)],
181     nameOfActor(actor));
182   logActivity('staff', verbs[status] + 'd ' + member.fullName, nameOfActor(actor));
183   markDataChanged();
184 }
```

```
183     return { ok: true, staff: member };
184 }
```

1. Binary search for the member; `canManageMember` applies the role rules (not yourself, Admins can't change the Owner, not a deleted account).
2. Only allowed moves pass; the old status is snapshotted for Undo.

assets/js/dsa/hashtable.js · lines 78–94

```
78  /**
79   * hashPut - store `value` under `key`; replaces the value if the key exists.
80   * Time O(1) average, O(n) worst (everything in one bucket) · Space O(1)
81   */
82  function hashPut(table, key, value) {
83      const bucket = table.buckets[hashText(key, table.buckets.length)]; // 1. hash the key → its bucket
84      const position = hashFindInBucket(bucket, key); // 2. already stored?
85      if (position !== -1) {
86          bucket[position].value = value; // yes → replace the value
87          return;
88      }
89      arrayAppend(bucket, { key: key, value: value }); // 3. no → add it to the bucket's chain
90      table.size = table.size + 1;
91      if (table.size / table.buckets.length > HASH_MAX_LOAD) { // 4. too full → about double the buckets
92          hashResize(table);
93      }
94  }
```

1. Hash the key to a bucket; if the key is already there, replace its value.
2. Otherwise add it to the bucket's chain; if the table is more than 0.75 full, `hashResize` rebuilds it with about twice as many buckets.

Admin Registration & Approval of Registration Admins

Screens	Sign-in → “Request staff access” (#/admin/register) · Admin → Registrations (#/admin/registrations)
Files	backend/registrations.js · views/admin/register.js · views/admin/registrations.js · backend/auth.js (sign-in messages)
Data	registrations (sorted by registrationId) · staffMembers · staffEmailIndex
DSA	Queue (approval order) · hash table (e-mail already a staff account?) · linear search (second pending request) · binary search · stack (Undo)

1 Business process

What task or problem from the client are you solving?

When E3 hires someone, the owner should not have to type the new employee’s details and hand over a temporary password. With **Admin Registration** the employee requests access themselves (name, work e-mail, mobile, the access they need, their own password). With **Approval**, an Owner or Admin reviews the requests **first come, first served**, approves the right people — which creates the staff account with the password they chose — and rejects the rest with a reason the requester sees when trying to sign in.

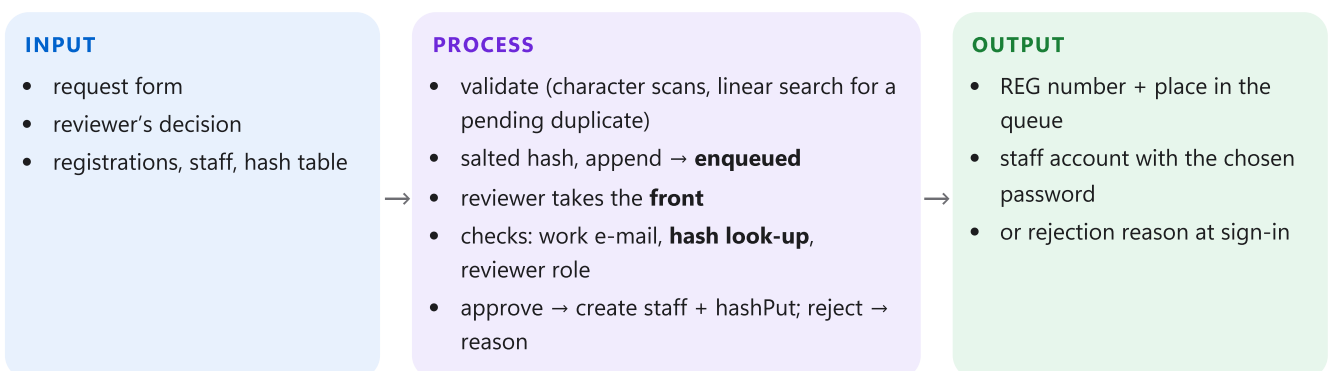
2 Input

What data is needed by the module?

Input	Rule
Full name	first and last name, ≤ 80 characters
Work e-mail	valid and ending with @e3fiberconnect.ph; no other pending request for it
Mobile	11 digits starting with 09
Access	Support or Admin (an Owner can never be self-registered)
Note for the reviewer	optional, ≤ 150 characters
Password + confirmation	8+ characters with a letter and a digit; both the same
Agreement	“this request is for my own E3 account” — required
Approval	the reviewer (Owner/Admin), the decision, and a reason when rejecting

3 Process

What data structure and/or algorithm did you use? How does it work step by step?



Registration

- 1** `validateRegistration` checks each field. It does *not* reveal whether the e-mail already has a staff account (privacy); it only blocks a second *pending* request for the same e-mail (linear search).
- 2** `submitRegistration` gives the next id (REG-000005), stores only a salted hash of the password, and **appends** the request — it joins the rear of the approval queue. The screen shows the place in line.

- 3 If the person tries to sign in before approval, sign-in finds the request and says it is still waiting (only with the right password).

Approval

- 1 **registrationQueue** enqueues the pending requests in arrival order; the page shows the **front** one (**queuePeek**) with its checks.
- 2 **registrationChecks**: work e-mail? no staff account with this e-mail yet (one **hash-table** look-up)? is the reviewer an Owner or Admin?
- 3 **Approve** (**approveRegistration**): binary search for the request, repeat the checks, create the staff record with the *same* password hash and salt, append it, **hashPut** the e-mail, mark the request Approved (it leaves the queue), push one Undo step with both changes.
- 4 **Reject** (**rejectRegistration**): choose a reason (plus a note for "Other"); the requester sees it at sign-in.

4 Output

What result does the module produce?

For the requester: "Request sent", the request number and "number N in the approval queue"; later a sign-in message (waiting / not approved + reason — except "Duplicate request", which is not repeated because it would hint that the e-mail belongs to staff) or a working staff account. For the reviewer: the approval queue with the front request and its checks, the list of all requests (Pending, Approved with the new staff id, Rejected with the reason), and a new member in the Staff list.

5 Complexity analysis

What is the time complexity? What is the space complexity?

Operation	Time	Space
Validate a request (pending-duplicate scan over r requests)	$O(r + k)$	$O(1)$
Submit (append)	$O(1)$ + the password hash	$O(1)$
Build the approval queue / peek the front	$O(r) / O(1)$	$O(p)$
Checks (hash look-up)	$O(1)$ average	$O(1)$
Approve (binary search + append + hashPut)	$O(\log r) + O(1) + O(1)$ average	$O(1)$
Reject	$O(\log r)$	$O(1)$
Sign-in message for a request	$O(r)$	$O(1)$

6 Justification

Why did you choose that data structure or algorithm? Why is it appropriate?

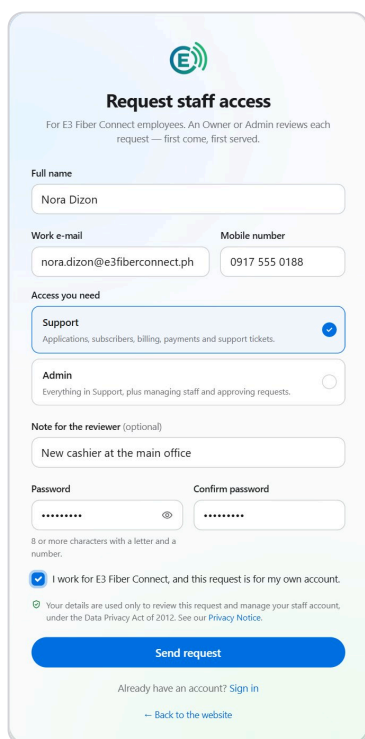
- **Queue**: requests are reviewed in the order they arrived — nobody is skipped; the front is always the oldest.
- **Hash table**: "does this e-mail already have a staff account?" is one $O(1)$ look-up in the same index the sign-in uses.
- **Linear search for a second pending request**: requests are few and short-lived, so a scan is simplest.
- **Password hash moved unchanged**: the approver never sees or sets the password; the new member signs in with the one they chose.
- **Privacy**: the public form never tells whether an e-mail exists; the status is only explained to someone who knows the password.

7 Demonstration

Show that the module works correctly — the input, the process and the output.

- 1 Sign-in → "Request staff access" → fill in Nora Dizon (Support) → Send → REG-000005, number 3 in the queue.

- 2 Try to sign in as Nora → “still waiting for an Owner or Admin”.
- 3 Sign in as the Owner → Registrations: the front is Lorenzo (REG-000003) with all checks passed → Approve → he appears in Staff and can sign in with lorenzo2026.



Request staff access
For E3 Fiber Connect employees. An Owner or Admin reviews each request — first come, first served.

Full name
Nora Dizon

Work e-mail
nora.dizon@e3fiberconnect.ph

Mobile number
0917 555 0188

Access you need
☒ **Support**
Applications, subscribers, billing, payments and support tickets.
 ☐ **Admin**
Everything in Support, plus managing staff and approving requests.

Note for the reviewer (optional)
New cashier at the main office

Password
8 or more characters with a letter and a number.

Confirm password

☒ I work for E3 Fiber Connect, and this request is for my own account.

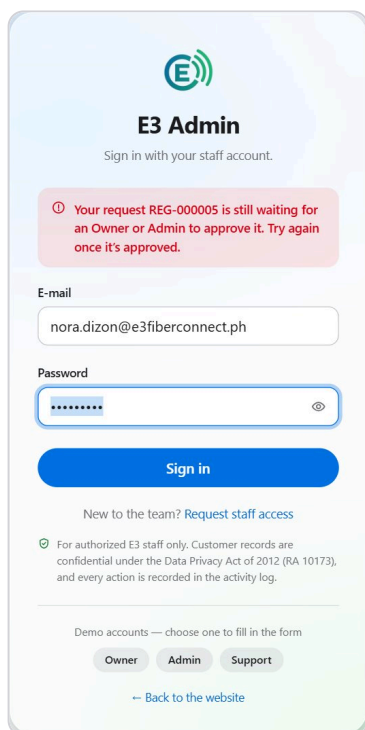
☒ Your details are used only to review this request and manage your staff account, under the Data Privacy Act of 2012. See our [Privacy Notice](#).

Send request

Already have an account? [Sign in](#)

[Back to the website](#)

Input: the request form.



E3 Admin
Sign in with your staff account.

⚠ Your request REG-000005 is still waiting for an Owner or Admin to approve it. Try again once it's approved.

E-mail
nora.dizon@e3fiberconnect.ph

Password

Sign in

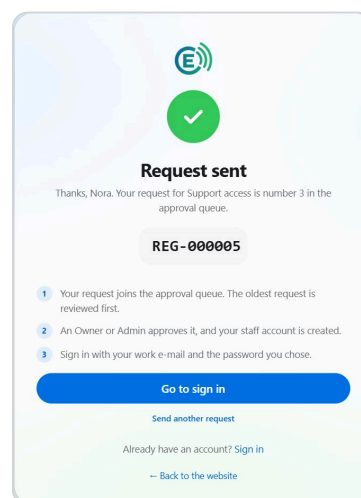
New to the team? [Request staff access](#)

☒ For authorized E3 staff only. Customer records are confidential under the Data Privacy Act of 2012 (RA 10173), and every action is recorded in the activity log.

Demo accounts — choose one to fill in the form
☐ Owner ☐ Admin ☐ Support

[Back to the website](#)

Signing in before approval.



Request sent
Thanks, Nora. Your request for Support access is number 3 in the approval queue.

REG-000005

- 1 Your request joins the approval queue. The oldest request is reviewed first.
- 2 An Owner or Admin approves it, and your staff account is created.
- 3 Sign in with your work e-mail and the password you chose.

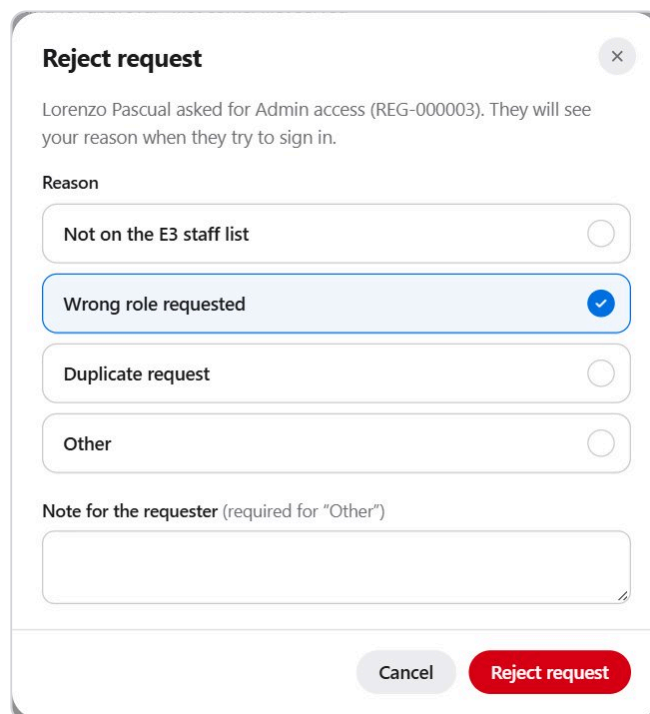
Go to sign in

[Send another request](#)

Already have an account? [Sign in](#)

[Back to the website](#)

Output: request number and place in the queue.



Reject request

Lorenzo Pascual asked for Admin access (REG-000003). They will see your reason when they try to sign in.

Reason

☐ Not on the E3 staff list
☒ **Wrong role requested**
☐ Duplicate request
☐ Other

Note for the requester (required for "Other")

Cancel **Reject request**

Rejecting needs a reason.

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OPERATIONS

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Payments 4

Support 2

SYSTEM

Staff

Registrations 3

Activity

Algorithms

MS

Maria Santos

Owner · STF-0001

Account

View website

Sign out

Customer data is confidential under the Data Privacy Act of 2012. Demo data lives in memory and starts fresh when the page reloads.

3 requests waiting for approval · first come, first served

Approval queue

New requests join the back of a first-in, first-out queue. The one at the front is reviewed first.

FRONT

REG-000003

REG-000004

REG-000005

REAR

NEXT TO REVIEW · 1 OF 3

LP

Lorenzo Pascual

lorenzo.pascual@e3fiberconnect.ph · 0918 555 0137

REG-000003 · sent yesterday

Admin access

New branch supervisor for the Poblacion office.

✓

Work e-mail

Ends with @e3fiberconnect.ph.

✓

No staff account yet

Not in the staff e-mail hash table (one look-up among 6 e-mails).

✓

You can approve it

Owners and Admins approve staff requests.

✓

Ready to approve — the account keeps the password they chose.

Reject

Approve

All 5

Pending 3

Approved 1

Rejected 1

Search name, e-mail, request

Newest first

The approval queue: the front request and its checks.

Trace — the approval queue from request to account (computed by our code)

Validation of a bad request

Field	Message
Full name	X Enter your first and last name.
Work e-mail	X Use your E3 work e-mail (...@e3fiberconnect.ph).
Mobile	X Enter an 11-digit mobile number like 0917 555 0199.
Access	X Choose the access you need.
Password	X Use at least 8 characters with a letter and a number.
Confirm	X The passwords don't match.
Agreement	X Please confirm that this request is for your own E3 account.

The approval queue (circular buffer, capacity 4)

Before: REG-000003 Lorenzo Pascual → REG-000004 Trisha Manalo (front = slot 0, count 2)

REG-000003 [0] front

REG-000004 [1]

[2]

[3]

Nora submits → **REG-000005** is enqueued at the rear (password stored as salted hash b7ad5ed5, salt REG-000005):

REG-000003 [0] front

REG-000004 [1]

REG-000005 [2]

[3]

Her sign-in before approval (right password): “Your request REG-000005 is still waiting for an Owner or Admin to approve it. Try again once it’s approved.” — with a wrong password: “The e-mail or password is incorrect.”.

Review of the front request by the Owner

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Check	Result	Detail
Work e-mail	✓ yes	Ends with @e3fiberconnect.ph.
No staff account yet	✓ yes	Not in the staff e-mail hash table (one look-up among 5 e-mails).
You can approve it	✓ yes	Owners and Admins approve staff requests.

Approve → staff account **STF-0006** for Lorenzo Pascual (Admin), same password hash as the request: ✓ yes. The request leaves the queue:



Lorenzo Pascual can now sign in with the password chosen at registration: ✓ signed in.

Code walkthrough

assets/js/backend/registrations.js · lines 87–118

```

87  /**
88   * submitRegistration - save a request with the next id (appended → still sorted).
89   * Time O(r) for the checks + O(1) append · Space O(1)
90   */
91  function submitRegistration(data) {
92    const errors = validateRegistration(data);
93    if (hasAnyErrors(errors)) {
94      return { ok: false, errors: errors };
95    }
96    counters.registration = counters.registration + 1;
97    const id = formatRegistrationId(counters.registration);
98    const request = {
99      registrationId: id,
100     fullName: collapseSpaces(data.fullName),
101     email: toLowerText(trimText(data.email)),
102     mobile: normalizeMobile(data.mobile),
103     role: data.role,
104     note: collapseSpaces(data.note),
105     passwordHash: hashPassword(data.password, id),
106     passwordSalt: id,
107     status: 'Pending',
108     submittedAt: nowISO(),
109     reviewedBy: '',
110     reviewedAt: null,
111     rejectReason: '',
112     staffId: '',
113   };
114   arrayAppend(registrations, request);
115   logActivity('staff', 'Staff registration ' + id + ' from ' + request.fullName + ' (' + request.role + ')',
request.fullName);
116   markDataChanged();
117   return { ok: true, registration: request };
118 }

```

1. Validate first; any message stops the request.
2. The next request id is also the password's salt; only hashPassword(password, id) is stored.
3. The request is appended (ids only grow, so the table stays sorted) and logged.

assets/js/backend/registrations.js · lines 120–133

```

120 /**
121  * registrationQueue - requests waiting for approval, oldest first (FIFO).
122  * The table is in arrival order, so one pass enqueues them.
123  * Time O(r) · Space O(p) for p pending requests
124  */
125  function registrationQueue() {
126    const queue = createQueue(4);
127    for (let i = 0; i < registrations.length; i++) {
128      if (registrations[i].status === 'Pending') {
129        enqueue(queue, registrations[i]);
130      }
131    }
132  }

```

```

131 }
132 return queue;
133 }

```

1. One pass over the requests in arrival order; each pending one is enqueued, so the front is the oldest.

assets/js/backend/registrations.js · lines 144–176

```

144 /**
145  * registrationChecks - what the reviewer should know before approving:
146  * a work e-mail, no staff account with that e-mail yet (hash table), and a
147  * reviewer who is allowed to approve. Returns { checks, passed, firstFailure }.
148  * Time O(k) · Space O(1)
149  */
150 function registrationChecks(request, actor) {
151   const checks = [];
152   const work = isWorkEmail(request.email);
153   arrayAppend(checks, { key: 'domain', label: 'Work e-mail', ok: work, text: work ? 'Ends with ' +
STAFF_EMAIL_DOMAIN + '.' : 'Not an ' + STAFF_EMAIL_DOMAIN + ' address.' });
154   const taken = isStaffEmailTaken(request.email);
155   arrayAppend(checks, {
156     key: 'email',
157     label: 'No staff account yet',
158     ok: !taken,
159     text: taken ? 'This e-mail already has a staff account - reject it as a duplicate.'
160       : 'Not in the staff e-mail hash table (one look-up among ' + pluralize(staffEmailIndex.size, 'e-mail') +
'.)',
161   });
162   const allowed = canManageStaff(actor);
163   arrayAppend(checks, {
164     key: 'reviewer',
165     label: 'You can approve it',
166     ok: allowed,
167     text: allowed ? 'Owners and Admins approve staff requests.' : 'Only an Owner or Admin can approve requests.',
168   });
169   let firstFailure = null;
170   for (let i = 0; i < checks.length; i++) {
171     if (!checks[i].ok && firstFailure === null) {
172       firstFailure = checks[i];
173     }
174   }
175   return { checks: checks, passed: firstFailure === null, firstFailure: firstFailure };
176 }

```

1. Three checks, each with a plain-language detail for the reviewer; the second is the hash-table look-up.
2. The first failed check becomes the reason the Approve button is disabled.

assets/js/backend/registrations.js · lines 178–224

```

178 /**
179  * approveRegistration - Pending → Approved. Creates the staff account with the
180  * role and the (hashed) password from the request, and puts the e-mail in the
181  * hash table. One undo step removes the account and re-opens the request.
182  * Time O(log n) + O(1) append + O(1) average hash put · Space O(1)
183  */
184 function approveRegistration(registrationId, actor) {
185   const request = findRegistration(registrationId); // 1. binary search by request id
186   if (!request) {
187     return { ok: false, error: 'Registration not found.' };
188   }
189   if (request.status !== 'Pending') {
190     return { ok: false, error: 'This request was already ' + toLowerText(request.status) + '.' };
191   }
192   const review = registrationChecks(request, actor); // 2. work e-mail, e-mail free (hash table),
reviewer allowed
193   if (!review.passed) {
194     return { ok: false, error: 'Can't approve - ' + review.firstFailure.text };
195   }
196   counters.staff = counters.staff + 1; // 3. build the staff account with the chosen
password hash
197   const id = formatStaffId(counters.staff);
198   const member = {

```

```

199     id: id,
200     fullName: request.fullName,
201     email: request.email,
202     role: request.role,
203     status: 'Active',
204     startDate: todayISO(),
205     lastSignIn: null,
206     passwordHash: request.passwordHash,
207     passwordSalt: request.passwordSalt,
208     mustChangePassword: false,
209   };
210   arrayAppend(staffMembers, member);           // 4. ids increase → the table stays sorted
211   hashPut(staffEmailIndex, member.email, id); // 5. index the e-mail so the new member can sign in
212   const before = snapshotFields(request, ['status', 'reviewedBy', 'reviewedAt', 'staffId']);
213   request.status = 'Approved';                 // 6. mark the request approved (it leaves the queue)
214   request.reviewedBy = nameOfActor(actor);
215   request.reviewedAt = nowISO();
216   request.staffId = id;
217   pushUndo('Approve ' + request.fullName + ' as ' + request.role, [
218     updateOperation('registrations', registrationId, before),
219     insertOperation('staffMembers', id),
220   ], nameOfActor(actor));
221   logActivity('staff', 'Approved registration ' + registrationId + ' - ' + request.fullName + ' joined as ' +
request.role + ' (' + id + ')', nameOfActor(actor));
222   markDataChanged();
223   return { ok: true, registration: request, staff: member };
224 }

```

1. Binary search, then refuse anything not pending or not passing the checks.
2. The staff record copies the request's password hash and salt, is appended, and its e-mail is put in the hash table.
3. The request is marked Approved with the new staff id; one Undo step reverses both changes.

Aaron Sebastian

Module 11 Billing: Client Billing List · Module 12 Billing: Bill Generation · Module 13 Payments
(Payments Validation)

60-second pitch

My modules handle the money. Every bill has an id like `BILL-202609-004872` — year, month, account — and the `bills` array is kept sorted by that id. So the **Client Billing List** finds one month's bills with a *binary search* to the first one plus a short scan, then groups them by urgency. **Bill Generation** bills every active subscriber on their connection day ("anniversary billing"), refuses duplicates with a binary search, and puts each new bill in place with a *sorted insert*; the whole batch is one Undo step. **Payments** takes the customers' reported payments in a first-in-first-out *queue* and runs five validation checks on the front one — including a *hash-table* look-up that catches a reference number used twice — before staff may confirm it.

Structures and algorithms I must be able to explain

Where	Structure / algorithm	Time	Space
One month's bills	Lower bound (binary search) + scan	$O(\log n + k)$	$O(k)$
Group by urgency	One pass into 4 arrays + insertion sort	$O(k + k^2)$	$O(k)$
Create one bill	Binary search (duplicate) + sorted insert	$O(n)$	$O(1)$
Generate a month	Loop over subscribers	$O(s \cdot n)$	$O(s)$
Payment validation order	Queue (circular buffer)	$O(n)$ build, $O(1)$ peek	$O(p)$
Re-used reference number	Hash table (METHOD:REFERENCE → ids)	$O(1)$ average	$O(n + b)$

Likely panel questions

Why does the bill id start with the year and month?

Because the array is sorted by id, all bills of one month sit next to each other. A binary search jumps to the first one and a short scan collects the rest — $O(\log n + k)$ instead of reading every bill.

What happens to someone connected on the 31st?

In a month without a 31st the period starts on that month's last day (e.g. February 28), then goes back to the 31st when it exists. The due date is always 15 days after the period starts.

How do you stop double billing?

The bill id is built from the month and the account, so a second bill for the same month has the same id — `findBill` (binary search) finds it and the bill is refused.

How do you detect a re-used reference number?

Every payment is stored in a hash table under "METHOD:REFERENCE". Validation looks that key up once; if a confirmed payment, or an earlier report still waiting, is listed there, check 5 fails. Only the *later* report is the duplicate — the first one is not blamed, and declined reports don't count.

Why doesn't the public form reject a used reference right away?

That would let anyone test other people's reference numbers. The customer's form only checks the format; the duplicate check is done by staff.

Why can't staff confirm a payment when a check fails?

`confirmPayment` runs the checks again itself and refuses — so a bill can't be marked paid for a wrong amount, a bill that is already paid, or a fake reference, even by mistake.

Billing: Client Billing List

Screens	Admin → Billing (#/admin/billing) — one month at a time
Files	backend/billing.js (billsForMonth, summarizeBills, receivablesSummary, earlierOverdueBills, filterBillRows, groupBills, billDueText) · dsa/search.js (rangeWithPrefix) · views/admin/billing.js
Data	bills sorted by billId "BILL-YYYYMM-ACCOUNT" · subscribers · the payments queue
DSA	Lower bound + range scan · one-pass grouping · insertion sort · linear search + string matching · binary search (subscriber names)

1 Business process

What task or problem from the client are you solving?

Each month the office must know how much is expected, how much was collected and how much is still outstanding, **who is overdue** (including debts from earlier months, which must never disappear from view), who is due within a week (to remind them), and which customer payments are waiting to be validated. Staff also need to find one customer's bill quickly and mark a bill paid when the customer pays at the office.

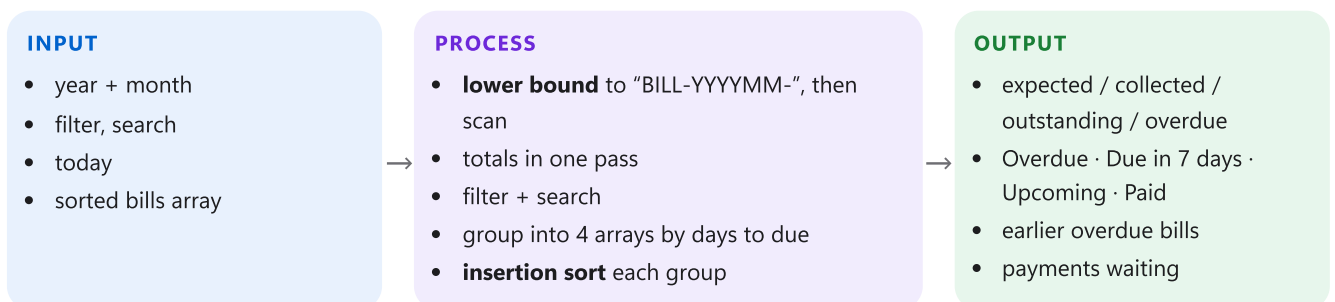
2 Input

What data is needed by the module?

Input	Values
Month	previous / next arrows, or "This month" (year + month)
Filter	All · Unpaid · Paid
Search text	subscriber name, account number or bill id
Today's date	decides overdue / due soon / upcoming
Stored data	bills (sorted by id), subscribers (for names), reported payments

3 Process

What data structure and/or algorithm did you use? How does it work step by step?



- Month's bills** (billsForMonth → rangeWithPrefix): a lower-bound binary search finds the first id $\geq$ "BILL-202609-"; a scan collects bills while the id still starts with that prefix.
- Totals** (summarizeBills): one pass adds every amount to "expected" and to "collected" or "outstanding". receivablesSummary adds up overdue bills of all months.
- Earlier overdue** (earlierOverdueBills): unpaid, past-due bills of earlier months, sorted by due date.
- Filter and search** (filterBillRows): keep Unpaid or Paid, add the subscriber's name (binary search by account), then string-search name / account / bill id.
- Group** (groupBills): one pass puts each bill in Paid, Overdue (due date passed), Due within 7 days, or Upcoming; each group is sorted by date with insertion sort.
- The **payments callout** shows the size and the front of the payment-validation queue (Module 13).

4 Output

What result does the module produce?

Four totals (expected, collected with the percentage, outstanding, overdue in all months), the “payments waiting for validation” callout, the list of overdue bills from earlier months, and the month’s bills in four groups with coloured due labels (“*N* days overdue” in red, “Due in *N* days” in orange, “Paid” + date in green) and a “Mark paid” button on each unpaid bill.

5 Complexity analysis

What is the time complexity? What is the space complexity?

Operation	Time	Space
One month’s bills (<i>n</i> bills, <i>k</i> in the month)	$O(\log n + k)$	$O(k)$
Month totals	$O(k)$	$O(1)$
Overdue in all months / earlier overdue (<i>e</i> of them)	$O(n) / O(n + e^2)$	$O(1) / O(e)$
Filter + names + search	$O(k \log s + k \cdot L \cdot m)$	$O(k)$
Group + sort each group	$O(k + k^2)$	$O(k)$
Mark one bill paid	$O(\log n)$	$O(1)$

6 Justification

Why did you choose that data structure or algorithm? Why is it appropriate?

- **Sorted array + range search:** putting year and month at the front of the id makes each month a contiguous block, so the page reads only that block ($O(\log n + k)$) instead of every bill ever issued ($O(n)$) — this matters more every month the business runs.
- **One-pass grouping:** each bill is compared with today once, and the groups match how staff work (collect overdue first, remind the due-soon ones).
- **Insertion sort per group:** the groups are small and almost in order already, the best case for insertion sort; it is also stable.
- **Totals computed on the fly** from the bills, so they are always correct after a payment or an Undo.

7 Demonstration

Show that the module works correctly — the input, the process and the output.

- 1 Admin → Billing: read the totals, the callout and the overdue panel.
- 2 Scroll to the groups; click “Unpaid”; type a name in the search; use the arrows to change month.
- 3 Press “Mark paid” on one bill → the totals change; press Undo.

Dashboard

CUSTOMERS

Applications 6

New application

Subscribers

OPERATIONS

Billing 3

Payments 4

Support 2

SYSTEM

Staff

Registrations 2

Activity

Algorithms

MS

Maria Santos

Owner · STF-0001

Account

View website

Sign out

Customer data is confidential under the Data Privacy Act of 2012. Demo data lives in memory and starts fresh when the page reloads.

Billing

Bills for September 2026 · each is due 15 days after its period starts

+ New bill

Generate bills

September 2026

This month

Expected

₱6,300

6 bills

Collected

₱1,800

29% of expected

Outstanding

₱4,500

4 unpaid bills

Overdue, all months

₱3,200

3 bills past due

4 payments waiting for validation

Next in the queue: Kristine Ramos · ₱800 via GCash, sent yesterday. Confirming a payment marks its bill paid here.

Validate payments

Still overdue from earlier months

Unpaid bills that are past their due date

Patricia Gonzales

July 2026 · ₱1,200 · 44 days overdue

Mark paid

Kristine Ramos

August 2026 · ₱800 · 18 days overdue

Mark paid

Patricia Gonzales

August 2026 · ₱1,200 · 13 days overdue

Mark paid

All

Unpaid

Paid

Search name, account, bill

Due within 7 days

1 bill · ₱1,200

Totals, payments waiting and overdue bills from earlier months.

Dashboard

CUSTOMERS

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Customer data is confidential under the Data Privacy Act of 2012. Demo data lives in memory and starts fresh when the page reloads.

Billing

1 bill · ₱1,200

All

Unpaid

Paid

Search name, account, bill

Due within 7 days

1 bill · ₱1,200

Subscriber	Period	Due	Amount	Status
Ramon Villareal E3-2026-004872	Sep 16 – Oct 15	Due in 7 days	₱1,200	Unpaid

Upcoming

3 bills · ₱3,300

Subscriber	Period	Due	Amount	Status
Kristine Ramos E3-2026-004874	Sep 22 – Oct 21	Due Oct 7	₱800	Unpaid
Mark Bautista E3-2026-004875	Sep 25 – Oct 24	Due Oct 10	₱1,000	Unpaid
Carlo Mendoza E3-2026-004877	Sep 30 – Oct 29	Due Oct 15	₱1,500	Unpaid

Paid

2 bills · ₱1,800

Subscriber	Period	Due	Amount	Status
Liza Fernandez E3-2026-004871	Sep 12 – Oct 11	Paid Sep 24	₱800	Paid
Angelica Cruz E3-2026-004878	Sep 2 – Oct 1	Paid Sep 15	₱1,000	Paid

The month's bills grouped by how soon they are due.

Trace — finding one month's bills (computed by our code)

All **21** bills are sorted by id BILL-YYYYMM-ACCOUNT. This month's ids start with **BILL-202609-**, so they sit next to each other (blue):

202607-004871 [0]	202607-004872 [1]	202607-004873 [2]	202607-004874 [3]	202607-004875 [4]	202607-004876 [5]	202607-004877 [6]
202608-004871 [7]	202608-004872 [8]	202608-004873 [9]	202608-004874 [10]	202608-004875 [11]	202608-004876 [12]	202608-004877 [13]
202608-004878 [14]	202609-004871 [15] first	202609-004872 [16]	202609-004874 [17]	202609-004875 [18]	202609-004877 [19]	202609-004878 [20]

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Step	low	high	middle	Bill id at middle	Decision
1	0	21	10	BILL-202608-004874	smaller → low = 11
2	11	21	16	BILL-202609-004872	not smaller → high = 16
3	11	16	13	BILL-202608-004877	smaller → low = 14
4	14	16	15	BILL-202609-004871	not smaller → high = 15
5	14	15	14	BILL-202608-004878	smaller → low = 15

Binary search finds the first bill of the month at index **15** in 5 steps; a short scan collects the **6** bills of the month (the code reports 11 comparisons in total instead of reading all 21).

Group (by days to due date)	Bills
Overdue	—
Due within 7 days	Ramon Villareal
Upcoming	Kristine Ramos, Mark Bautista, Carlo Mendoza
Paid	Liza Fernandez, Angelica Cruz

Totals in one pass: expected ₱6,300 · collected ₱1,800 · outstanding ₱4,500 (4 unpaid of 6).

Code walkthrough

assets/js/backend/billing.js · lines 152–159

```

152  /**
153   * billsForMonth – one month's bills. Bill ids start with "BILL-YYYYMM-", so a
154   * binary search jumps to the first one and a short scan collects the rest.
155   * Time O(log n + k) · Space O(k)
156   */
157  function billsForMonth(year, month) {
158    return rangeWithPrefix(bills, 'billId', 'BILL-' + year + pad2(month) + '-');
159  }

```

1. The prefix "BILL-" + year + two-digit month is exactly how every bill id of that month begins.

assets/js/dsa/search.js · lines 176–197

```

176  /**
177   * rangeWithPrefix – every record whose `field` starts with `prefix`, from an
178   * array sorted by `field`. Binary search jumps to the first candidate, then a
179   * short scan collects the neighbours. (Bill ids start with "BILL-YYYYMM-",
180   * so this finds one month's bills without reading the whole table.)
181   * Time O(log n + k) for k matches · Space O(k)
182   */
183  function rangeWithPrefix(sortedRecords, field, prefix) {
184    const matches = [];
185    let position = lowerBound(sortedRecords, field, prefix);
186    let comparisons = dsaLastRun.comparisons; // 1. the binary-search steps
187    while (position < sortedRecords.length && textStartsWith(sortedRecords[position][field], prefix)) {
188      arrayAppend(matches, sortedRecords[position]); // 2. collect while the prefix matches
189      position++;
190      comparisons++;
191    }
192    if (position < sortedRecords.length) {
193      comparisons++; // 3. the check that ended the scan (none at the array's
194    }
195    recordRun('Binary search + scan', comparisons, 0);
196    return matches;
197  }

```

1. `lowerBound` (binary search) finds the first id that is $\geq$ the prefix — the first bill of the month.
2. The loop keeps adding bills while their id still starts with the prefix, then stops at the first one that doesn't.

3. The comparisons are recorded (binary-search steps + scan) and shown on the Algorithms page.

assets/js/backend/billing.js · lines 230–261

```
230 /**
231  * groupBills - bucket bills into four arrays by comparing the due date with
232  * today; each bucket is then sorted by due date (insertion sort).
233  * Time O(n + k²) · Space O(n)
234  */
235 function groupBills(list, today) {
236   const overdue = [];
237   const dueSoon = [];
238   const upcoming = [];
239   const paid = [];
240   for (let i = 0; i < list.length; i++) {           // 1. one pass over the month's bills
241     const bill = list[i];
242     if (bill.status === 'Paid') {
243       arrayAppend(paid, bill);
244     } else {
245       const daysLeft = daysBetweenISO(today, bill.dueDate); // 2. days until (or since) the due date
246       if (daysLeft < 0) {
247         arrayAppend(overdue, bill);
248       } else if (daysLeft <= 7) {
249         arrayAppend(dueSoon, bill);
250       } else {
251         arrayAppend(upcoming, bill);
252       }
253     }
254   }
255   return {                                           // 3. sort each group by date (insertion sort)
256     overdue: insertionSort(overdue, 'dueDate', 'asc'),
257     dueSoon: insertionSort(dueSoon, 'dueDate', 'asc'),
258     upcoming: insertionSort(upcoming, 'dueDate', 'asc'),
259     paid: insertionSort(paid, 'paidOn', 'desc'),
260   };
261 }
```

1. One pass: paid bills go to paid; for the others daysBetweenISO gives the days to the due date (negative = overdue, 0–7 = due soon).
2. Each group is then sorted by date with insertion sort (paid ones newest first).

assets/js/backend/billing.js · lines 263–277

```
263 /** summarizeBills - expected / collected / outstanding totals in one pass. O(n) */
264 function summarizeBills(list) {
265   const totals = { expected: 0, collected: 0, outstanding: 0, count: list.length, paidCount: 0, unpaidCount: 0 };
266   for (let i = 0; i < list.length; i++) {
267     totals.expected += list[i].amount;
268     if (list[i].status === 'Paid') {
269       totals.collected += list[i].amount;
270       totals.paidCount++;
271     } else {
272       totals.outstanding += list[i].amount;
273       totals.unpaidCount++;
274     }
275   }
276   return totals;
277 }
```

1. One loop adds each amount to “expected” and to “collected” (paid) or “outstanding” (unpaid), and counts them.

Billing: Bill Generation

Screens	Billing → “Generate bills” (all active subscribers) and “New bill” (one subscriber)
Files	backend/billing.js (billingPeriod, createBill, generateMonthlyBills, billableSubscribers) · backend/utils.js (calendar maths) · views/admin/billing.js
Data	subscribers (status, plan, since) · bills (sorted by id) · the Undo stack
DSA	Binary search (subscriber, existing bill) · sorted insert (lower bound + shift) · days-from-civil date arithmetic · stack (one Undo for the batch)

1 Business process

What task or problem from the client are you solving?

Every active subscriber must receive one bill per month, for the right amount (their plan’s price), for the right period, with a due date — and never twice for the same month. E3 bills on each customer’s **connection day** (“anniversary billing”): the period starts on the day of the month they were connected, ends the day before that date next month, and is due 15 days after it starts. Suspended and terminated accounts are not billed. Staff can bill everyone at once or one subscriber, up to one month ahead, and undo the whole batch.

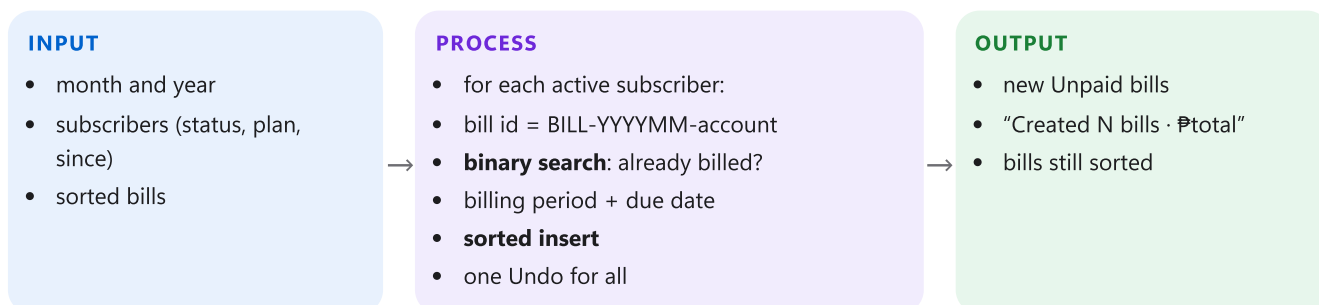
2 Input

What data is needed by the module?

Input	Detail
Month and year	from the month switch or the “New bill” sheet; at most one month ahead
Subscriber (New bill only)	chosen from the active subscribers; already-billed ones are disabled
Each subscriber’s data	status (must be Active), plan / custom price, connection date (since)
Existing bills	to refuse a second bill for the same month
Signed-in staff	recorded in the log and the Undo step

3 Process

What data structure and/or algorithm did you use? How does it work step by step?



- 1 generateMonthlyBills visits every subscriber and skips those that are not Active or were not connected yet in that month (isBillableIn).
- 2 createBill: binary search for the subscriber; check the month is valid, not before they were connected, and not more than one month ahead.
- 3 Build the id BILL-202610-004872 and **binary search** for it — if it exists, this subscriber is already billed (skipped, or an error for “New bill”).
- 4 billingPeriod: the period starts on the connection day (the month’s last day if the month is shorter), ends the day before the next start, and is due 15 days after the start — all with our day-number calendar maths.
- 5 The amount is the plan price (or the custom price); the bill is put in id order with **sortedInsert** (lower bound + shift).

6 All the new bills are collected into **one** Undo step and one activity line with the total.

4 Output

What result does the module produce?

New bill records { billId, accountNo, billingYear, billingMonth, periodStart, periodEnd, dueDate, amount, status: "Unpaid", paidOn: null, paymentRef: "", createdAt } in the sorted bills array; the message "Created N bills · Ptotal " with Undo; the month's totals and groups updated; or a clear message when there is nothing to do ("Every active subscriber already has a ... bill.").

5 Complexity analysis

What is the time complexity? What is the space complexity?

Operation	Time	Space
billingPeriod (day-number arithmetic)	$O(1)$	$O(1)$
createBill: find subscriber + duplicate check	$O(\log s) + O(\log n)$	$O(1)$
createBill: sorted insert	$O(\log n) + O(n) \text{ shift} = O(n)$	$O(1)$
generateMonthlyBills (s subscribers, n bills)	$O(s \cdot n)$	$O(s)$ for the Undo operations

6 Justification

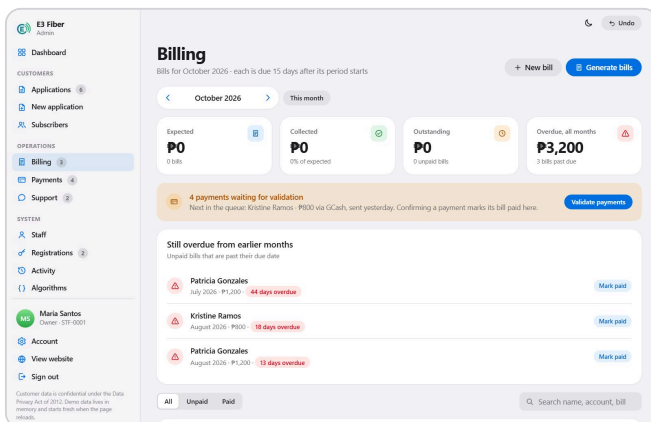
Why did you choose that data structure or algorithm? Why is it appropriate?

- **The id encodes the uniqueness rule** (one bill per account per month), so the duplicate check is a single binary search instead of scanning all bills.
- **Sorted insert** keeps the bills sorted by id, which the billing list (Module 11), Pay Bills and payment validation depend on. Bills are created in account order, so most inserts land near the end of their month block — few shifts.
- **Day-number calendar maths** (days-from-civil) makes "+15 days" and "next month" exact, including leap years and short months, without the Date object doing hidden work.
- **One Undo step for the batch** (a stack entry holding many insert operations): a mistaken "Generate" is reversed in one click.

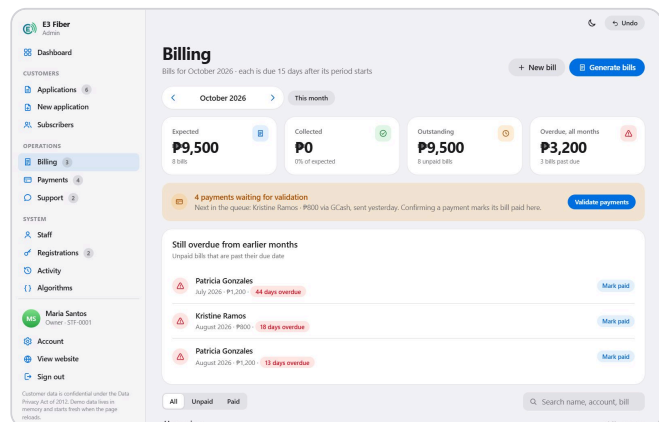
7 Demonstration

Show that the module works correctly — the input, the process and the output.

- 1 Billing → next month (empty) → "Generate bills" → the confirmation says how many bills will be created → Generate.
- 2 The month now has one bill per active subscriber; press "Generate bills" again → "Every active subscriber already has a ... bill."
- 3 "New bill": already-billed subscribers are disabled.



Input: next month, no bills yet.



Output: one bill per active subscriber.

Generate 8 bills?

Creates a October 2026 bill for each active subscriber who doesn't have one yet. You can undo this in one step.

Cancel

Generate bills

Confirmation before generating (one Undo step).

New bill

Month

September

Year

2026

Q

Search active subscribers

Angelica Cruz

Already billed for September 2026

Carlo Mendoza

Already billed for September 2026

Kristine Ramos

Already billed for September 2026

Liza Fernandez

Already billed for September 2026

Maricel Dizon

E3-2026-004881 · Power - ₱1,200

Mark Bautista

Already billed for September 2026

Cancel

Create bill

“New bill” — already-billed accounts are disabled.

Trace — periods and a generation run (computed by our code)

Connected on	Bill for	Period	Due (start + 15 days)	Why
2026-08-15	September 2026	2026-09-15 → 2026-10-14	2026-09-30	normal month: same day each month
2026-01-31	February 2026	2026-02-28 → 2026-03-30	2026-03-15	February has no 31st → its last day is used

Generate bills for 2026-10: 6 bills created, total ₱6,300. Bills table: 21 → 27 records, still sorted by id: **✓ verified**. Skipped (not active): Jerome Santiago (Terminated), Patricia Gonzales (Suspended).

Example bill	Value
Subscriber (connected on)	Liza Fernandez (2026-07-12)
Bill id	BILL-202610-004871
Period	2026-10-12 → 2026-11-11
Due date	2026-10-27
Amount (plan price)	₱800

Clicking *Generate* again: “Every active subscriber already has a bill for October 2026.” (each bill id was found by binary search). Undo stack top: “Generate 6 bills for October 2026” holding **6 insert operations** — one Undo removes all of them.

Code walkthrough

assets/js/backend/billing.js · lines 23–40

```

23  /**
24   * billingPeriod - start, end and due date of a subscriber's bill for a month.
25   * Joined on the 31st? Short months use their last day instead.
26   * Time O(1) · Space O(1)
27   */
28  function billingPeriod(subscriber, year, month) {
29      const anchorDay = parseISODate(subscriber.since).day;           // 1. the day of the month they were
connected
30      const startDay = anchorDay <= daysInMonth(year, month) ? anchorDay : daysInMonth(year, month); // 2. short
month? use its last day
31      const next = addMonths(year, month, 1);
32      const nextDay = anchorDay <= daysInMonth(next.year, next.month) ? anchorDay : daysInMonth(next.year,
next.month);
33      const periodStart = isoDate(year, month, startDay);
34      const nextStart = isoDate(next.year, next.month, nextDay);       // 3. the next period's start ...

```

```

35     return { // 4. ... minus one day ends this one; due 15
days after the start
36         periodStart: periodStart,
37         periodEnd: addDaysISO(nextStart, -1),
38         dueDate: addDaysISO(periodStart, BILL_DUE_AFTER_DAYS),
39     };
40 }

```

1. The anchor is the day of the month the customer was connected.
2. If this month (or next month) is too short for that day, its last day is used instead.
3. The period ends the day before the next period starts; the due date is 15 days after the start (addDaysISO uses our day numbers).

assets/js/backend/billing.js · lines 53–105

```

53  /**
54   * createBill - issue one month's bill for an ACTIVE subscriber.
55   * The new bill is "put" in id order with sortedInsert.
56   * options.quiet → no own undo/log entry (used by generateMonthlyBills).
57   * Time O(log n) checks + O(n) insert · Space O(1)
58   */
59  function createBill(accountNo, year, month, actor, options) {
60      const quiet = options && options.quiet;
61      const subscriber = findSubscriber(accountNo); // 1. binary search for the account
62      if (!subscriber) {
63          return { ok: false, error: 'Subscriber not found.' };
64      }
65      if (subscriber.status !== 'Active') { // 2. only active accounts are billed
66          return { ok: false, error: subscriber.fullName + ' is ' + toLowerText(subscriber.status) + ' - only active
accounts are billed.' };
67      }
68      if (!(month >= 1 && month <= 12) || !(year >= 2000 && year <= 2100)) {
69          return { ok: false, error: 'Choose a valid month.' };
70      }
71      const since = parseISODate(subscriber.since);
72      if (year * 12 + month < since.year * 12 + since.month) {
73          return { ok: false, error: subscriber.fullName + ' was connected on ' + formatDate(subscriber.since) + '.' };
74      }
75      const clock = readClock();
76      if (year * 12 + month > clock.year * 12 + clock.month + 1) {
77          return { ok: false, error: 'Bills can be created up to one month ahead.' };
78      }
79      const billId = makeBillId(year, month, accountNo); // 3. e.g. BILL-202609-004872
80      if (findBill(billId)) { // 4. binary search: already billed this month?
81          return { ok: false, error: subscriber.fullName + ' already has a bill for ' + formatMonthYear(year, month) +
'.' };
82      }
83      const period = billingPeriod(subscriber, year, month); // 5. period and due date (anniversary billing)
84      const bill = {
85          billId: billId,
86          accountNo: accountNo,
87          billingYear: year,
88          billingMonth: month,
89          periodStart: period.periodStart,
90          periodEnd: period.periodEnd,
91          dueDate: period.dueDate,
92          amount: planPrice(subscriber.planId, subscriber.customPrice),
93          status: 'Unpaid',
94          paidOn: null,
95          paymentRef: '',
96          createdAt: nowISO(),
97      };
98      sortedInsert(bills, 'billId', bill); // 6. "put" it in id order (binary search +
shift)
99      if (!quiet) {
100          pushUndo('Create ' + formatMonthYear(year, month) + ' bill for ' + subscriber.fullName,
[insertOperation('bills', billId)], nameOfActor(actor));
101          logActivity('billing', 'Created ' + formatMonthYear(year, month) + ' bill for ' + subscriber.fullName + ' ('
+ formatPeso(bill.amount) + ')', nameOfActor(actor));
102      }
103      markDataChanged();

```

```

104     return { ok: true, bill: bill };
105 }

```

1. Steps 1–2 and the checks after them: the subscriber must exist (binary search) and be Active; the month must be valid, not before the connection, and at most one month ahead.
2. Steps 3–4: build the id and refuse it if it already exists (binary search).
3. Steps 5–6: compute the period, build the Unpaid bill with the plan price, and put it in id order. When called by “Generate”, quiet skips the per-bill Undo so the batch gets one step.

assets/js/backend/billing.js · lines 113–150

```

113  /**
114   * generateMonthlyBills – bill every active subscriber who has no bill for the
115   * month yet. All the new bills form ONE undo step.
116   * Time O(s · n) for s subscribers · Space O(s)
117   */
118  function generateMonthlyBills(year, month, actor) {
119      const operations = [];
120      let created = 0;
121      let total = 0;
122      let firstError = '';
123      let billable = 0;
124      for (let i = 0; i < subscribers.length; i++) {                // 1. visit every subscriber
125          if (!isBillableIn(subscribers[i], year, month)) {        // 2. skip inactive accounts and months before
they were connected
126              continue;
127          }
128          billable++;
129          const result = createBill(subscribers[i].accountNo, year, month, actor, { quiet: true }); // 3. refuses if
already billed
130          if (result.ok) {
131              created++;
132              total += result.bill.amount;
133              arrayAppend(operations, insertOperation('bills', result.bill.billId)); // 4. remember it for one Undo
134          } else if (firstError === '' && !hasBill(subscribers[i].accountNo, year, month)) {
135              firstError = result.error;                                // a real problem (e.g. too far ahead), not
"already billed"
136          }
137      }
138      if (created === 0) {
139          let error = 'Every active subscriber already has a bill for ' + formatMonthYear(year, month) + '.';
140          if (firstError !== '') {
141              error = firstError;
142          } else if (billable === 0) {
143              error = 'No active subscriber was connected yet in ' + formatMonthYear(year, month) + '.';
144          }
145          return { ok: false, created: 0, total: 0, error: error };
146      }
147      pushUndo('Generate ' + pluralize(created, 'bill') + ' for ' + formatMonthYear(year, month), operations,
nameOfActor(actor));
148      logActivity('billing', 'Generated ' + pluralize(created, 'bill') + ' for ' + formatMonthYear(year, month) + '
(' + formatPeso(total) + ')', nameOfActor(actor));
149      return { ok: true, created: created, total: total, error: '' };
150  }

```

1. Visit every subscriber, skip the inactive ones, and try to bill each; already-billed ones simply return “not ok”.
2. Every created bill adds one insert operation. If none was created, the message gives the real reason: the first problem found (e.g. more than one month ahead), nobody connected yet that month, or everyone already billed.
3. pushUndo stores all the operations as one step; the log shows the count and total.

Payments (Payments Validation)

Screens	Customer: Pay Bills (#/pay) · Staff: Payments (#/admin/payments) with the validation queue
Files	backend/payments.js (reportPayment, paymentsQueue, validatePayment, confirmPayment, declinePayment, listPayments) · views/public/pay.js · views/admin/payments.js
Data	payments (sorted by id) · paymentReferenceIndex (hash table) · bills
DSA	Queue (validation order) · binary search (bill) · character scan (reference format) · hash table (re-used reference) · stack (Undo)

1 Business process

What task or problem from the client are you solving?

Customers pay through GCash, Maya, a bank or a payment centre and then tell E3 they paid. Before a bill is marked paid, staff must be sure the report is genuine: it is for a real bill of that account, the bill is not already paid, the amount is exactly the bill, the reference number looks like one from that payment method, and **the same reference was not used before** (a common mistake — or fraud). Reports are handled in the order they arrived. Confirmed payments settle the bill; declined ones show the customer a reason.

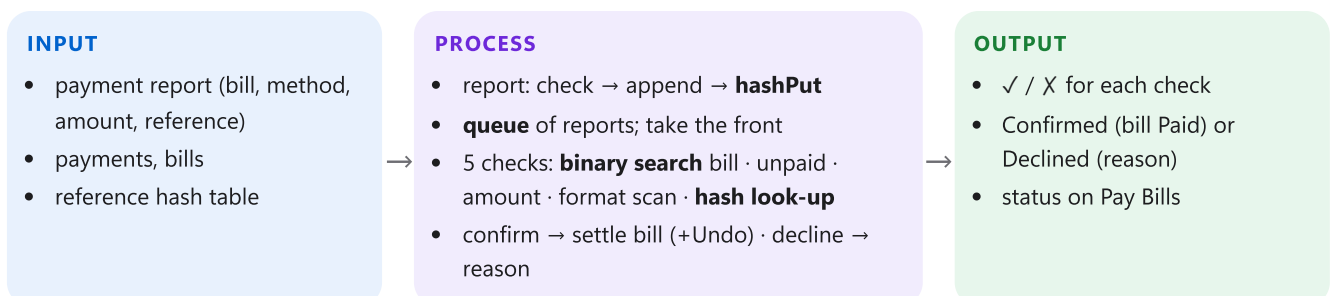
2 Input

What data is needed by the module?

Who	Input	Rule
Customer (Pay Bills)	Account number + registered mobile	both must match (the error never says which part was wrong)
	Bill paid	one of their unpaid bills not already being checked
	Method	GCash · Maya · Bank transfer · Over the counter
	Amount paid	₱1 – ₱100,000 (filled in from the bill)
	Reference number	GCash 13 digits · Maya 12 letters/digits · bank or counter 8–20 letters/digits
Staff (Payments)	The front of the queue (or any row → “Validate”)	—
	Decision	Confirm (only if every check passes) or Decline with a reason (≥ 5 characters)

3 Process

What data structure and/or algorithm did you use? How does it work step by step?



Customer reports a payment (reportPayment): check the bill, the method, the reference *format* and the amount; append the payment (ids only grow → sorted) and add its reference to the hash table under “METHOD:REFERENCE”. It joins the rear of the validation queue.

Staff validate (paymentsQueue → queuePeek → validatePayment):

- 1 Bill belongs to the account** — binary search for the bill id; its account must match.

- 2 **Bill is still unpaid** — e.g. not settled at the office in the meantime.
- 3 **Amount matches the bill** — one comparison; the text says how much short or over.
- 4 **Reference format** — scan the characters: right length for the method, digits only for GCash.
- 5 **Reference not used before** — one hash-table look-up of "METHOD:REFERENCE"; a confirmed payment there, or an earlier report still waiting, means this one is a duplicate (declined reports and later reports don't count).
- 6 **Confirm** (`confirmPayment`) re-runs the checks and refuses if any fails; otherwise the payment becomes Confirmed and the bill Paid — one Undo step reverses both. **Decline** suggests a customer-friendly reason from the first failed check.

4 Output

What result does the module produce?

For staff: the validation card with ✓/X and a plain explanation for each check, the verdict (all checks passed — safe to confirm, or how many failed — decline it with a reason), Confirm enabled only when all pass, and the list of all payments (filter, search, sort). For the customer: the report under "Payment reports" as For verification, then Confirmed (the bill is paid) or Declined with the reason.

5 Complexity analysis

What is the time complexity? What is the space complexity?

Operation	Time	Space
Report a payment (pending check over n payments, then append + <code>hashPut</code>)	$O(n) + O(1)$ average	$O(1)$
Build the validation queue / peek the front	$O(n) / O(1)$	$O(p)$
Check 1 — find the bill (binary search over B bills)	$O(\log B)$	$O(1)$
Checks 2 and 3	$O(1)$	$O(1)$
Check 4 — format (k characters)	$O(k)$	$O(1)$
Check 5 — hash look-up (d earlier uses of the reference)	$O(1)$ average (+ $O(d \log n)$)	$O(1)$
validatePayment overall	$O(\log B + k)$	$O(1)$
Confirm (validate + settle the bill)	$O(\log B + k)$	$O(1)$

6 Justification

Why did you choose that data structure or algorithm? Why is it appropriate?

- **Queue:** payments are validated in the order customers reported them — first come, first served — and the front is available in $O(1)$.
- **Hash table for references:** checking "was this reference used?" by scanning every payment would be $O(n)$ for each validation and grow forever; the hash table answers in $O(1)$ on average. The key includes the method, so a GCash and a bank number that happen to look alike don't clash.
- **Binary search for the bill:** bills are sorted by id (Module 12), so finding the reported bill is $O(\log B)$.
- **Character scan** for the format: exact rules per method without regular expressions.
- **Checks repeated inside Confirm:** the rule is enforced by the backend, not just by a disabled button.
- **Privacy:** the public form never reveals whether a reference exists; only staff see duplicates.

7 Demonstration

Show that the module works correctly — the input, the process and the output.

- 1 Pay Bills → "Fill in" the demo account → GCash with a short reference → the format error; fix it → Send → the report appears as For verification.
- 2 Staff → Payments: the front (Kristine) passes all 5 checks → Confirm → her bill is paid.

- 3 Open the others: Liza's bill was already paid at the office, Patricia sent ₱200 short, Mark re-used Angelica's GCash reference (found by the hash table) → Decline with the suggested reason.

Report a payment
Paid through GCash, Maya, a bank or at our office? Send the details and we'll confirm within one business day.

GCash or Maya
Send to E3 Fiber Connect using the number printed on your bill.

Bank transfer
Use the bank details printed on your bill.

Over the counter
E3 office, Santa Maria - Mon-Sat, 9 AM - 5 PM.

Bill you paid
October 2026 - ₱800
Due Oct 27, 2026

Paid through
GCash

Amount paid (₱)
800

Reference number
1234 5678

GCash reference numbers have 13 digits e.g. 5012 873 246 119.
GCash reference numbers have 13 digits.

We use these details only to verify your payment. See our [Privacy Notice](#).

Kristine Ramos
PAY-000002 - August 2026 bill of ₱800 - sent yesterday
GCash - 7204519833102

Bill belongs to the account
Found the August 2026 bill of E3-2026-004874 with binary search (3 comparisons).

Bill is still unpaid
Unpaid - due Sep 6, 2026.

Amount matches the bill
₱800 paid for a ₱800 bill.

Reference number format
GCash references have 13 digits — 7204519833102 fits.

Reference number not used before
No earlier or confirmed payment used it — one hash-table look-up among 5 references on record.

All 5 checks passed — safe to confirm.

[Decline](#) [Confirm payment](#)

Staff: the front payment passes all 5 checks.

Customer input: the GCash format rule.

Validate PAY-000005

Mark Bautista
PAY-000005 - September 2026 bill of ₱1,000 - sent 1 hr ago
GCash - 5012873246119

Bill belongs to the account
Found the September 2026 bill of E3-2026-004875 with binary search (5 comparisons).

Bill is still unpaid
Unpaid - due Oct 10, 2026.

Amount matches the bill
₱1,000 paid for a ₱1,000 bill.

Reference number format
GCash references have 13 digits — 5012873246119 fits.

Reference number not used before
Already used by PAY-000001 (Angelica Cruz, confirmed).

1 check failed — decline it with a reason.

[Close](#) [Decline](#) [Confirm payment](#)

Check 5 fails: reference already used (hash table).

Validate PAY-000004

Patricia Gonzales
PAY-000004 - August 2026 bill of ₱1,200 - sent yesterday
GCash - 8830146527015

Bill belongs to the account
Found the August 2026 bill of E3-2026-004876 with binary search (4 comparisons).

Bill is still unpaid
Unpaid - due Sep 11, 2026.

Amount matches the bill
Paid ₱1,000 for a ₱1,200 bill — ₱200 short.

Reference number format
GCash references have 13 digits — 8830146527015 fits.

Reference number not used before
No earlier or confirmed payment used it — one hash-table look-up among 5 references on record.

1 check failed — decline it with a reason.

[Close](#) [Decline](#) [Confirm payment](#)

Check 3 fails: ₱200 short.

Validate PAY-000006

Liza Fernandez
PAY-000006 - October 2026 bill of ₱800 - sent just now
GCash - 6130552918447

Bill belongs to the account
Found the October 2026 bill of E3-2026-004871 with binary search (2 comparisons).

Bill is still unpaid
Unpaid - due Oct 27, 2026.

Amount matches the bill
₱800 paid for a ₱800 bill.

Reference number format
GCash references have 13 digits — 6130552918447 fits.

Reference number not used before
No earlier or confirmed payment used it — one hash-table look-up among 5 references on record.

All 5 checks passed — safe to confirm.

[Close](#) [Decline](#) [Confirm payment](#)

Check 2 fails: the bill was already paid at the office.

Decline payment

Liza Fernandez reported ₱800 via GCash (ref 6130552918447). They will see your reason on the Pay Bills page.

Reason

e.g. Reference number not found in GCash

[Cancel](#) [Decline payment](#)

Decline: the reason is suggested from the failed check.

Trace — the queue, the five checks and the hash table (computed by our code)

The validation queue (circular buffer, front = oldest report):

PAY-000002
[0] front

PAY-000003
[1]

PAY-000004
[2]

PAY-000005
[3]

Payment	1. Bill belongs to the account	2. Bill is still unpaid	3. Amount matches the bill	4. Reference number format	5. Reference number not used before	Verdict
PAY-000002 Kristine Ramos GCash · ₱800	✓ yes	✓ yes	✓ yes	✓ yes	✓ yes	✓ Confirm
PAY-000003 Liza Fernandez Maya · ₱800	✓ yes	✗ no	✓ yes	✓ yes	✓ yes	✗ Decline
PAY-000004 Patricia Gonzales GCash · ₱1,000	✓ yes	✓ yes	✗ no	✓ yes	✓ yes	✗ Decline
PAY-000005 Mark Bautista GCash · ₱1,000	✓ yes	✓ yes	✓ yes	✓ yes	✗ no	✗ Decline

Payment	The failed check says	Suggested reason for the customer
PAY-000003 Liza Fernandez	Already paid on Sep 24, 2026 (Office payment).	"This bill was already paid, so this payment was not applied. Please contact us if you paid twice."
PAY-000004 Patricia Gonzales	Paid ₱1,000 for a ₱1,200 bill — ₱200 short.	"You sent ₱1,000 but the bill is ₱1,200. Please visit our office so we can settle the difference."
PAY-000005 Mark Bautista	Already used by PAY-000001 (Angelica Cruz, confirmed).	"This reference number was already used for another payment. Please check your receipt."

The reference hash table (key METHOD:REFERENCE → payment ids):

Bucket	Key	Payment ids
3	GCASH:7204519833102	PAY-000002
12	MAYA:4F7K2M9Q1T8B	PAY-000003
13	GCASH:8830146527015	PAY-000004
15	GCASH:5012873246119	PAY-000001, PAY-000005 ✗ used twice

Reference formats (check 4): GCash — 13 digits (e.g. 5012 873 246 119); Maya — 12 letters or digits (e.g. 4F7K 2M9Q 1T8B); Bank transfer — 8–20 letters or digits (e.g. FT26 0918 8830); Over the counter — 8–20 letters or digits (e.g. BC26 0918 0457).

Code walkthrough

assets/js/backend/payments.js · lines 147–212

```
147 /**
148  * validatePayment - the five checks shown on the Payments page. Nothing is
149  * changed; the result says which checks passed and why.
150  * Returns { checks: [{ key, label, ok, text }], passed, failed, firstFailure, bill }.
151  * Time O(log n + k) · Space O(1)
```

```

152  */
153  function validatePayment(payment) {
154      const checks = [];
155
156      const bill = findBill(payment.billId); // 1 · binary search
157      const billSteps = dsaLastRun.comparisons;
158      const belongs = bill !== null && bill.accountNo === payment.accountNo;
159      const billName = bill ? formatMonthYear(bill.billingYear, bill.billingMonth) + ' bill' : 'bill ' +
payment.billId;
160      arrayAppend(checks, {
161          key: 'bill',
162          label: 'Bill belongs to the account',
163          ok: belongs,
164          text: belongs ? 'Found the ' + billName + ' of ' + payment.accountNo + ' with binary search (' +
pluralize(billSteps, 'comparison') + ').'
165              : 'There is no ' + billName + ' for account ' + payment.accountNo + '.',
166      });
167
168      const paidByThis = belongs && bill.status === 'Paid' && bill.paymentRef === payment.paymentId;
169      const unpaid = belongs && (bill.status === 'Unpaid' || paidByThis); // 2 · one field (a confirmed payment
settled it itself)
170      let unpaidText = 'No bill to check.';
171      if (paidByThis) {
172          unpaidText = 'Was unpaid – this payment settled it on ' + formatDate(bill.paidOn) + '.';
173      } else if (belongs) {
174          unpaidText = unpaid ? 'Unpaid · due ' + formatDate(bill.dueDate) + '.'
175              : 'Already paid on ' + formatDate(bill.paidOn) + (bill.paymentRef ? ' (' + bill.paymentRef + ')' : '') +
'.';
176      }
177      arrayAppend(checks, { key: 'unpaid', label: 'Bill is still unpaid', ok: unpaid, text: unpaidText });
178
179      const amountOk = belongs && payment.amount === bill.amount; // 3 · one comparison
180      let amountText = 'No bill amount to compare.';
181      if (belongs) {
182          const difference = payment.amount - bill.amount;
183          amountText = amountOk ? formatPeso(payment.amount) + ' paid for a ' + formatPeso(bill.amount) + ' bill.'
184              : 'Paid ' + formatPeso(payment.amount) + ' for a ' + formatPeso(bill.amount) + ' bill – '
185              + formatPeso(difference < 0 ? -difference : difference) + (difference < 0 ? ' short.' : ' too much. ');
186      }
187      arrayAppend(checks, { key: 'amount', label: 'Amount matches the bill', ok: amountOk, text: amountText });
188
189      const format = checkReferenceFormat(payment.method, normalizePaymentReference(payment.referenceCode)); // 4 ·
scan
190      arrayAppend(checks, { key: 'format', label: 'Reference number format', ok: format.ok, text: format.text });
191
192      const duplicate = findDuplicatePayment(payment); // 5 · hash table
193      arrayAppend(checks, {
194          key: 'duplicate',
195          label: 'Reference number not used before',
196          ok: duplicate === null,
197          text: duplicate === null ? 'No earlier or confirmed payment used it – one hash-table look-up among ' +
pluralize(paymentReferenceIndex.size, 'reference') + ' on record.'
198              : 'Already used by ' + duplicate.paymentId + ' (' + duplicate.fullName + ', ' +
toLowerText(duplicate.status) + ').'
199      });
200
201      let failed = 0;
202      let firstFailure = null;
203      for (let i = 0; i < checks.length; i++) {
204          if (!checks[i].ok) {
205              failed++;
206              if (firstFailure === null) {
207                  firstFailure = checks[i];
208              }
209          }
210      }
211      return { checks: checks, passed: failed === 0, failed: failed, firstFailure: firstFailure, bill: bill };
212  }

```

1. Check 1: findBill is a binary search; dsaLastRun.comparisons right after it is shown in the check's text.
2. Checks 2–3 read the bill's status and compare the amounts (the text says how much short or over). A payment that was already confirmed still passes check 2, because the bill was settled by that same payment (bill.paymentRef).

3. Check 4 normalises the reference (spaces and dashes removed, letters in capitals) and scans it with the method's rule.
4. Check 5 asks the hash table for another payment with the same method and reference.
5. The last loop counts failures and remembers the first one (used for the verdict and the suggested decline reason).

assets/js/backend/payments.js · lines 108–121

```
108  /**
109   * indexPaymentReference - add a payment to the reference hash table. The value
110   * stored under a key is the list of payment ids that used that reference.
111   * Time O(1) average · Space O(1)
112   */
113   function indexPaymentReference(payment) {
114     const key = paymentReferenceKey(payment.method, payment.referenceCode);
115     const ids = hashGet(paymentReferenceIndex, key);
116     if (ids === null) {
117       hashPut(paymentReferenceIndex, key, [payment.paymentId]);
118     } else {
119       arrayAppend(ids, payment.paymentId);
120     }
121   }
```

1. The key is "METHOD:REFERENCE", e.g. GCASH:5012873246119.
2. The value is a small list of payment ids: a new key gets a new list; an existing key gets the id appended — so a second use is visible in one look-up.

assets/js/backend/payments.js · lines 123–145

```
123  /**
124   * findDuplicatePayment - a payment that used the same method and reference number
125   * BEFORE this one: an already confirmed payment, or an earlier report that is still
126   * waiting. Declined reports don't count, and neither do later reports (the later one
127   * is the duplicate, not this one). One hash-table look-up, then a binary search per id.
128   * Time O(1) average + O(d log n) for d other uses (d is almost always 0) · Space O(1)
129   */
130   function findDuplicatePayment(payment) {
131     const ids = hashGet(paymentReferenceIndex, paymentReferenceKey(payment.method, payment.referenceCode));
132     if (ids === null) {
133       return null; // 1. reference never seen before
134     }
135     for (let i = 0; i < ids.length; i++) {
136       if (ids[i] !== payment.paymentId) {
137         const other = findPayment(ids[i]); // 2. every other use of it (binary search)
138         if (other && (other.status === 'Confirmed'
139           || (other.status === 'For verification' && other.paymentId < payment.paymentId))) {
140           return other; // 3. confirmed, or reported earlier → duplicate
141         }
142       }
143     }
144     return null;
145   }
```

1. One hashGet; no list → never used.
2. Otherwise every other id in the list is found by binary search; one that is Confirmed, or still waiting with a *smaller* id (reported earlier), makes this payment the duplicate.

assets/js/backend/payments.js · lines 327–361

```
327  /**
328   * confirmPayment - accept the payment and mark its bill paid. Refused unless
329   * all five validation checks pass. One undo step reverses both changes.
330   * Time O(log n + k) · Space O(1)
331   */
332   function confirmPayment(paymentId, actor) {
333     if (!actor) {
334       return { ok: false, error: 'Please sign in again.' };
335     }
336     const payment = findPayment(paymentId);
337     if (!payment) {
338       return { ok: false, error: 'Payment not found.' };
339     }
340   }
```

```

340   if (payment.status !== 'For verification') {
341     return { ok: false, error: 'This payment was already ' + toLowerText(payment.status) + '.' };
342   }
343   const validation = validatePayment(payment);
344   if (!validation.passed) {
345     return { ok: false, error: 'Can't confirm - "' + validation.firstFailure.label + '" failed. Decline it
instead.', validation: validation };
346   }
347   const bill = validation.bill;
348   const beforePayment = snapshotFields(payment, ['status', 'reviewedBy', 'reviewedAt']);
349   const beforeBill = snapshotFields(bill, ['status', 'paidOn', 'paymentRef']);
350   payment.status = 'Confirmed';
351   payment.reviewedBy = nameOfActor(actor);
352   payment.reviewedAt = nowISO();
353   settleBill(bill.billId, actor, payment.paymentId, { quiet: true });
354   pushUndo('Confirm payment ' + paymentId + ' (' + payment.fullName + ')', [
355     updateOperation('payments', paymentId, beforePayment),
356     updateOperation('bills', bill.billId, beforeBill),
357   ], nameOfActor(actor));
358   logActivity('payment', 'Confirmed payment ' + paymentId + ' - ' + payment.fullName + ' paid ' +
formatPeso(payment.amount), nameOfActor(actor));
359   markDataChanged();
360   return { ok: true, payment: payment, bill: bill };
361 }

```

1. Refuse when no staff member is signed in, or the payment is unknown or already handled; then run `validatePayment` again — if any check fails, nothing changes.
2. Snapshot the payment and the bill, mark the payment Confirmed and settle the bill, then push one Undo step containing both changes.

APPENDIX A

Complexity summary — all 13 modules

The main operation of each module. n = records in the table, k = results or text length, s = subscribers, r = staff requests, p = items waiting in a queue, b = hash buckets. Details are on each module's page.

#	Module	Member	Main structure / algorithm	Time	Space
1	Plan Selection	Joshua	Linear search (recommender) · insertion sort (best case) · selection sort (cards)	$O(n) \cdot O(n^2)$ cards	$O(n)$
2	Application	Joshua	Character scans · binary search (barangay) · linear duplicate check · append	$O(n)$	$O(1)$ extra
3	Application Tracking	Joshua	Binary search on the reference number	$O(\log n)$	$O(1)$
4	Application List	Justin	Linear filter · naive string matching · insertion / selection / bubble sort · review queue	$O(n^2)$ worst sort · $O(1)$ peek	$O(n)$
5	Walk-in Override	Justin	Validation · linear duplicate check · append · Undo stack	$O(n)$	$O(1)$
6	Create Account	Justin	Binary search · lower bound + shift (sorted insert)	$O(n)$	$O(1)$ extra
7	Admin Login	Riceerich	Hash table (djb2, chaining) · binary search · salted password hash	$O(1)$ average	$O(1)$; table $O(n + b)$
8	Accounts Management	Riceerich	Filter + search + sort · binary search · oldest-bill queue	$O(\log n)$ find · $O(n^2)$ worst sort	$O(n)$
9	Admin/Staff Creation & Management	Riceerich	Hash table (unique e-mail) · append · binary search	$O(1)$ average add · $O(\log n)$ actions	$O(1)$
10	Admin Registration & Approval	Riceerich	Approval queue · hash look-up · binary search · append + hashPut	$O(1)$ peek · $O(\log r)$ approve	$O(p)$
11	Billing: Client Billing List	Aaron	Lower bound + range scan · one-pass grouping · insertion sort	$O(\log n + k)$	$O(k)$
12	Billing: Bill Generation	Aaron	Binary search (duplicate) · sorted insert · one Undo step	$O(s \cdot n)$	$O(s)$
13	Payments Validation	Aaron	Queue · binary search · character scan · hash table	$O(\log n + k)$	$O(1)$

Reading Big-O in one minute

What it means

Big-O describes how the work **grows** when the data grows, ignoring fixed costs. $O(\log n)$ grows very slowly (double the data → one more step); $O(n)$ grows in step with the data; $O(n^2)$ grows with the square (double the data → four times the work).

Time vs. space

Time counts the steps (comparisons, moves). **Space** counts the *extra* memory the function needs besides its input — e.g. a sort that copies the list is $O(n)$ space, a binary search is $O(1)$.

General questions — any member may be asked

Why is the whole website one HTML page?

Our data lives in JavaScript arrays, and arrays exist only while the page is open — they cannot be carried to another HTML file. So every screen is a section of one `index.html`, and the router shows the section named after the # in the address. The arrays stay alive while you move between screens.

Where is the data saved? What happens when you refresh?

Only in memory (the arrays in `backend/database.js`). There is no `localStorage`, cookie or server. Refreshing the page loads the sample data again — this is expected for the project rule, and it also makes every demonstration start from the same state.

Why didn't you use `push`, `sort`, `indexOf`, `filter` and the other built-ins?

It was a project rule, and it shows we understand what they do. Each one has our own version in `assets/js/dsa/`: `arrayAppend` (`push`), `arrayInsertAt` / `arrayRemoveAt` (`splice`), `linearSearch` (`indexOf` / `find`), `linearSearchAll` (`filter`), `insertionSort` and others (`sort`), `textFind` (`includes`), `trimText`, `splitText` and `toLowerCaseText` — and no regular expressions.

What does “procedural, not OOP” mean in your code?

There are no classes, no `this` and no methods. Data is plain records (e.g. a queue is `{ items, front, count, capacity }`) and every operation is a separate function that receives the record: `enqueue(queue, item)`, `stackPush(stack, item)`, `hashPut(table, key, value)`.

Aren't $O(n^2)$ sorts too slow?

Not for our data: the lists have tens to hundreds of records, and the tables are usually already in order, which is insertion sort's best case — $O(n)$. They are also simple to verify, and insertion and bubble sort are stable. The *Benchmark lab* on the Algorithms page runs the three sorts on the same random data and shows the exact comparison counts, so the difference can be seen live.

Why keep the arrays sorted instead of sorting when needed?

Records are looked up far more often than they are added. Keeping each array sorted by its key (append when ids only grow, sorted insert otherwise) lets every look-up be a binary search — $O(\log n)$ — instead of a linear search, and each month of bills becomes one contiguous block (Module 11).

Why hash tables, when you already have binary search?

Binary search only works on the key the array is sorted by. Staff are sorted by id, but sign-in searches by e-mail; payments are sorted by id, but the duplicate check searches by reference number; wrong sign-in tries are counted per e-mail. A hash table indexes that second key and answers in $O(1)$ on average, without a second sorted copy that would cost $O(n)$ per insert.

How does Undo work?

It is a stack. Every admin action pushes one step holding its operations (insert / update / remove) and the “before” values. Undo pops the newest step and reverses its operations. One step can hold many operations — generating a month of bills, or confirming a payment (payment + bill). Only the last 25 steps are kept, and the stack is cleared at sign-in and sign-out so staff only ever undo their own work. Customers' actions are not on the stack, so Undo refuses a step that would break one — e.g. deleting a bill a customer already reported a payment for, or re-opening a declined payment after the customer sent a new report. Each toast's Undo only reverses the action that toast announced.

How secure is it?

It is a class project that runs entirely in the browser, so we are honest about the limits. What we did: passwords are stored only as a salted hash (FNV-1a repeated 400 times — a demo, not a real password hash), five wrong passwords in a row for one e-mail (counted in a hash table) pause sign-in for 30 seconds, the sign-in error never says whether the e-mail or the password was wrong — nor, through the registration messages, which e-mails belong to staff — staff screens need a signed-in account, and everything shown on screen is escaped so typed text can't run as code. A real deployment would need a server and database, a proper password hash (bcrypt or Argon2) and HTTPS, because anything in the browser can be read with developer tools.

How do you prove the complexities?

Every search and sort records its comparisons and moves; the Algorithms page shows the last runs, the live sizes of every structure and the benchmark. The traces in this reviewer were produced by running our actual functions on the sample data.

Did you use the Date object?

Only to read the clock: `readClock` reads today's date from the computer, and `Date.now()` / `performance.now()` give timestamps for the sign-in pause, the activity log and the sort stopwatch. All date maths — adding days, next month, days between two dates, leap years — is our own day-number arithmetic in `backend/utils.js`.

What would you improve with more time?

A real server and database so data survives a refresh; sending e-mail or SMS for application updates and bills; online payment through the payment provider's API instead of reported references; and a balanced tree or database index once the tables grow to many thousands of records.

Glossary

Array	Items stored in numbered slots (0, 1, 2 ...); reading slot i is O(1). Our “tables”.
Key	The field an array is sorted and searched by (reference number, account number, bill id ...).
Linear search	Check items one by one until a match. O(n); works on unsorted data.
Binary search	On a sorted array, compare with the middle and keep one half. O(log n).
Lower bound	Binary search for the first position whose key is ≥ a value — where a new item belongs, or where a range starts.
Sorted insert	Lower bound + shift the later items one slot right. Keeps the array sorted. O(n).
Naive string matching	Try the search text at every position of the field. O(L · m).
Stable sort	Equal items keep their original order (insertion and bubble sort are stable; selection sort is not).
Stack (LIFO)	Last in, first out: push and pop at the top. Used for Undo, Back and open dialogs.
Queue (FIFO)	First in, first out: enqueue at the rear, dequeue at the front. Used for review, payments, staff requests, support, oldest bill.
Circular buffer	A queue in a fixed array whose rear wraps to the start, so nothing is shifted: O(1) enqueue and dequeue.
Hash table	A hash function turns a key into a bucket number, so a look-up checks one short list. O(1) on average.
djb2	Our string hash function: start at 5381, then hash × 33 + character for every character.
Collision / chaining	Two keys in the same bucket; we keep them in a chain (list) inside that bucket.
Load factor	Keys ÷ buckets. Above 0.75 the table grows to 2b + 1 buckets and re-inserts every key.
Salt	Extra text mixed into a password before hashing, so equal passwords get different hashes.
Big-O	How the work grows with the size of the data (worst case unless stated).

File map — who explains what

Member	Backend (rules)	Screens (views)
Joshua	backend/plans.js · backend/applications.js (submit, track)	views/public/plans.js · apply.js · track.js · submitted.js
Justin	backend/applications.js (list, review queue, completeInstallation) · backend/subscribers.js (create account) · backend/history.js	views/admin/applications.js · application-detail.js · new-application.js
Riceerich	backend/auth.js · backend/staff.js · backend/registrations.js · backend/subscribers.js (list, status, plan)	views/admin/login.js · subscribers.js · subscriber-detail.js · staff.js · register.js · registrations.js
Aaron	backend/billing.js · backend/payments.js	views/admin/billing.js · payments.js · views/public/pay.js
Everyone	dsa/arrays.js · strings.js · search.js · sort.js · stack.js · queue.js · hashtable.js · stats.js · backend/database.js · utils.js	ui/router.js · sheets.js · toasts.js · forms.js · app.js

Shared screens for the demonstration

E3 Fiber

Admin

Dashboard

CUSTOMERS

Applications

6

New application

Subscribers

OPERATIONS

Billing

2

Payments

4

Support

2

SYSTEM

Staff

Registrations

2

Activity

Algorithms

MS

Maria Santos

Owner · STF-0001

Account

View website

Sign out

Algorithms

Undo

Algorithms

How this website stores and processes its data — every search, sort, stack and queue is written by hand.

Data structures in use right now

Live sizes from the running website

Arrays — the database

Eight tables, each kept sorted by its key so binary search works.

applications 18

subscribers 10

bills 29

payments 6

tickets 4

staff 7

registrations 5

activity 39

Stack — Undo

3 of 25 actions. Push on every change, pop on Undo, peek for the button label.

Confirm payment PAY-000002 (Kristine Ramos)

Generate 8 bills for October 2026

Approve Lorenzo Pascual as Admin

Stack — Back button

28 screens remembered with their scroll position.

Activity

Payments

Pay bills

Billing

Staff

Registrations

Queue — Review

Pending applications, oldest at the front (circular buffer).

004883

004884

004885

004886

004887

004888

Queue — Support

Open tickets; “Serve next” dequeues the front.

TKT-000101

TKT-000102

Queue — Payment validation

Customer reports, oldest at the front; the front one is checked first.

PAY-000003

PAY-000004

PAY-000005

PAY-000006

Hash table — Staff e-mails

7 e-mails in 17 buckets · load 0.41 · longest chain 2 · 1 collision.

djb2 hash

separate chaining

resize at 0.75

Hash table — Payment references

5 references in 17 buckets · load 0.29 · longest chain 1. Finds a re-used reference in one look-up.

key “METHOD:REFERENCE”

value: payment ids

Time and space complexity

n = number of records; m = length of the search text

E3 Filter

Admin

Dashboard

Applications 4

New application

Subscribers

Billing 2

Payments 4

Support 2

Staff

Registrations 2

Activity

Algorithms

Algorithms

Benchmark lab

Same random data for every algorithm; counts are exact, times depend on your computer.

Records: 100 1,000 3,000

Run benchmark

Searching 1,000 records — average comparisons per look-up

Linear search

494 · O(n)

Binary search

9 · O(log n)

Sorting 1,000 records — comparisons

Insertion sort

251,483 · 8.4 ms

Selection sort

496,500 · 20.6 ms

Bubble sort

498,510 · 20.8 ms

Moves (page or shift): Insertion sort 250,491 · Selection sort 992 · Bubble sort 250,491. Binary search needs the data sorted first; selection sort always makes n(n-1)/2 comparisons.

Recent operations

The last 40 searches and sorts you triggered (a ring-buffer queue)

Algorithm	Where	n	Comparisons	Moves	Time
Bubble sort	Benchmark	1,000	498,510	250,491	20.6 ms
Selection sort	Benchmark	1,000	499,500	992	20.6 ms
Insertion sort	Benchmark	1,000	251,483	250,491	8.4 ms
Binary search	Benchmark (avg of 200)	1,000	9	0	< 0.01 ms

E3 Filter

Admin

Dashboard

Applications 4

New application

Subscribers

Billing 2

Payments 4

Support 2

Staff

Registrations 2

Activity

Algorithms

Activity

Everything that happened since the page was opened, plus the sample history.

Activity log

Newest first

Confirmed payment PAY-000002 — Kristine Ramos paid \$800

Maria Santos · Just now

Payment PAY-000006 reported by Liza Fernandez (\$800 via GCash)

Customer · Just now

Generated 8 bills for October 2026 (#9,500)

Maria Santos · Just now

Approved registration REG-000003 — Lorenzo Pascual joined as Admin

STF-0007 · Just now

Maria Santos signed in

Maria Santos · Just now

Staff registration REG-000005 from Nora Dixon (Support)

Nora Dixon · Just now

Juan Dela Cruz signed out

Juan Dela Cruz · Just now

Added Paula Ignacio as Support

Juan Dela Cruz · Just now

Installed E3-2026-004880 — new subscriber Noel Garcia

Undo stack

3 of 25 · top · newest

Last in, first out: Undo always reverses the action on top. Only the latest 25 actions are kept — when the stack is full the bottom one is dropped.

Confirm payment PAY-000002 (Kristine Ramos) [Top](#)

Generate 8 bills for October 2026

Approve Lorenzo Pascual as Admin

Activity: the log of every action and "Undo last action" (the Undo stack).

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